

THE EXACT MAXIMUM GROWTH FACTOR FOR COMPLETE PIVOTING ON REAL 5×5 MATRICES

QIANLI MA AND CHAO WU

ABSTRACT. Let $\rho_5^{\mathbb{R}}$ be the largest element growth possible in exact Gaussian elimination with complete pivoting on real 5×5 matrices, with every legal choice at a pivot tie allowed. We prove

$$\rho_5^{\mathbb{R}} = \alpha = 4.132517078632472854223346853277 \dots,$$

where α is the algebraic lower bound identified by Chen, Edelman, and Urschel, specified by their degree-61 polynomial and a rational isolating interval. Our contribution is the global upper bound. An exact inverse-elimination representation retains all initial and intermediate entry bounds of the same matrix. A constructive contraction preserves the final-pivot height while reducing the problem to two saturated tail geometries. On the negative-diagonal geometry, three feasible paths eliminate six variables and give an attained fibre maximum over seventeen shared parameters. Rational certificates control the remaining domains, while a constrained algebraic maximum argument and feasible transports resolve regions meeting the sharp value. The proof combines these analytic reductions with exact verification of a finite, complete cover. We provide a partial Lean formalization together with publicly available exact certificates and verification code.

1. THE PROBLEM AND THE RESULT

Complete pivoting selects a largest-modulus entry of the current Schur complement at every step of Gaussian elimination. Although this rule controls local multipliers, successive Schur updates can generate entries larger than those in the input. Sharp growth factors in small dimensions provide concrete tests of what global information is retained, or lost, by general pivot-growth estimates.

The values in dimensions three and four are $9/4$ and 4 , respectively; classical work includes Cryer [4] and Cohen [3]. Numerical evidence for a fifth-pivot value near 4.1325 was reported by Day and Peterson [5] and subsequently reproduced by Gould [6]. Chen, Edelman, and Urschel [2] identified the specific degree-61 algebraic value α and proved a lower bound by a completely pivoted matrix. Their Theorem 2.2 proves optimality within a prescribed equality family and supplies the lower-bound direction used here; their Theorem 3.5 gives the global upper bound 4.84 . Theorem 1.2 establishes the equality in their Conjecture 2.1. The constant, its defining

polynomial, and the equality pattern are due to Chen, Edelman, and Urschel; the advance here is the matching global upper bound.

The present result concerns a fixed dimension and the real field. It is separate from estimates as the dimension tends to infinity, including the recent results of Bisain, Edelman, and Urschel [1] and Shah and Urschel [8]. Background on growth and its relation to numerical error can be found in Day and Peterson [5] and Higham and Higham [7].

Definition 1.1. For $0 \neq A \in \mathbb{R}^{n \times n}$, let π denote a legal complete-pivoting path in exact arithmetic. Write $A_\pi^{(j)}$ for the trailing Schur matrix at stage j , before its leading pivot is eliminated. Define

$$\rho(A, \pi) = \frac{\max_j \|A_\pi^{(j)}\|_{\max}}{\|A\|_{\max}}, \quad \rho_n^{\mathbb{R}} = \sup_{A \neq 0} \sup_{\pi \text{ legal}} \rho(A, \pi).$$

If the current pivot is zero, the trailing matrix is zero and elimination terminates; all subsequent pivot magnitudes are taken to be zero.

We normalize $\|A\|_{\max} = 1$. The first pivot magnitude is then one, and complete pivoting gives

$$\rho(A, \pi) = \max_{1 \leq j \leq n} p_j, \tag{1}$$

where p_j is the magnitude of the chosen pivot. Row and column permutations may be absorbed into the original matrix. Thus it suffices to prove the upper bound for every matrix that is completely pivoted in the fixed diagonal order. This does not assume uniqueness of a largest entry, nor equality of the results of different tied paths.

Theorem 1.2. *For exact complete-pivoting Gaussian elimination on real 5×5 matrices, $\rho_5^{\mathbb{R}} = \alpha$. Here α is the root specified in Section 2. The upper bound holds for every legal pivot path, including paths with ties and inputs for which elimination terminates early.*

Why the algebraic candidate does not settle the problem. An attaining matrix supplies one legal elimination path with growth α . Even a proof that this matrix is optimal among all matrices with the same active equalities leaves other equality patterns and interior feasible points to be controlled. This is the distinction between the restricted-family result of [2, Theorem 2.2] and the global conclusion here. Their work also gives a global upper bound of 4.84, using interval covers; the question addressed in the present paper is how to reach the sharp value without assuming the candidate's equality pattern.

A second obstacle is that the successive Schur complements are coupled. An upper bound on one entry may be attained by a different completion from the one attaining another bound. Combining such separate extrema can create a tail that no input matrix produces. We retain the entire input matrix in an inverse-elimination representation, and every analytic motion below is a motion of that same complete matrix. In particular, the six-variable

maximization is attained by a feasible completion; it is not a collection of independently optimized capacities.

The sharp value presents a further difficulty for strict numerical exclusion. Let $\gamma < \alpha$ be the rational threshold fixed below. A region containing an attaining matrix cannot be ruled out under the hypothesis $F \geq \gamma$. Nor can a strict rational estimate $F \leq \alpha_- < \alpha$ cover that matrix. Such regions require a separate exact-height argument or a feasible transport to a domain already bounded by α . This explains the two types of terminal used in the proof: some exclude high points, while others retain feasible points and prove that they are safe at the sharp value.

The architecture of the upper bound. The proof is organized around the following reductions. Here F is the magnitude of the fifth pivot of a normalized matrix.

- (1) *Remove earlier-stage growth.* The bounds 1, 2, 9/4, 4 on the first four pivots hold along every legal path, including ties and early termination. A counterexample to the theorem must therefore have $F > \alpha$.
- (2) *Reach a saturated tail without losing height.* Two contractions of actual rows and columns send every positive fifth-pivot value to a diagonal-saturated or cross-saturated tail. Signs and tied boundary cases reduce these to the complete X domain or the proper negative-diagonal domain. No rank or generic-position assumption is introduced into this final division of the problem.
- (3) *Control the cross-saturated domain.* Four overlapping parameter ranges cover all of X . The ranges away from the peak admit rational exclusions. Near the peak, a canonical representative theorem, finite localization, and a constrained algebraic maximum argument give the exact bound. This result is established before it is used in the negative-diagonal proof.
- (4) *Maximize a complete negative-diagonal fibre.* At fixed seventeen shared parameters, three feasible paths maximize six remaining variables. Their endpoint realizes the maximum of the whole fibre. The question is thereby reduced to bounding one explicitly defined capacity on the space of complete frameworks.
- (5) *Cover the remaining parameter domain.* Finite certificates use exact subdivisions, necessary inequalities, qualified contractions, and the two terminal types described above. A complete source is followed through these operations to a justified terminal. The original-root composition accounts for every residual branch, including branches inherited through changes of coordinates.

The finite computations carry specific parts of Steps 3 and 5. The written argument explains both the domains they must cover and why their acceptance rules imply the required real inequalities. The supplement specifies the exact proof objects used for those obligations. One can therefore audit the reduction and the verifier logic from the text; checking every large proof

object additionally requires the supplied exact-arithmetic programs or an independent implementation of their rules.

Objects used in the certificate argument. A *physical source* is one reconstructed real matrix together with the legal pivot path and all inequalities imposed on that same matrix. A *framework* records the seventeen shared parameters of a negative-diagonal source; its fibre consists of feasible completions, not independently chosen bounds for six tail variables. A *certificate region* is a closed rational box together with its inherited conditions and coordinate transformations. A statement about such a region applies to physical sources only through the proved source-to-model inclusion. Definition 7.4 makes the negative-diagonal fibre precise. Here *complete* means that all physical conditions are retained; it does not mean that a certificate has already been verified.

Mathematical obligation	Reduction established in the text	Finite evidence consumed
Arbitrary legal real input	Early-pivot bounds, complete lift, same-height saturation	None for these reductions
Complete X domain	Four-range union; qualified canonical and local height rules	Exclusion and localization certificates on the stated sources
Negative-diagonal fibre	An attained six-variable maximum with full physical bands	None for the fibre formula
All high frameworks	Source-preserving rules and mixed-height coverage	Complete root, continuation, and safety certificates
Sharp value	Upper bound together with the known attained lower bound	Exact identities binding the algebraic constant

2. THE CONSTANT AND AN ATTAINED LOWER BOUND

Let

$$P_{61}(g) = \sum_{i=0}^{61} c_i g^i \quad (2)$$

be the polynomial in Chen, Edelman, and Urschel [2, Eq. (2.15)]. Its leading coefficient is 59049 and its constant coefficient is

$$-400768351683901408958416200638054732473753927680.$$

The Supplementary Index, S1, identifies the full ascending coefficient list and exact decimal rational endpoints $L < U$, with $U - L = 10^{-200}$, isolating one real root. Throughout, α means this root and $\alpha_- = L$. There is also a short exact identification with the published constant. All 62 integer coefficients

agree with the published polynomial. For

$$Q(t) = (1+t)^{61} P_{61}\left(\frac{4+5t}{1+t}\right),$$

the coefficients of degrees $0, \dots, 8$ are negative and those of degrees $9, \dots, 61$ are positive. Descartes' rule of signs therefore gives at most one positive root of Q , hence at most one root of P_{61} in $(4, 5)$. Exact evaluation gives $P_{61}(L) < 0 < P_{61}(U)$. Consequently α is the unique root in $(4, 5)$. The exact coefficient comparison, transformed coefficients, and endpoint signs are recorded in the Supplementary Index, S1. For orientation,

$$\begin{aligned} 4.132517078632472854223346853277 &< \alpha, \\ \alpha &< 4.132517078632472854223346853278. \end{aligned}$$

All certificate comparisons use the full rational endpoints, not these truncated decimals. We also use

$$\Lambda = \frac{1653}{400}, \quad \gamma = \frac{4132517}{10^6}, \quad 4 < \Lambda < \gamma < \alpha_- < \alpha. \quad (3)$$

The lower bound and an attaining completely pivoted matrix are [2, Theorem 2.2]. The coefficient comparison and root identification above identify their constant with the present α . Only this established lower-bound direction is needed here.

The Supplementary Index, S1, also identifies an independent exact reconstruction of the same algebraic candidate. Its archived equations, rational reconstruction functions, nonzero-denominator checks, and complete-pivoting inequalities allow the witness to be checked separately. We keep those longer calculations with the supplementary data so that the main text can develop the new global upper bound. Neither the candidate's existence nor the root isolation alone gives that upper bound.

3. AN ELEMENTARY REDUCTION TO THE FIFTH PIVOT

To bound the growth of a five-dimensional elimination, it is not enough to bound its fifth pivot unless all earlier pivots are already safe. We therefore give the elementary bounds used both for that reduction and for the parameter domains later in the proof. They also settle paths that terminate before the fifth stage.

Lemma 3.1. *For a normalized real 3×3 matrix completely pivoted in the indicated order, let its pivot magnitudes be $1, t, g$. Then*

$$0 \leq t \leq 2, \quad g \leq \Phi(t), \quad \Phi(t) = \begin{cases} 2t, & 0 \leq t \leq 1, \\ t(3-t), & 1 \leq t \leq 2. \end{cases}$$

Each branch is sharp. In particular, $g \leq 9/4$.

Proof. The first Schur update is a difference of two quantities of magnitude at most one, so $t \leq 2$. If $t = 0$, its whole trailing block is zero. Otherwise the last update gives $g \leq 2t$, proving the first branch.

For $1 < t \leq 2$, normalize the first row and column to be nonnegative by independent sign changes. A positive entry of the first Schur complement is at most one, so the second pivot is negative. Write the tail as

$$\begin{pmatrix} -t & \gamma_0 \\ \eta_0 & \zeta_0 \end{pmatrix}.$$

Let the relevant first-row and first-column coordinates be $x, u, y, v \in [0, 1]$, and set

$$c = t - 1, \quad s = xv, \quad z = uy, \quad r = uv, \quad w = xy.$$

The original entry at the pivot is $w - t$, whence $w \geq c$. Furthermore $sz = wr$ and $r \geq sz$. Intersecting the original entry bounds with the Schur pivot bounds gives

$$\begin{aligned} \gamma_0 &\in [-1 - \min(c, s), 1 - s], \\ \eta_0 &\in [-1 - \min(c, z), 1 - z], \\ \zeta_0 &\in [-1 - \min(c, r), 1 - r]. \end{aligned} \tag{4}$$

We claim that, with $M = t^2(3 - t)$,

$$t(1 - r) + (1 + \min(c, s))(1 + \min(c, z)) \leq M, \tag{5}$$

$$t(1 + \min(c, r)) + (1 + \min(c, s))(1 - z) \leq M. \tag{6}$$

For (5), replace r by sz . When $s, z \leq c$ the resulting function increases in either variable, since its corresponding derivative is $1 - cz$ or $1 - cs$; its maximum is at $s = z = c$. If $s \leq c \leq z$, it is $t[2 + s(1 - z)] \leq t[2 + c(1 - c)] = M$. The other mixed case is symmetric. If $s, z \geq c$, the bound is $t(1 - c^2) + t^2 = M$.

For (6), the shared-product relations imply $z \geq cr$, and also $z \geq r$ whenever $s \leq c$. The latter includes $s = 0$: then $sz = wr$ and $w \geq c > 0$ force $r = 0$. Thus no division by s is needed. Separate four closed cases. If $r \geq c, s \geq c$, then $z \geq cr \geq c^2$, and the expression is at most $t^2 + t(1 - c^2) = M$. If $r \geq c, s \leq c$, then $z \geq r \geq c$, and the bound $t^2 + t(1 - c) = 2t \leq M$ suffices. If $r \leq c, s \leq c$, the expression is at most

$$t(1 + r) + (1 + s)(1 - r) \leq 2t \leq M;$$

the first difference is $(c - s)(1 - r) \geq 0$, and $M - 2t = tc(1 - c) \geq 0$. Finally, if $r \leq c, s \geq c$, then $r - z \leq r(1 - c) \leq c(1 - c)$. The expression is at most $t[2 + c(1 - c)] = M$. Interchanging s, z gives the symmetric form of (6).

Put $A_s = 1 + \min(c, s)$, $B_s = 1 - s$, and similarly for z, r . By (4),

$$t\zeta_0 + \gamma_0\eta_0 \leq tB_r + \max(A_sA_z, B_sB_z) \leq M.$$

The first product uses (5); the second is at most $t + 1 \leq M$. On the other side,

$$-(t\zeta_0 + \gamma_0\eta_0) \leq tA_r + \max(A_sB_z, B_sA_z) \leq M$$

by (6) and its symmetric version. Thus the magnitude of the tail determinant is at most M , and division by t proves $g \leq t(3 - t)$.

Sharp examples are

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & t & t \\ 0 & -t & t \end{pmatrix} \quad (0 \leq t \leq 1), \quad \begin{pmatrix} 1 & 1 & c \\ 1 & -c & -1 \\ c & -1 & 1 \end{pmatrix}, \quad c = t - 1 \quad (1 \leq t \leq 2).$$

Finally $9/4 - t(3 - t) = (t - 3/2)^2$. $\square$

Corollary 3.2. *The first four pivot magnitudes of a normalized real completely pivoted matrix of order at most five satisfy*

$$p_1 = 1, \quad p_2 \leq 2, \quad p_3 \leq 9/4, \quad p_4 \leq 4.$$

Consequently $\rho(A, \pi) > \alpha$ implies $p_5 > \alpha$.

Proof. Any leading submatrix in the chosen pivot order is completely pivoted, and a Schur tail divided by its first pivot magnitude is normalized and completely pivoted. Put $x = p_2$, $y = p_3/p_2$ when these denominators are nonzero. Lemma 3.1 gives

$$xy \leq \Phi(x), \quad p_4 \leq x\Phi(y).$$

If $x \leq 1$, the second bound is at most $9/4$. If $1 \leq x \leq 3/2$, it is at most $27/8$. If $3/2 \leq x \leq 2$, then $y \leq 3 - x \leq 3/2$. The function Φ is nondecreasing on $[0, 3/2]$, so

$$p_4 \leq x\Phi(3 - x) = x^2(3 - x) \leq 4,$$

using $4 - x^2(3 - x) = (2 - x)^2(x + 1)$. Zero pivots imply zero subsequent growth. Equation (1) and $\alpha > 4$ finish the proof. $\square$

4. A COMPLETE TWO-STEP LIFT

The representation in this section is the point at which entry bounds from different stages are coupled. Starting from the third Schur block, we undo the preceding two elimination steps and retain the original matrix as well. Consequently every row, column, and auxiliary product used later has a simultaneous physical interpretation.

All subsequent variables describe one common matrix. Let $u, x, v, q \in \mathbb{R}^3$, $p, k > 0$, and $e, \beta \in \mathbb{R}$. Set

$$M = \begin{pmatrix} 1 & -e & v^T \\ \beta & p - e\beta & (q + \beta v)^T \\ u & px - eu & D + xq^T + uv^T \end{pmatrix}, \quad D = kC, \quad C_{00} = 1. \quad (7)$$

Subscripts on 3×3 blocks run from zero to two. Its first two Schur tails are

$$M^{(2)} = \begin{pmatrix} p & q^T \\ px & D + xq^T \end{pmatrix}, \quad M^{(3)} = D. \quad (8)$$

Proposition 4.1. *Suppose C is completely pivoted in the fixed order, with $\|C\|_{\max} = 1$. Then (7) is normalized and completely pivoted, with first three pivot magnitudes $1, p, k$, if and only if*

$$|e|, |\beta|, \|u\|_{\infty}, \|v\|_{\infty}, \|x\|_{\infty} \leq 1, \quad (9)$$

$$|p - e\beta|, \|px - eu\|_\infty, \|q + \beta v\|_\infty \leq 1, \quad (10)$$

$$\|q\|_\infty \leq p, \quad \|D + xq^T\|_{\max} \leq p, \quad (11)$$

$$\|D + xq^T + uv^T\|_{\max} \leq 1. \quad (12)$$

Every normalized fixed-order completely pivoted matrix with three nonzero leading pivots has this representation after row-sign normalization of those pivots.

Proof. The identities (8) show sufficiency: the first matrix has entries bounded by one, the first tail by p , and the second tail is kC . Conversely recover e, β, u, v from the initial matrix, then p, x, q from its first Schur tail. The remaining tail is uniquely D ; division by its positive leading pivot gives C . The inequalities are precisely the initial and first-stage bounds. No independent completion is chosen for different inequalities. $\square$

The alternative notation $q = py$ makes the bottom block $kC + uv^T + pxy^T$. The actual variable q , rather than y , will be used in the seventeen-dimensional reduction. Write

$$C = \begin{pmatrix} 1 & a & b \\ c & ca + \rho & cb + \sigma \\ d & da + \tau & db + \omega \end{pmatrix}. \quad (13)$$

The fixed-order core conditions are $\|C\|_{\max} \leq 1$ and $|\sigma|, |\tau|, |\omega| \leq |\rho|$. For $\rho \neq 0$, set

$$h(C) = \omega - \tau\sigma/\rho, \quad F = k|h(C)|.$$

If $\rho = 0$, the final tail is zero; set $h(C) = F = 0$. Lemma 3.1 also yields

$$k \leq 9/4, \quad |h(C)| \leq 9/4, \quad k \leq 2p, \quad |h(C)| \leq 2|\rho|. \quad (14)$$

All these quantities belong to the same source.

5. CONSTRUCTIVE SATURATION OF THE LAST TAIL

The next reduction explains why two tail geometries suffice. It constructs a new complete matrix with the same last-pivot height from each input; it does not infer the candidate's tight constraints from numerical experiments. The construction uses contractions of actual rows and columns, which lets us check all earlier pivoting conditions at once.

The next reduction is an existence statement about representatives. It does not assert that every maximizer is already saturated.

Lemma 5.1 (Same-height diagonal saturation). *Every complete lift (7) with $F > 0$ has another complete lift with the same p, k, F whose normalized tail satisfies either*

$$D : \quad |\omega| = |\rho|, \quad |\sigma|, |\tau| \leq |\rho|,$$

or

$$X : \quad |\sigma| = |\tau| = |\rho|, \quad |\omega| \leq |\rho|.$$

The construction is valid on every legal tied pivot path.

Proof. For $0 \leq \lambda, \mu \leq 1$, let $R_\lambda = \text{diag}(1, \lambda, 1)$, $K_\mu = \text{diag}(1, \mu, 1)$. Replace

$$C \mapsto R_\lambda C K_\mu, \quad (u, x) \mapsto (R_\lambda u, R_\lambda x), \quad (v, q) \mapsto (K_\mu v, K_\mu q).$$

The new full matrix is $\text{diag}(I_2, R_\lambda) M \text{diag}(I_2, K_\mu)$. Its initial and first-tail entries are only contracted in magnitude, while the pivots $1, p, k$ do not change. The core tail transforms as

$$(\rho, \sigma, \tau, \omega) \mapsto (\lambda\mu\rho, \lambda\sigma, \mu\tau, \omega). \quad (15)$$

Since $F > 0$, $\rho \neq 0$. Put

$$S_0 = |\sigma/\rho|, \quad T_0 = |\tau/\rho|, \quad W_0 = |\omega/\rho|, \quad S_0, T_0, W_0 \in [0, 1].$$

If $W_0 \geq S_0 T_0$, choose $\lambda = \max(T_0, W_0)$, $\mu = W_0/\lambda$. In this case $W_0 = 0$ would force $S_0 T_0 = 0$, hence $\omega = 0$ and $\sigma\tau = 0$. This would give $F = k|\omega - \tau\sigma/\rho| = 0$, contrary to the hypothesis. Thus $W_0 > 0$, the division is valid, and $\lambda\mu > 0$. Now $\lambda \geq T_0$, $\mu \geq S_0$, and $\lambda\mu = W_0$, so the new completely pivoted tail lies on D . If $W_0 \leq S_0 T_0$, take $\lambda = T_0$, $\mu = S_0$. Here $S_0 T_0 = 0$ would also force $W_0 = 0$ and $F = 0$, so $\lambda\mu = S_0 T_0 > 0$; the new tail lies on X . At the positive-product junction $W_0 = S_0 T_0 > 0$, the constructions agree. Finally,

$$\omega - \frac{(\mu\tau)(\lambda\sigma)}{\lambda\mu\rho} = \omega - \frac{\tau\sigma}{\rho}.$$

This preserves F and all the complete early-stage conditions. No exchange of unequal pivots is introduced. $\square$

Zero fifth pivots are already safe by Corollary 3.2. They need not have a zero fourth-stage tail: a nonzero rank-one 2×2 tail can have zero last pivot. In contrast, a zero fourth pivot under complete pivoting does force the entire current tail to vanish. The lemma requires neither a zero-target extension nor equality of the two full-matrix constructions at a zero-product junction.

Actual sign changes on the final rows and columns reduce every X -tail to

$$H = \begin{pmatrix} r & r \\ r & w \end{pmatrix}, \quad r > 0, \quad |w| \leq r, \quad F = r - w. \quad (16)$$

Here H is the *actual* fourth-stage tail, whose entries are k times the normalized core-tail entries. Indeed, for $\begin{pmatrix} \epsilon_1 r & \epsilon_2 r \\ \epsilon_3 r & W \end{pmatrix}$, multiply rows by (ϵ_1, ϵ_3) and columns by $(1, \epsilon_1 \epsilon_2)$. This gives (16), with $w = \epsilon_1 \epsilon_2 \epsilon_3 W$, and preserves all absolute matrix bounds. Similarly every D -tail has a form

$$H = \begin{pmatrix} r & s \\ t & \chi r \end{pmatrix}, \quad r > 0, \quad 0 \leq s, t \leq r, \quad \chi \in \{-1, 1\}. \quad (17)$$

If $\chi = 1$, its height is $r - st/r \leq r \leq 4$; a zero arm also gives height $r \leq 4$. If $s = r$ or $t = r$, exchanging the corresponding tied final columns or rows and

normalizing signs gives (16) at the same height. The sole additional domain is therefore

$$H = \begin{pmatrix} r & s \\ t & -r \end{pmatrix}, \quad 0 < s, t < r, \quad F = r + \frac{st}{r}. \quad (18)$$

We call this the proper negative diagonal domain.

6. THE COMPLETE CROSS-SATURATED BOUND

For (16), the actual core matrix is

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r + cB \\ dk & r + dA & w + dB \end{pmatrix}. \quad (19)$$

All of (9)–(12) and $\|D\|_{\max} \leq k$ are retained. No rank, cofactor-sign, or auxiliary entry condition is assumed.

Proposition 6.1 (Certified complete X bound). *Every real complete lift with (19), $p, k, r > 0$, and $|w| \leq r$, satisfies $F = r - w \leq \alpha$.*

The proposition is assembled from four groups of certified source domains. Appendix A and the Supplementary Index, S4, identify the component certificates and their source domains.

Complete source domain	Certified bound
$0 < k \leq 2$	$F < 1653/400$
$k \geq 21/10$	$F < \gamma$
$2 < k \leq 21/10, r > k$	$F < \gamma$
$\gamma/2 \leq k \leq 21/10, r \leq k, F \geq \gamma$	$F \leq \alpha$

Assembly. The first three rows leave $2 < k < 21/10, r \leq k$. If $F < \gamma$, (3) suffices. Otherwise $\gamma \leq F \leq 2r \leq 2k$, hence $k \geq \gamma/2$, and the last row applies. The boundary $r = k$ lies in the last row; $k = 2, 21/10$ have closed coverage. Internal sign and transpose normalizations are actual matrix operations imposing no extra rank or cofactor hypothesis. $\square$

The small- k certificate has 268817 nodes, 134409 contradiction leaves, and no open leaf. The accepted two low- r roots have the following counts:

Root	Nodes	Contradiction leaves	α -safe leaves	Open
I	3450095	1725047	1	0
II	5920895	2873490	86958	0

The counts identify accepted objects, rather than replacing their acceptance rules and domain maps. An α -safe leaf can contain feasible points; it is not an infeasibility certificate at γ .

Order of the analytic inputs. The strict- and weak-room statements, their full physical predicates, and the qualified 22-coordinate wall entrance are given in Section 6.2. They use the independent canonical-height theorem below. The 24-coordinate two-gap transport of Lemma 8.1 instead uses the complete X bound; it is not an input to this proposition.

6.1. The exact height in the canonical near-peak domain. The near-peak part of the argument requires an algebraic height bound. A terminal estimate of the form $F \leq \alpha_- < \alpha$ cannot cover a matrix whose height is exactly α . We therefore explain the separate argument which supplies the exact bound, before using it in local transports and in the complete X -domain certificate.

Let $\mathcal{K}$ be the positive canonical chamber in Definition B.1. Its matrices have the complete physical representation of Section 4, with $e = 1$, $u = (1, V, T)^T$, $P_1 = -1$, $0 < V < 1$, the stated positive receiver coordinates and six strictly positive cofactors. All initial, intermediate, and core entry bounds remain part of the definition. No tail contact, equality of row scales, or equality-family hypothesis is added to this chamber.

Theorem 6.2 (Canonical exact height). *Every complete physical matrix in $\mathcal{K}$ satisfies $F \leq \alpha$. More precisely, every admissible target representative of Definition B.22 has height at most α . The representative and reconstruction results in Propositions B.21 and B.23 then give the assertion for every matrix in $\mathcal{K}$.*

Proof. We give the argument here; Appendix B specifies the complete source predicate, reconstruction formulas, eight constraints, and finite verification statements used in the proof. Set

$$f_0 = \frac{4132517}{10^6}, \quad f_1 = \frac{103313}{25000}, \quad f_0 < \alpha < f_1.$$

Suppose a source has height greater than α . The target reduction provides, at the same seven-coordinate head, a complete representative of any chosen height f between α and the source height, with $f < 5$. It does not assert that the source itself already has the representative's contacts.

In the row-scale coordinates write

$$p = 1 + s, \quad r = 1 + R, \quad F = 2(1 + R), \quad z = (\beta, V, s, Y, Z, v, h, b, R).$$

The variables t, E, X, k, g, j, a and the complete matrix are recovered from this one tuple by (79). The equivalence in Proposition B.23 retains both column capacities and the strict scale qualifications; it is not a relaxation obtained by completing different columns separately.

First, a finite rational certificate puts every height- f_0 representative into the six-dimensional box Q in (92). Comparing target representatives at the same old head is legitimate because their tail receiver satisfies the identity

$$v(R) = \frac{R + Y - t(1 + X)}{t(1 - \beta X) - \Gamma}, \quad \Gamma = V - \beta Y.$$

On the height- f_0 representatives the denominator is greater than $1/5$. Consequently, increasing the target from f_0 to f_1 changes v by less than 10^{-5} , while the other five localization coordinates stay fixed. A separate finite certificate excludes height f_1 in this enlarged box. Every potentially supercritical representative therefore has $F < f_1$ and lies in the nine-dimensional box $\mathcal{B}$ of (95). This step is what connects a fixed-target localization certificate to an upper bound for all possible source heights.

Second, let $G_1, \dots, G_8$ be the physical slacks in (96). On $\mathcal{B}$, all the other strict qualifications have uniform positive margins. The exact reconstruction theorem thus identifies the high-value completion set with

$$\{z \in \mathcal{B} : G_1(z), \dots, G_8(z) \geq 0\}.$$

It is compact. Under the assumed counterexample, its maximum height is greater than α ; no separate numerical witness is needed for this existence argument. For fixed s , the Jacobian

$$J = \frac{\partial(G_1, \dots, G_8)}{\partial(\beta, V, Y, Z, v, h, b, R)}$$

is invertible throughout $\mathcal{B}$. The uniform rational certificate in Proposition B.27 gives

$$\frac{\partial R}{\partial G_i} < -\frac{1}{250} \quad (1 \leq i \leq 8).$$

If a maximizing point had $G_i > 0$, the local inverse-function coordinates would allow that slack to decrease, with s and all other slacks fixed. This would preserve the physical inequalities and strictly increase the height. A short such arc is an arc of the original completion model; the proved global localization prevents an artificial box boundary from invalidating the argument. Hence a maximizing point satisfies all eight equalities.

Third, these equalities force

$$\begin{aligned} t = X = Z = 1, \quad h = T = s, \quad b = k = r + Ea, \\ O_{01} = -1, \quad S_{01} = -p, \quad O_{21} = 1. \end{aligned}$$

Only at this stage is the contact family used. With $\epsilon = r - 2$, its remaining equation is the explicitly defined polynomial $G(s, \epsilon) = 0$ in (102). The same invertible eight-constraint Jacobian makes the contact family a local smooth curve parametrized by s . At its maximum, $\epsilon'(s) = 0$, so differentiation gives $G_s(s, \epsilon) = 0$. This conclusion does not require a separate assumption that $G_\epsilon \neq 0$.

Finally, exact elimination gives

$$\text{Res}_s(G, G_s) = \frac{\epsilon^{34}(\epsilon - 2)(\epsilon - 1)^2(\epsilon + 1)^7(\epsilon^2 - 2\epsilon + 2)^2}{131072} P_{61}(4 + 2\epsilon).$$

Every factor outside P_{61} is nonzero in the proved height interval. By Section 2, the polynomial P_{61} has exactly one root in $(4, 5)$, namely α , and therefore exactly one in $[f_0, f_1]$. Thus the maximum must equal α , contrary to the assumed larger height. $\square$

The proof uses a constrained maximum argument before it invokes stationarity and root identification. It therefore does not infer a neighborhood bound from an isolated critical point, nor does it use the restricted-family optimum in Chen–Edelman–Urschel [2], Theorem 2.2, as a substitute for reaching that family. The source reductions and the finite certificates have distinct roles, all specified in Appendix B. The local X -transports first reach an actual positive-room matrix, to which this canonical result and the explicit room-reduction results apply. The complete X bound subsequently serves the negative-diagonal proof; it is not an input to the canonical-height argument.

6.2. The positive-room interfaces and the complete source cover.

Here the variables are those of the complete lift (7), with the core (19). Thus $v, q \in \mathbb{R}^3$ are the actual receiver vectors, and $P_j = \beta v_j + q_j$. Write $u = (U, V, T)^T$, $x = (X, Y, Z)^T$, and define

$$\begin{aligned} \delta &= U - \beta X, & \Gamma &= V - \beta Y, & \nu &= \beta Z - T, \\ \tau &= UY - XV, & \Delta &= UZ - TX, & \mu &= VZ - TY. \end{aligned} \quad (20)$$

These are room cofactors; Γ is distinct from the height threshold γ , and τ here is distinct from the later tail-gap coordinate. A *strict positive-room source*, denoted $\mathcal{R}_0^+$, satisfies all of (9)–(12), $\|D\|_{\max} \leq k$, and

$$\begin{aligned} p, k, r &> 0, & |w| &\leq r, & A, B, d &\leq 0, \\ \beta, U, V, T, X, Y, Z &> 0, & \delta, \Gamma, \nu, \tau, \Delta, \mu &> 0. \end{aligned} \quad (21)$$

Here “strict” refers to the seven receiver signs and six cofactor signs; physical pivot ties and the weak bounds on A, B, d are retained. The physical bands retain every upper bound on these coordinates. The *weak positive room* $\mathcal{R}_0^{\geq 0}$ replaces only the seven receiver and six cofactor positivity conditions by nonnegativity. It retains $p, k, r > 0$, $A, B, d \leq 0$, and all physical bands. Neither room imposes an additional rank hypothesis.

Proposition 6.3 (Strict-room reduction and weak-room closure). (i) *Every source in $\mathcal{R}_0^+$ with $F \geq 33/8$ has a complete physical representative in $\mathcal{K}$ of Definition B.1 with exactly the same height. Consequently every source in $\mathcal{R}_0^+$ satisfies $F \leq \alpha$.*

(ii) *Every source in $\mathcal{R}_0^{\geq 0}$ is a limit of complete strict-room sources whose heights converge to its height. Consequently it also satisfies $F \leq \alpha$, including when receiver coordinates, receiver vectors, or cofactors vanish.*

Reduction and closure inputs. For (i), the two-face normalization preserves the complete physical conditions and F , and sends every strict-room source with $F > 4$ to one of

$$e = 1, P_1 = -1, \quad \text{or} \quad e = 1, P_2 = 1, v_2 < 0.$$

It changes only e, β , preserving D, p, u, x, v, q and the strict cofactor signs. The second face, including its boundary $v_2 = 0$, has $F < 33/8$. On the first

face the feasible shear

$$p' = p + \lambda, \quad \beta' = \beta + \lambda, \quad u' = u + \lambda x, \quad q' = q - \lambda v,$$

$$\lambda = \min \left\{ 1 - \beta, \frac{1 - U}{X}, \frac{1 - V}{Y}, \frac{1 - T}{Z} \right\}.$$

preserves $D, F, e = 1, P_1 = -1$, every physical band and all six cofactors. To identify the endpoint, $F > 4$ and $|w| \leq r$ give $r > 2$, while $r + dA \leq k$ and $dA \geq 0$ give $k \geq r > 2$. Put $O_{00} = k + Xq_0 + Uv_0$. The physical bands and $\delta > 0$ imply

$$1 + U + (1 - \beta)X - k = (1 - O_{00}) + X(1 + P_0) + \delta(1 + v_0) \geq 0.$$

Thus $k \leq 3 - \beta$, and every such high-value source has $\beta < 1$. This also applies to the physical shear endpoint, so $\beta' < 1$. Moreover $\nu' = \beta'Z - T' > 0$, with $\beta', Z \leq 1$, gives $T' < 1$. At least one of the four bounds defining λ is attained; hence the endpoint has $U' = 1$ or $V' = 1$. This includes tied minima and $\lambda = 0$. The latter face also has $F < 33/8$; thus a source with $F \geq 33/8$ reaches $U' = 1, 0 < V' < 1$. Here p is allowed to change in the shear. The exact dictionary to Definition B.1 is

$$t = c, \quad E = -d, \quad a = -A, \quad b = -B,$$

$$\hat{v} = (v_0, v_1, v_2)^T, \quad \hat{q} = (q_0, -1 - \beta v_1, q_2)^T.$$

The canonical scalars g, h, j are respectively $-v_0, v_1, -q_0$; the scalars denoted v, q there are v_2, q_2 here. This is a single complete matrix at each stage. Theorem 6.2 now gives the bound; sources below $33/8$ already lie below α . The original two-face, tail-face and middle-face statements are V10, Sections 2–6; V11, Theorem T11; and V12, Section 8, respectively, as indexed in Supplementary Index S4.

For (ii), V27, Theorem 4.1, first applies a positive diagonal contraction that preserves all weak signs and gives strict margins in the physical bands changed by the subsequent regularization. Its height is $\lambda_0^2 F$, with $0 < \lambda_0 < 1$ tending to one. It then regularizes the four ordered nonnegative rays

$$(U, X), \quad (V, Y), \quad (\beta, 1), \quad (T, Z),$$

whose six ordered determinants are precisely (20). The regularization makes all receiver coordinates and all six determinants positive, including at zero vectors, while retaining the contracted p, D, e, v, q and hence its height. Reconstructing (7) preserves the physical bands by their strict margins. Taking the contraction and regularization parameters to their limits gives the stated complete physical approximation and the bound from (i). This closure uses all six weak cofactors. Three adjacent weak minors, or the nonnegative version of a strict ten-term entry test, do not suffice. No same-height strict-room representative is asserted for an arbitrary weak-room source. $\square$

Lemma 6.4 (Qualified complete X -to-room wall entrance). *Use the ordered native coordinates*

$$z = (k, r, w, A, B, c, d, p, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2).$$

Let $\bar{c}_j$, $j = 0, 1, 2, 3$, be the first 22 coordinates of the rational centres in Appendix C.2, in the declared ξ -order of (35) rather than the grouped display order of the centre tables. If a complete physical X source satisfies

$$\|z - \bar{c}_j\|_\infty + 3(r + w) < \frac{1}{1250}, \quad (22)$$

there is a complete physical path to $r + w = 0$, followed by a signed permutation of the last two rows and columns, whose complete endpoint $\tilde{z}$ belongs to $\mathcal{R}_0^+$ and satisfies

$$F(\tilde{z}) \geq F(z) + \frac{r + w}{10}.$$

Thus $F(z) \leq \alpha - (r + w)/10$.

Wall-flow input. The 22-coordinate V35 certificate and its Round47 flow-tube statement in Supplementary Index S4 keep the source's p and twenty selected physical slacks fixed, decrease $r + w$ at unit speed, and certify speed less than 3 and height gain at least $1/10$. All other physical slacks stay positive on the radius- $1/1250$ outer box. Condition (22) keeps the entire path strictly inside this box, so the ordinary differential equation continues to the wall. Only there is the signed tail permutation applied. Its sixteen certified strict conditions are the seven positive receiver coordinates, $-A, -B, -d > 0$, and all six positive cofactors; these imply (21). Signed permutations preserve the physical bands and height. Proposition 6.3 then applies. When $r + w = 0$, the path has zero length and the same endpoint argument applies. The use of the centres above merely identifies the common rational data: this proof uses the original 22-coordinate wall certificate. $\square$

Coverage and direction of dependence. The wall entrance is a qualified local exit. The complete low- r trees also have weak-room exits using Proposition 6.3(ii), direct rational height exits, and exact contradiction leaves. Their original-root cover retains all other signs, zero receivers and cofactors, and physical ties; these are not assumed to enter a positive room. The two complete low- r roots, together with the first three domains of Proposition 6.1, supply the complete cover. In particular $r = k$ remains in the low- r root, and the interfaces $k = 2, 21/10$ are covered as stated in its assembly. Earlier zero pivots have zero subsequent growth by Section 3.

The dependency is the canonical-height theorem, then the room bounds, then the wall exits and complete X cover. The 24-coordinate two-gap transport of Lemma 8.1 instead preserves $r + w$, removes the two cross gaps, and uses Proposition 6.1 to bound its endpoint. The later V43 guard-preserving flow also uses that proposition. Neither is an input to the room bounds or the complete X certificate.

7. EXACT REDUCTION OF THE NEGATIVE DIAGONAL DOMAIN

Replace (19) by

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & s + cB \\ dk & t + dA & -r + dB \end{pmatrix}. \quad (23)$$

We initially allow $0 \leq s, t \leq r$, so the safe boundary is included. A *complete source* means (7), (9)–(12), $\|D\|_{\max} \leq k$, and these tail conditions. The normalized first Schur tail is a real 4×4 completely pivoted matrix. Corollary 3.2 gives

$$F \leq 4p. \quad (24)$$

Hence $F > 4$ requires $p > 1$. The head bound implies $e\beta \geq p - 1 > 0$. A simultaneous sign flip of the second row and column gives $e, \beta > 0$. Simultaneously reversing all three remaining rows and columns leaves D fixed and reverses x , allowing $x_0 \geq 0$. Simultaneously reversing just the last two rows and columns reverses A, B, c, d , leaves x_0 fixed, and leaves the last tail fixed, allowing $A \leq 0$. No sign is imposed on the other receivers or on c, d .

Fix the seventeen-coordinate common framework

$$\vartheta = (k, A, B, c, d, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2). \quad (25)$$

The purpose of keeping this common framework fixed is to separate two questions: which tail values are attainable by one complete matrix, and how large the best such tail can be. The following three steps answer both questions exactly. In contrast to independent entry bounds, the resulting capacities are realized simultaneously.

We maximize over the six remaining actual variables (β, p, e, r, s, t) , in the normalized range $p \geq 1$, $0 \leq e, \beta \leq 1$. Every high source has such a representative by the preceding argument. The constructions give feasible paths, not independent necessary relaxations.

7.1. The right-prefix interval. Define

$$I_\beta = [0, 1] \cap \bigcap_{j=0}^2 \{b : |q_j + bv_j| \leq 1\}.$$

If $v_j = 0$, this condition is $|q_j| \leq 1$; otherwise its endpoints are $(-1 - q_j)/v_j$ and $(1 - q_j)/v_j$, sorted in increasing order. An empty interval rules out the fibre. When it is nonempty, set $\bar{\beta} = \max I_\beta$.

Lemma 7.1. *For a complete source with $p \geq 1$, $0 \leq e, \beta \leq 1$, increasing β linearly to $\bar{\beta}$, while fixing the other variables, preserves complete feasibility and F .*

Proof. The right-prefix bands define the convex interval I_β . The head upper bound $p - e\beta \leq 1$ improves, and its lower bound is automatic because $p \geq 1$ and $e\beta \leq 1$. No other condition or height depends on β . $\square$

7.2. The attained p, e capacity. If $\bar{\beta} = 0$, the head forces $p = 1$, safe by (24). Suppose $\bar{\beta} > 0$. Form the head row $(a_0, b_0) = (1, \bar{\beta})$. For every $x_i \neq 0$, add the row

$$a_i = |x_i| > 0, \quad b_i = \operatorname{sgn}(x_i)u_i.$$

When $x_i = 0$, its prefix bound is automatic. There are at most four rows, with $0 < a_i \leq 1$, $|b_i| \leq 1$. The complete remaining prefix system is

$$a_i p - b_i e \leq 1, \quad 0 \leq e \leq 1, \quad p \geq 1. \quad (26)$$

The reverse bound is automatic from $a_i p \geq 0$ and $b_i e \leq 1$. Define capacities for $b_i \geq 0$ and for $b_i > 0 > b_j$, respectively:

$$C_i = \frac{1 + b_i}{a_i}, \quad C_{ij} = \frac{b_i - b_j}{a_j b_i - a_i b_j}.$$

All denominators are positive. A single row bounds p by using $e \leq 1$; a positive and a negative row bound it by requiring their lower and upper limits for e to intersect. There are no other obstructions in this two-dimensional polygon. Put

$$P = \min(C_i, C_{ij}), \quad e_* = \max\left(0, \max_{b_i > 0} \frac{a_i P - 1}{b_i}\right). \quad (27)$$

Lemma 7.2. *The maximum p in (26) is P ; (P, e_*) is feasible. Furthermore $1 \leq P \leq 2$, and at most six capacities occur.*

Proof. Writing $\ell_i = 1 - a_i p + b_i e \geq 0$, feasibility implies

$$1 + b_i - a_i p = \ell_i + b_i(1 - e) \geq 0 \quad (b_i \geq 0),$$

and

$$b_i - b_j - (a_j b_i - a_i b_j)p = b_i \ell_j - b_j \ell_i \geq 0 \quad (b_i > 0 > b_j).$$

Thus every capacity is necessary. Conversely positive b_i give lower bounds for e , negative b_j give upper bounds, and zero b_i constrain only p . The C_i make all lower bounds at most one; the C_{ij} make each lower bound at most each upper bound. A negative row additionally requires a nonnegative upper bound, namely $p \leq 1/a_j$. Its pairing with the head implies this:

$$\frac{\bar{\beta} - b_j}{a_j \bar{\beta} - b_j} \leq \frac{1}{a_j},$$

since $a_j \leq 1$, $b_j < 0$. The head lower bound $(P - 1)/\bar{\beta}$ is nonnegative. Therefore (27) satisfies every row. Each capacity is at least one, and the head capacity is $1 + \bar{\beta} \leq 2$. If the numbers of positive, zero, and negative rows are m, z, n , then their number of capacities is $m + z + mn$, where $m \geq 1$, $m + z + n \leq 4$, and hence at most six. $\square$

After Lemma 7.1, move (p, e) along the segment to (P, e_*) . The prefix polygon is convex and p increases. The actual matrices

$$D, \quad S = D + xq^T, \quad O = D + xq^T + uv^T$$

stay fixed. Thus $|q_j| \leq p$, $|S_{ij}| \leq p$ improve, and all other bands remain valid. This second path preserves F ; it does not require e itself to vary monotonically.

7.3. The attained tail maximum. At $p = P$, form joint intervals for the entries of D :

$$\begin{aligned}\ell_{ij} &= \max(-k, -P - x_i q_j, -1 - x_i q_j - u_i v_j), \\ U_{ij} &= \min(k, P - x_i q_j, 1 - x_i q_j - u_i v_j).\end{aligned}\tag{28}$$

The five entries in the zeroth row or column of D are fixed. Check their intervals and $|q_j| \leq P$. Failure makes the fibre empty: decreasing p cannot repair a failed stage bound, while original-matrix and core bounds do not depend on p . For the other entries put

$$\begin{aligned}r_- &= \max(0, \ell_{11} - cA, dB - U_{22}), & R &= \min(U_{11} - cA, dB - \ell_{22}), \\ s_- &= \max(0, \ell_{12} - cB), & s_+ &= U_{12} - cB, \\ t_- &= \max(0, \ell_{21} - dA), & t_+ &= U_{21} - dA, \\ S_* &= \min(R, s_+), & T_* &= \min(R, t_+).\end{aligned}$$

Theorem 7.3 (Complete six-variable fibre maximum). *For a nonempty high-value fibre at fixed ϑ , the construction yields a complete source with*

$$(\beta, p, e, r, s, t) = (\bar{\beta}, P, e_*, R, S_*, T_*),$$

and

$$\mathcal{B}(\vartheta) = R + \frac{S_* T_*}{R} = \max_{\text{complete six-variable fibre}} F.\tag{29}$$

The tail fibre is nonempty exactly when the fixed checks pass and

$$R > 0, \quad R \geq r_-, \quad S_* \geq s_-, \quad T_* \geq t_-.$$

Every original source reaches the maximum by three complete feasible paths with nondecreasing height.

Proof. The first two paths have been established. At their endpoint the tail feasible set is exactly

$$r_- \leq r \leq R, \quad s_- \leq s \leq s_+, \quad t_- \leq t \leq t_+, \quad r > 0, \quad 0 \leq s, t \leq r.$$

The intervals are computed simultaneously from the same u, x, v, q . Nonemptiness implies $R \geq r > 0$, $S_* \geq s$, $T_* \geq t$; then (R, S_*, T_*) satisfies every interval. The stated conditions are therefore necessary and sufficient. The endpoint dominates every feasible tail coordinatewise. Coordinatewise domination would not suffice for an arbitrary nonlinear objective: here the complete-pivoting condition $st \leq r^2$ supplies the needed sign. The following identity makes that dependence explicit. For any feasible tail,

$$Rr(\mathcal{B} - F) = (R - r)(Rr - st) + rt(S_* - s) + rS_*(T_* - t) \geq 0.\tag{30}$$

The factors have the required signs because $R \geq r$ and $st \leq r^2$. The segment to this maximum stays in the convex feasible set, with all three coordinates nondecreasing. Along it,

$$\partial_r F = 1 - st/r^2 \geq 0, \quad \partial_s F = t/r \geq 0, \quad \partial_t F = s/r \geq 0.$$

This proves the last path and the attained maximum. $\square$

Every hypothetical $F > \alpha$ source has its framework in the closed box

$$\begin{aligned} \frac{4\gamma}{9} \leq k \leq \frac{9}{4}, \quad -\frac{9}{4} \leq A \leq 0, \quad -\frac{9}{4} \leq B \leq \frac{9}{4}, \quad |c|, |d| \leq 1, \\ |u_i|, |v_i| \leq 1, \quad 0 \leq x_0 \leq 1, \quad |x_1|, |x_2| \leq 1, \quad |q_i| \leq 2. \end{aligned} \quad (31)$$

The lower bound on k follows from $F \leq (9/4)k$, and the other bounds from the complete bands and actual sign normalizations. These endpoints follow from universal source bounds; the coverage claim does not depend on which boxes were tried during discovery. A larger closed box would retain coverage but could make exclusion harder. The root retains the complete fibre conditions, not merely the box. In particular it does not discard the proper negative domain with $k \leq 2$. An additional shared head budget $pk \leq 4$, when used, follows from the leading 3×3 determinant: its magnitude is pk , and the maximum determinant on the real unit-entry cube is 4. For completeness, multilinearity reduces that maximum to sign matrices; their 3×3 determinants are multiples of four, while the Hadamard bound is $3\sqrt{3} < 8$.

All intersections, minima, and maxima include ties. Zero coefficients are handled before division. The point $R = 0$ cannot support a high source and is rejected as an empty positive-tail fibre. No early generic-position restriction is needed.

7.4. The complete source carried by a certificate. It is useful to state explicitly what the root box represents. A point of the box alone need not admit a complete matrix, and an auxiliary product variable alone need not equal a product of physical coordinates. The following definition fixes the mathematical source throughout the computational argument.

Definition 7.4 (Complete fibre and high framework). For ϑ in (25), let $\mathcal{F}(\vartheta)$ be the set of tuples $\zeta = (\beta, p, e, r, s, t)$ such that

$$p \geq 1, \quad k, r > 0, \quad 0 \leq e, \beta \leq 1, \quad 0 \leq s, t \leq r,$$

and the matrix defined by (7) and (23) satisfies all of (9)–(12) and $\|D\|_{\max} \leq k$. Let Q_0 denote the closed box (31). Define

$$\mathcal{H}_\gamma = \left\{ \vartheta \in Q_0 : \exists \zeta \in \mathcal{F}(\vartheta), \quad r + \frac{st}{r} \geq \gamma \right\}.$$

All inequalities in this definition refer to the same reconstructed matrix. Zero tails, whose height is zero, do not belong to this high domain.

Proposition 7.5 (Root semantics). *Every normalized proper negative-diagonal source with $F > \alpha$ has, after the actual sign changes above, a framework in $\mathcal{H}_\gamma$. For every $\vartheta \in \mathcal{H}_\gamma$, Theorem 7.3 constructs a member $\zeta_*(\vartheta) \in \mathcal{F}(\vartheta)$ with*

$$F(\vartheta, \zeta_*(\vartheta)) = \mathcal{B}(\vartheta) = \max_{\zeta \in \mathcal{F}(\vartheta)} F(\vartheta, \zeta).$$

Thus the negative-diagonal computation must establish $\mathcal{B}(\vartheta) \leq \alpha$ for all $\vartheta \in \mathcal{H}_\gamma$.

Proof. The sign normalizations preserve height and feasibility. The bound $F \leq 4p$ and $\alpha > 4$ give $p > 1$, and (31) contains the normalized framework. Since $F > \alpha > \gamma$, this framework belongs to $\mathcal{H}_\gamma$. The attained endpoint and equality with the fibre maximum follow from Theorem 7.3. All divisions and selector choices in that theorem are qualified by its nonempty high-fibre hypothesis. $\square$

The explicit interval construction is also a decision procedure for these qualifications: it checks the right-prefix interval, the prefix capacities, the five fixed core entries, and the simultaneous tail intervals. Empty intersections are rejected; equal endpoints and tied minima are included. A zero coefficient is handled by its original inequality before any division. If a formula requires a sign that is not established over a box, its alternatives must remain covered. These cases are part of the source semantics, rather than discretionary choices of the search algorithm.

For clarity, a necessary polynomial model may describe a larger set than $\mathcal{H}_\gamma$. This is legitimate only after proving that every relevant complete source supplies all its variables simultaneously and satisfies every model row. Excluding the larger set then excludes that source. A feasible-transport terminal requires more: its image must be an actual complete source, because a bound for a relaxed auxiliary point does not by itself bound a physical matrix.

Operation	Required mathematical relation
Introduce products	One complete point supplies each shared product with its exact value.
Form necessary rows	Every high source in the current region satisfies those same rows.
Intersect a box	Every source lost by the intersection is already impossible or safe.
Select a capacity formula	Its signs and selector conditions hold, or all alternatives are retained.
Change coordinates	The same physical source is reconstructed, or a complete image of no smaller height is supplied.
Attach a subproof	Its full input source, including the inherited transformation, agrees with the residual obligation.

8. RATIONAL CERTIFICATES AND MIXED-HEIGHT COVERAGE

The remaining finite proof obligation on (31) is that every complete framework has $\mathcal{B}(\vartheta) \leq \alpha$. Discovery proposes subdivisions and terminal certificates. Acceptance checks them by exact rational arithmetic on their complete declared sources. Floating-point optimization may guide discovery but is not an acceptance criterion.

A *root box* is a closed rational region before subdivision; its *child boxes* cover the parent box. An *open terminal* is a region for which that tree has not yet established either exclusion above γ or safety at α . A *verified terminal* has one of these two conclusions, with its full inherited source conditions. Grafting a subcertificate discharges an open terminal only when its region and inherited conditions agree with that terminal. The words *accepted*, *frozen*, and *receipt* below refer to recorded verifier results and data identities, not to additional mathematical axioms.

8.1. Necessary outer systems. Each actual source gives all coordinates and products simultaneously. Necessary polynomial inequalities may be relaxed over a rational box, provided the relaxation contains this same source. For example, for $L_x \leq x \leq U_x$, $L_y \leq y \leq U_y$, the shared product $z = xy$ satisfies

$$\begin{aligned} z &\geq L_x y + L_y x - L_x L_y, & z &\geq U_x y + U_y x - U_x U_y, \\ z &\leq L_x y + U_y x - L_x U_y, & z &\leq U_x y + L_y x - U_x L_y. \end{aligned}$$

A product appearing in several rows is not assigned independent physical values. Enclosures are outward. Sign-sensitive division or use of a selected min/max formula requires qualification on the entire box, or subdivision that retains every alternative.

If the necessary rows are $a_j \cdot z \leq b_j$, exact nonnegative weights give

$$a = \sum_j \lambda_j a_j, \quad B = \sum_j \lambda_j b_j, \quad a \cdot z \leq B, \quad \lambda_j \geq 0.$$

A box lower bound for $a \cdot z$ greater than B rules out the source. The acceptor checks all weights, row identities, endpoints, and strict comparisons as rationals. An interval contradiction is the one-expression case.

These are inequalities for arbitrary real points in the box. The rational endpoints and weights make the finite arithmetic exact; they do not restrict the matrix entries to rational numbers. Thus a verified strict separation excludes a continuum of real sources at once. The finite computation is a proof of a universal implication on a region, rather than a sampling of matrices from that region. The actual height can be introduced as a shared coordinate through

$$rF - r^2 - st = 0,$$

using both inequality directions. It must not be replaced by the X -specific objective $r - w$.

8.2. A real source-bound contradiction certificate. We give an actual terminal from the final negative-diagonal computation. The source being excluded consists of the canonical high sources at this target satisfying $F \geq \gamma$, with the complete parent and ancestor restrictions retained. It is not the entire geometric box. The frozen construction supplies 24 rationally bounded source coordinates and 43 shared products, hence a lifted vector $z \in \mathbb{R}^{67}$, together with 692 necessary linear inequalities. Every physical source under consideration has a lift satisfying these inequalities and all 67 coordinate bounds. The identifiers locate this source in the frozen certificate:

Identifier	Value
Original residual parent	435003
Anchor path	10011101
Local terminal path	100111011001101111

The certificate selects 49 rows with positive integer weights; all unselected weights are zero. Two of its actual rows are

Row index	Necessary inequality	Weight
2	$-z_0 - z_4 \leq 0$	31 317 836
9	$-z_0 + z_2 + z_{36} \leq 0$	229 060 761

Indices here are zero-based. The first 24 lifted coordinates are the source coordinates; in this dictionary $z_{36} = z_5 z_6$. The two displayed rows are an extract, not a two-row certificate: the following calculation uses all 49 selected rows.

Write their exact weighted sum as $a \cdot z \leq B$. If $[L_i, U_i]$ is the bound for coordinate i , then

$$m = \sum_{i=0}^{66} \begin{cases} a_i L_i, & a_i \geq 0, \\ a_i U_i, & a_i < 0 \end{cases} \quad \text{satisfies} \quad m \leq a \cdot z. \quad (32)$$

For example, the actual first coordinate has

$$[L_0, U_0] = \left[\frac{142554217571}{68719476736}, \frac{142566885919}{68719476736} \right],$$

$$a_0 = -\frac{237458354877801951091349245551}{1267650600228229401496703205376}.$$

Thus its contribution to m uses U_0 . Summing the 67 contributions and subtracting the exact weighted right-hand side gives

$$m - B = \frac{1005495384602754603811064507351680469029685784938386479}{54445178707350154154139937189082913832960000000000} > 0. \quad (33)$$

Consequently $m \leq a \cdot z \leq B < m$ is impossible. This is an exact rational contradiction for the specified high source. Its margin is approximately 1.84680335×10^3 in the displayed weight normalization. It depends on that normalization and is not a bound on $\alpha - F$.

The complete row list, weights, coordinate bounds and weighted sums are supplied in the real example identified in Appendix D. Its source record

also identifies the frozen parent, projection, ancestor trace and target image. These bindings are needed to establish that the rows and bounds apply to this source; the arithmetic in (33) alone does not establish that fact.

This terminal discharges one position in the anchor's complete 54-terminal cover. The other positions are retained, including those proved safe by a height bound or feasible transport instead of a contradiction. By Lemma 8.2, their composition proves that complete anchor safe at α . The anchor then discharges the last of 240 constituent sources of parent 435003; the other 239 sources retain their accepted coverage. Finally, that parent is bound to one of the four open positions of the preserved original root. Thus the example illustrates both parts of certificate acceptance: an exact local inequality and a complete, source-preserving route back to the original domain.

8.3. Height certificates and feasible transports. Suppose $F = h \cdot z$, after introducing a height coordinate if necessary. For $t > 0$, let $c = a - th$ and

$$m = \sum_i \min(c_i L_i, c_i U_i).$$

Then

$$tF = a \cdot z - c \cdot z \leq B - m. \quad (34)$$

Thus $B - m \leq t\alpha_-$ proves $F \leq \alpha$. It does not establish infeasibility.

Other safe terminals use feasible transformations to a domain already bounded by α . If every source z in a box has a complete image $T(z)$ with $F(T(z)) \geq F(z)$, and $F(T(z)) \leq \alpha$, that box is safe. The transformation must preserve the actual coupled matrix bands, rather than changing unrelated abstract capacities.

The following local transport gives a quantitative safe terminal near cross-saturated tails. Its parameter bounds hold on full-dimensional boxes; the construction is not restricted to an equality family. Write the complete source coordinates as

$$\begin{aligned} \xi = (k, r, w, A, B, c, d, p, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, \\ v_0, v_1, v_2, q_0, q_1, q_2, \sigma, \tau) \in \mathbb{R}^{24}, \end{aligned} \quad (35)$$

where $\sigma = r - s$ and $\tau = r - t$. Let $y \in \mathbb{R}^{23}$ be the ordered tuple obtained by deleting p from ξ . Thus ξ is specified by (p, y) , with the ordering in (35) understood. The actual tail and its height expression are

$$H(\xi) = \begin{pmatrix} r & r - \sigma \\ r - \tau & w \end{pmatrix}, \quad F(\xi) = \frac{(r - \sigma)(r - \tau)}{r} - w. \quad (36)$$

A complete physical source satisfies the original entry and successive Schur-complement bounds, as well as $\sigma, \tau, r - \sigma, r - \tau \geq 0$. Appendix C specifies these bounds as 104 polynomial slacks and gives their actual matrix reconstruction. In particular, the proper negative-diagonal domain $w = -r$, $\sigma, \tau > 0$ is a subdomain of the sources considered below; it is not an additional assumption of the transport.

For each $j \in \{0, 1, 2, 3\}$, select the twenty physical slacks $\ell_{j,0}, \dots, \ell_{j,19}$ listed in Table 3, and set

$$\Phi_{j,p}(y) = (\ell_{j,0}(\xi), \dots, \ell_{j,19}(\xi), r + w, \sigma, \tau), \quad J_j(\xi) = D_y \Phi_{j,p}(y). \quad (37)$$

Holding p fixed in this derivative does not restrict its allowed value to a centre coordinate. All four rational centres c_j have the same p -coordinate

$$p_* = \frac{726612331}{500000000}.$$

The estimates below hold on each entire closed box

$$Q_j = \{\xi : \|\xi - c_j\|_\infty \leq R\}, \quad R = \frac{1}{1250}, \quad (38)$$

and hence uniformly for the full parameter interval

$$p \in \left[\frac{726212331}{500000000}, \frac{727012331}{500000000} \right]. \quad (39)$$

The rational centres, the selected slacks, and the finite verification identities are specified in Appendix C.

Lemma 8.1 (Uniform local transport to a complete X source). *Let ξ_0 be a complete physical source, with gaps $\sigma_0, \tau_0 \geq 0$. Suppose that, for one of the four centres,*

$$d_0 + 3(\sigma_0 + \tau_0) < R, \quad d_0 = \|\xi_0 - c_j\|_\infty. \quad (40)$$

There is a piecewise continuously differentiable path of complete physical sources from ξ_0 to a complete X source ξ_ inside Q_j . Along this path, p , $r + w$, and the twenty selected slacks retain their values at ξ_0 . Moreover,*

$$F(\xi_*) - F(\xi_0) \geq \frac{\sigma_0 + \tau_0}{10}.$$

Consequently, Proposition 6.1 gives

$$F(\xi_0) \leq \alpha - \frac{\sigma_0 + \tau_0}{10}. \quad (41)$$

The selected slacks need not vanish at the source.

Proof. We give the uniform estimates and the continuation argument explicitly. For each j , the rational preconditioner C_j satisfies, on all of Q_j ,

$$\|I - C_j J_j(\xi)\|_\infty \leq \eta := \frac{396725479558446119}{10^{19}} < \frac{1}{25}. \quad (42)$$

Here the matrix norm is the induced infinity norm. Thus $J_j(\xi)$ is invertible. If e_σ, e_τ denote the last two coordinate vectors in the range of $\Phi_{j,p}$, put

$$v_\sigma(\xi) = J_j(\xi)^{-1} e_\sigma, \quad v_\tau(\xi) = J_j(\xi)^{-1} e_\tau.$$

The same whole-box rational estimates give

$$\begin{aligned} \|v_\sigma\|_\infty, \|v_\tau\|_\infty &\leq \frac{24566153053000000000}{9603274520441553881} < 3, \\ D_y F v_\sigma &\leq -\frac{1}{10}, \quad D_y F v_\tau \leq -\frac{1}{10}. \end{aligned} \quad (43)$$

Every one of the 81 slacks not used as a component of $\Phi_{j,p}$ is bounded below throughout Q_j by

$$\mu := \frac{913752034886059}{10^{18}} > 0. \quad (44)$$

In particular the first four pivots $1, p, k, r$ stay nonzero. Also, throughout these boxes,

$$r \geq r_{\min} := \frac{2065458539}{10^9} > 0, \quad w \leq -r_{\min} < 0.$$

Together with the physical inequalities $r - \sigma, r - \tau \geq 0$, this makes F in (36) positive, so that it is the magnitude of the final pivot of the reconstructed matrix. Appendix C derives (42)–(44) from explicit polynomial coefficients and rational matrices, without using a numerical trajectory.

Fix $p = p(\xi_0)$. First solve the autonomous equation

$$\dot{y} = -v_\sigma(p, y), \quad y(0) = y_0. \quad (45)$$

Since $J_j \dot{y} = -e_\sigma$, differentiation of (37) shows that the selected slacks and $\delta = r + w$ are constant, while $\sigma(t) = \sigma_0 - t$ and $\tau(t) = \tau_0$. We stop at time σ_0 ; if $\sigma_0 = 0$, this first segment is empty. From its endpoint, solve

$$\dot{y} = -v_\tau(p, y) \quad (46)$$

for time τ_0 . On the second segment, $\sigma = 0$ and $\tau(t) = \tau_0 - t$, with the same twenty slacks and δ constant. Neither gap crosses zero before its prescribed stopping time.

For completeness, these stopping times are reached by actual solutions. The vector fields are smooth wherever J_j is invertible, and this open set contains Q_j . On either segment, for total elapsed time at most $T = \sigma_0 + \tau_0$, the displacement from ξ_0 is at most $3T$ in infinity norm. Therefore every part of a solution before its prescribed endpoint lies in the compact inner box

$$K = \{\xi : \|\xi - c_j\|_\infty \leq d_0 + 3T\} \subset \text{int } Q_j.$$

This estimate first holds up to any possible first exit and precludes that exit by (40). The usual continuation theorem for a smooth finite-dimensional vector field then rules out a finite maximal existence time before either prescribed endpoint: the trajectory remains in a compact subset of the field's domain. The parameter coordinate remains exactly fixed during this argument.

The twenty selected physical slacks remain nonnegative because they retain their initial values. The same is true of δ , while the two gaps remain nonnegative by their explicit evolution. The other 81 source slacks remain strictly positive by (44). Thus the complete entry and pivot-stage bounds hold simultaneously for $M(\xi(t))$ at every point of both segments. Pivot ties are allowed in these bounds; the construction does not exchange unequal pivots or divide by a vanishing pivot.

At the endpoint both gaps vanish, so the actual tail is $\begin{pmatrix} r & r \\ r & w \end{pmatrix}$. This is a complete X source, whether or not $\delta = 0$. Finally, the minus signs in (45)

and (46), together with (43), give $\dot{F} \geq 1/10$ on each nonempty segment. Integration gives the stated height gain, and Proposition 6.1 bounds the endpoint by α . $\square$

For a proper negative-diagonal source the conclusion reads $F \leq \alpha - (2r - s - t)/10$. More generally, every physical source in a closed inner box $\|\xi - c_j\|_\infty \leq 1/10000$, with nonnegative gaps, satisfies (40), since

$$\frac{1}{10000} + 3 \left(\frac{1}{10000} + \frac{1}{10000} \right) = \frac{7}{10000} < \frac{1}{1250}.$$

The strict inequality in (40) is part of the terminal rule. A source box is accepted by this rule only when its entire enclosure satisfies that strict distance-and-time bound. Equality at this outer guard is not accepted by a limiting argument; it remains with the other sources to be subdivided or covered by a separate valid terminal. Zero gaps themselves are permitted, as the proof omits the corresponding zero-length segment. These four local regions are not asserted to cover the whole proper negative-diagonal domain. Further safe transformations obey the same source and height-direction requirements.

8.4. Complete composition.

Lemma 8.2 (Mixed threshold and algebraic coverage). *Let $\gamma < \alpha$. Let a rational root Q_0 contain a canonical image of every hypothetical source with $F > \alpha$, with height at least that of the source. Suppose a finite acyclic rooted proof graph covers Q_0 by exact subdivisions, qualified contractions, and source-preserving coordinate changes that do not decrease the height of a hypothetical source. Suppose every terminal establishes either:*

- (1) *no complete source with $F \geq \gamma$ exists in its region; or*
- (2) *every complete source in its region has $F \leq \alpha$.*

If every child is accounted for and each terminal source is bound to its declared root path, then no original source has $F > \alpha$.

Proof. A hypothetical source has a canonical image in Q_0 with no smaller height, still exceeding $\alpha > \gamma$. At a subdivision it belongs to a closed child; overlaps on boundaries are harmless. A qualified contraction has already certified its discarded region impossible or safe. A coordinate change maps the same source into its declared image at no smaller height. Following these relations through the finite complete graph reaches a terminal and contradicts either terminal assertion. $\square$

Previously accepted subproofs may be composed by this lemma, but only on the same mathematical sources. The input includes the mathematical domain, not only its dimension or a previously recorded success label. The implementation binds model, parent, path, endpoints, transformation prefix, rule implementation, and proof payload. The final traversal reconstructs the original root boxes and verifies every grafted subproof's source. Hashes

establish byte identity; mathematical validity comes from the accepted sub-proof and its declared rule.

9. THE COMPLETE NEGATIVE-DIAGONAL CERTIFICATE

Proposition 9.1 (Certified complete negative-diagonal bound). *Every complete real source with (23), $p, k, r > 0$, and $0 < s, t < r$, satisfies*

$$F = r + st/r \leq \alpha.$$

The computation uses the full seventeen-variable root (31) with Theorem 7.3’s exact framework semantics. Its model contains every hypothetical $F > \alpha$ source after the three actual paths; it imposes no selected tail-contact equation on all canonical images. The Supplementary Index, parts S5–S7, identifies the model, acceptance rules, and complete certificate data.

9.1. Base certificate and remaining source domains. Exact rational verification of the base certificate accepted 749693 nodes: 374846 splits, 374839 contradiction terminals, four safe terminals, and four open terminals. Its 750 enclosed continuation certificates had no internal open terminal. The four remaining domains are closed dyadic boxes reconstructed from the original seventeen-dimensional root. Denote them by Q_1, Q_2, Q_3, Q_4 , in the order specified in the Supplementary Index, S7. The base certificate proves the required bound outside their union.

Complete certificates for Q_1, Q_3, Q_4 cover, respectively, 111, 159, and 56 constituent sources. The certificate for Q_2 has 240 constituent sources; 239 are covered by its component certificates and the remaining one by the local completion described below. Every constituent source retains its complete ancestor transformation prefix.

9.2. Local completion. The remaining source in Q_2 is covered by a binary subtree with 54 distinct, prefix-free terminal paths. Their union exhausts that subtree. The terminal justifications are

Terminal justification	Count
Contradiction certificates	48
Height and feasible-transport certificates	3
Canonical-source threshold exclusions	3
Total	54

Each canonical-source threshold exclusion covers both qualified charts of its declared source. The three height or transport certificates establish safety at α . The mathematical verifiers and source-prefix checks were executed for this local completion. It covers one constituent source of Q_2 ; full coverage of Q_2 also uses the other 239 sources.

9.3. Composition at the original root. The local completion is attached to its exact source in Q_2 , preserving the already verified complement. The four complete certificates are then attached to the four open terminals of the base certificate. The composition traverses the actual base-root structure and reconstructs its entire set of open boxes. For each attachment it checks the original model, root path, box endpoints, and full transformation prefix against the corresponding complete certificate. The reconstructed boxes coincide with Q_1, Q_2, Q_3, Q_4 . Thus every open base-root source is covered by a matching complete proof, and the assembled certificate has no uncovered source.

The constituent-source counts above total $111 + 240 + 159 + 56 = 566$. These are internal obligations of the four remaining parents, not a count of original parents. They discharge four original-parent obligations; together with the previously covered 465, they give the original-parent coverage count 469/469. Appendix D makes this distinction explicit and describes the source-by-source correspondence. This count provides a consistency check; coverage follows from the complete root traversal and exact source comparisons. The Supplementary Index, S7, identifies the paths, endpoints, certificate identities, and composition results needed to check these bindings.

Proof of Proposition 9.1. A hypothetical $F > \alpha$ source has $p > 1$ by (24). Its actual sign normalization and Theorem 7.3's paths produce a complete framework in (31) with $\mathcal{B} > \alpha$. The accepted base proof and the four full-parent completions form the finite complete cover just described. Each contradiction terminal excludes this high source, and each safe terminal bounds its height by α . All original root paths and transformed sources are accounted for. Lemma 8.2 therefore excludes the hypothetical source. $\square$

Remark 9.2 (Verification and composition). The final composer reuses identified, previously accepted complete-root and parent receipts. It verifies their producer and payload identities and exact source bindings; it does not rerun every terminal's mathematics. The base tree retains four open records, each discharged by an attached complete certificate. The records retain their original meaning in this composition. Likewise a source safe at α need not be infeasible at the lower threshold γ .

10. COMPLETION OF THE GLOBAL PROOF

Proof of Theorem 1.2. The lower bound is [2, Theorem 2.2]; Section 2 describes the exact corresponding witness in the archive.

For the upper bound, fix an arbitrary nonzero real matrix and arbitrary legal complete-pivoting path. Normalize its maximum initial entry to one and absorb the path's permutations into the matrix. If a pivot vanishes, its whole trailing block vanishes. Corollary 3.2 bounds all growth through the fourth stage by $4 < \alpha$. It remains to exclude a fifth-pivot value larger than α .

Proposition 4.1 gives the complete two-step lift. Lemma 5.1 gives a same-height complete source on the D or X saturated boundary. Actual sign changes put every X source into (16), bounded by Proposition 6.1. For a D source, (17) with positive sign or a zero arm has height at most $r \leq 4$; a tied arm goes to X . Every remaining source belongs to (18), so Proposition 9.1 applies. No fifth pivot exceeds α . Since the matrix and legal path were arbitrary, $\rho_5^{\mathbb{R}} \leq \alpha$, proving equality. $\square$

11. REPRODUCIBILITY AND SCOPE

The proof can be read at two complementary levels. The analytic reductions and local acceptance rules specify why the conclusion follows from the stated finite checks. The published exact data make those checks independently executable. Reading the proof and inspecting the rules therefore exposes the mathematical obligations; it does not replace checking every large certificate instance.

The computational proof separates discovery, verification, and composition. Discovery constructs cuts, subdivisions, and feasible transformations. Verification checks finite rational inequalities and qualified analytic rules on complete sources. Composition establishes that verified pieces cover the original domain with matching source data. This separation allows component proofs to be checked and reused independently.

The separately available Supplementary Index is organized into eight parts, S1–S8, listed in Appendix A. It identifies the polynomial and algebraic witness, analytic reductions, complete X certificates, negative-diagonal model, terminal rules, and final coverage graph. The index points to archived proofs, executable verifiers, and certificate payloads; it is not itself the complete proof-data archive. It specifies exact file identities, source domains, verification entry points, and the scope of each recorded run. Large certificate payloads are separate from the readable documentation; full verification requires those payloads and all referenced component dependencies.

The preserved base-root run used the exact **Fraction** backend on Linux with eight worker processes, and recorded 22815.35 seconds of elapsed wall time (about 6.34 hours). The separate final 54-terminal anchor recorded 184.37 seconds. Neither value measures the whole proof: the former excludes the additional parent and transport acceptances, and the latter checks only the named anchor. They are elapsed times, not summed process CPU times or estimates of the discovery cost. The retained acceptance records fix the inputs, backend and worker count; they do not supply a portable performance benchmark with a recorded processor model and peak memory usage.

Later parent and root compositions reuse identified mathematical acceptance records while checking exact source bindings. An independent fresh reproduction must execute the mathematical producers of all required records, then compose the resulting conclusions. A review that instead checks the

preserved records and their bindings should be described as a record-and-composition audit. Both procedures have well-defined scopes; the latter is not a fresh execution of the former. Appendix D.4 gives the order and acceptance conditions. A file hash establishes identity, not the truth of a mathematical conclusion.

Fresh negative-diagonal replay. On 30 September 2026, a separate directory on the existing Linux host was used to re-execute the four-parent mathematical plan and the original root. All 570 planned producers passed their prescribed acceptance conditions: 512 algebraic cuts, 46 branch checks, seven production projections, four whole-anchor complements, and one joint anchor. The complements intentionally retained their prescribed open paths; their acceptance alone did not close a parent. The 566 distinct target sources, in groups of 111, 240, 159 and 56, were covered before the four parent paths and rational boxes were matched back to the actual root. The resulting effective open count was zero. The root used sixteen **Fraction** workers and retained its four expected open positions. Its internal elapsed time was 12952.26 seconds; the launcher’s elapsed interval, including overhead, was 12965.71 seconds (about 3.60 hours). Supplementary Index S7–S8 identifies the sealed plan, executed drivers and complete receipt archive. This establishes fresh negative-diagonal finite coverage relative to retained upstream analytic and complete X inputs; those inputs were not re-executed in this run. It is neither a new-machine replay nor an independent implementation of the acceptors, and does not enlarge the Lean scope stated below.

A companion Lean development formalizes the algebraic constant and its attainment, key analytic reductions, and a conditional final theorem. The latter retains two explicit global safety hypotheses, for the physical X domain and the original B -root capacity endpoints (named **XGlobalSafety** and **RootEndpointSafety**, respectively). The present proof addresses these obligations through the written analytic arguments, exact certificates, and source-composition logic; their complete discharge in Lean, including the remaining model and coverage connections, is not claimed. The Python/C++ verifier implementations are not formally verified. This is a computer-assisted proof with partial Lean formalization. Component verifications and final composition are distinguished in the Supplementary Index, S8; composition does not re-execute every earlier verifier.

Lean project cold rebuild. On 12 September 2026, the frozen first-phase Lean project was rebuilt on Linux from an empty project-object directory, using Lean 4.30.0 and pinned third-party dependencies. The recorded rebuild passed all 634 product modules, three additional example/audit modules, and 47 named axiom checks. It used an explicit dependency-order driver, with third-party caches reused and supplemented as needed. The final theorem’s two safety parameters were checked and remain explicit. The Supplementary Index, S8, records the source identity, successful build route, and scope of the retained evidence.

Data and code availability. The companion Lean source, exact verifier code, certificate archives, and step-by-step verification documentation are available at <https://github.com/hkjtsgmc79-boop/rho5-proof>. The fixed release v1.0.1 provides the certificate downloads, SHA-256 manifest, file sizes, computing requirements, and recorded verification scopes: <https://github.com/hkjtsgmc79-boop/rho5-proof/releases/tag/v1.0.1>. The project cold rebuild is distinct from a fresh-machine rebuild of Lean and all dependencies, and from a new end-to-end mathematical replay of the large certificates. The original v1.0.1 manuscript and Supplementary Index PDFs are archived at <https://doi.org/10.5281/zenodo.22727368>. The present v1.0.2 manuscript and Supplementary Index are archived at <https://doi.org/10.5281/zenodo.23057256>.

The theorem determines the maximum value and supplies an attaining matrix. Classification of all maximizers, uniqueness modulo symmetry, and the analogous complex-matrix value remain separate questions.

ACKNOWLEDGMENTS

AI models and agent systems played an important role in this work, contributing substantially to the development and verification of the proof.

Qianli Ma led and coordinated the research program, managed its progress and budget, and made the final decisions on the selection, continuation, or termination of research approaches. He organized the allocation of scientific tasks and responsibilities among the research agents.

The models and tools used included GPT, Claude, Wujie AI AGENT, and Deepseek Harness. The authors retain responsibility for the mathematical claims, supporting evidence, and final manuscript.

APPENDIX A. PROOF MODULES AND SUPPLEMENTARY MATERIAL

The Supplementary Index, parts S1–S8, maps the analytic and computational dependencies by their mathematical role. The theorem statements in the text specify the domains on which each component is used.

Supplementary unit	Mathematical content and role
S1: Constant and attainment	The degree-61 polynomial, exact isolating interval, identity with the constant of [2], and supplementary algebraic reconstruction of an attaining matrix. The lower bound is [2, Theorem 2.2].
S2: Scalar bounds and complete coordinates	The conditional three-pivot envelope, the first-four-pivot bound, the complete two-step lift, and the same-height D/X saturation proved in Sections 3–5. The saturation statement used here requires positive fifth-pivot height.

Supplementary unit	Mathematical content and role
S3: Canonical representative and height	Qualified representative existence, exact row-scale reconstruction, and canonical localization and height certificates, established independently of the final negative-diagonal bound. Six complete exclusion trees contain 228542 nodes and 114274 contradiction leaves. The subsequent 152467-record localization uses 208274 contraction certificates; its three terminal boxes are covered by the algebraic height theorem.
S4: Complete cross-saturated domain	The four source domains and exact sign and transpose maps of Proposition 6.1, including their shared boundaries. The low- r component uses S3. All statements concern complete real matrices; no rank, cofactor-sign, or auxiliary entry assumption is added.
S5: Negative-diagonal framework	The three complete feasible paths, attained six-variable maximum, and seventeen-variable root of Section 7. Zero factors, capacity ties, and the proper negative-diagonal range $k \leq 2$ are retained.
S6: Exact acceptance rules	Rational interval and nonnegative-combination certificates, qualified contractions, algebraic-height rules, and complete feasible transports. Contradictions above γ and safety at α have the distinct semantics of Lemma 8.2.
S7: Complete negative-diagonal cover	The accepted base proof, complete parent proofs, and exact source bindings of Section 9. The final 54-terminal source completes the last parent's 240/240 responsibilities and the total 469/469 original parents. Four source-bound completions cover the base tree's four retained open records; no original source region remains uncovered.
S8: Reproduction and dependency index	Model and payload identities, acceptance records, executable dependencies, and source-composition maps. The final composition's 357 direct dependencies are supplemented by their recursive dependencies; the direct list alone is not the complete proof archive.

The dependency direction is S3 to S4, and then S4 to the cross-saturated safety rules used in S6–S7; the final negative-diagonal cover is not an input to S3 or S4. Counts identify accepted objects, while their rules and exact source bindings establish validity and coverage. An α -safe region may contain feasible matrices.

Component acceptance and final composition are separate operations. The latter reuses identified acceptance records and checks their source bindings; it does not rerun every component's mathematics. Accordingly, the completed composition is not claimed to be a new from-scratch replay of the entire dependency chain. The supplementary index distinguishes the

recorded acceptance runs, the checks performed during composition, and the materials required for independent reproduction.

APPENDIX B. CANONICAL REPRESENTATIVES AND THE EXACT NEAR-PEAK HEIGHT

B.1. The original source and the precise reduction inputs. This appendix separates three operations: choosing a representative of a prescribed attainable height; reconstructing that representative from one set of row-scale coordinates; and bounding the resulting height by a constrained maximum argument. All three operations are developed below. The common-column budgets, wall recovery, and prescribed-target construction have explicit analytic proofs; the safe-tail, stationary-source, exceptional-contact, and localization exclusions are identified separately as finite computational inputs. This ordering prevents the final X or negative-diagonal bound from being used to justify an earlier near-peak terminal.

Reading map. The deduction proceeds in the order below. Each representative or reconstruction retains a common physical source and its stated qualifications. The last column distinguishes analytic arguments from finite certificate inputs; their source inclusions are proved before acceptance is used.

Sections	Conclusion used next	Method and finite inputs
B.2	High-value budgets and simultaneous first/middle columns	Analytic recovery; thirteen fixed safe-tail positivity checks.
B.3–B.5	Same-head wall maximum and qualified head entrance	Tail reconstruction and maximum argument; C19 and difference exclusions.
B.6–B.7	Representative at a prescribed height; endpoint equivalence	Exceptional-contact exclusions, compactness and minimum attainment.
B.8	One complete row-scale coordinate system	Analytic reconstruction in both directions.
B.9–B.10	Localization and a compact model near the sharp height	Finite localization, qualified target transport and upper-target exclusion.
B.11–B.12	Forced contacts at a maximum and the exact height α	Uniform Jacobian bound, feasible local variation and algebraic elimination.

Section B.13 is the source and certificate crosswalk, not a further deduction. The chain runs from canonical source to exact height; neither the later complete X bound nor the final negative-diagonal theorem is an input to it.

All indices of the 3×3 blocks and their vectors are $0, 1, 2$. The letter s in this appendix means $p - 1$, and R means the scalar $F/2 - 1$. They are not the tail coordinate s of the negative-diagonal fibre or the real field $\mathbb{R}$.

We write $\Gamma = V - \beta Y$ for a cofactor, reserving the threshold γ of the main paper for $4132517/10^6$.

Definition B.1 (Complete canonical source). Take real parameters

$$(t, E, \beta, V, T, X, Y, Z, p, k, r, w, a, b, g, h, v, j, q)$$

and set

$$\begin{aligned} u &= (1, V, T)^T, & x &= (X, Y, Z)^T, \\ \hat{v} &= (-g, h, v)^T, & \hat{q} &= (-j, -1 - \beta h, q)^T, \\ D &= \begin{pmatrix} k & -a & -b \\ tk & r - ta & r - tb \\ -Ek & r + Ea & w + Eb \end{pmatrix}, & S &= D + x\hat{q}^T, \quad O = S + u\hat{v}^T, \\ P &= \beta\hat{v} + \hat{q}, & L &= px - u, \\ M &= \begin{pmatrix} 1 & -1 & \hat{v}^T \\ \beta & p - \beta & P^T \\ u & L & O \end{pmatrix}, & F &= r - w. \end{aligned} \quad (47)$$

The source belongs to $\mathcal{K}$ when

$$\begin{aligned} p, k, r > 0, \quad |t| \leq 1, \quad 0 \leq E \leq 1, \quad 0 < \beta, T, X, Y, Z \leq 1, \quad 0 < V < 1, \\ 0 \leq a, b \leq k, \quad |w| \leq r, \\ |p - \beta| \leq 1, \quad |L_i| \leq 1, \quad |\hat{v}_j| \leq 1, \quad |P_j| \leq 1, \quad |O_{ij}| \leq 1, \\ |\hat{q}_j| \leq p, \quad |S_{ij}| \leq p, \quad |D_{ij}| \leq k, \end{aligned} \quad (48)$$

for every indicated index, and all six cofactors

$$\begin{aligned} \delta &= 1 - \beta X, \quad \Gamma = V - \beta Y, \quad \nu = \beta Z - T, \\ \tau &= Y - XV, \quad \Delta = Z - TX, \quad \mu = VZ - TY \end{aligned} \quad (49)$$

are strictly positive. In particular $P_1 = -1$ is built into the representation. The signs of g, h, v, j, q are not extra assumptions in this definition.

These inequalities retain the complete initial and successive Schur bounds. Eliminating the first two pivots gives the receiver/core blocks with pivots p, k ; eliminating k gives $\begin{pmatrix} r & r \\ r & w \end{pmatrix}$. The last signed pivot is $w - r = -F$. In the positive high-value regime no division at a zero pivot is made. Sources with an earlier zero pivot have already been handled before entry to this chamber.

B.2. High-value budgets and simultaneous column reconstruction. The reductions in this subsection keep the seven-coordinate head $\eta = (t, \beta, V, T, X, Y, Z)$ fixed. Whenever the row scale E is fixed as well, the first, middle, and last columns always use the same E, p, k . The sign conclusions below are consequences of a complete source; they are not imposed on an enlarged independent-column relaxation.

Lemma B.2 (Elementary budgets of a complete high-value source). *Every source of Definition B.1 with $F > 4$ satisfies*

$$2 < r \leq k < 3, \quad 0 < g, j, h \leq 1, \quad j < 1, \quad 1 < p < 1 + \beta < 2, \quad 0 < a < p, \quad (50)$$

$$0 < t \leq 1, \quad X > \frac{1}{1 + \beta} > \frac{1}{2}, \quad Ek < 1 - T, \quad r < \frac{5}{2}, \quad r + 2E + T < 3, \quad (51)$$

$$\Delta > \tau, \quad t > V, \quad Q_o := t\delta - \Gamma > 0, \quad \Omega := t\Delta + E\tau - \mu > 0, \quad \zeta := t\nu - E\Gamma > 0. \quad (52)$$

Moreover $q > 0$. If $F \geq 33/8$, the positive-receiver conclusion of Lemma B.4 also gives $v > 0$, and the quantitative bounds are

$$3 - r - 2E - T > \frac{1}{320}, \quad \Delta - \tau > \frac{1}{32}, \quad t - V > \frac{1}{48}, \quad q > \frac{41}{480}, \quad P_2 > \frac{61}{480}. \quad (53)$$

Proof. Since $|w| \leq r$ and $D_{21} = r + Ea \leq k$, we have $r \geq F/2 > 2$ and $k \geq r + Ea$. The identity $O_{00} = k + XP_0 - \delta g$, together with $P_0 \geq -1$, implies $g > 0$. If $j \leq 0$, then $O_{00} = k - Xj - g \geq k - 1 > 1$, a contradiction. Similarly $O_{21} = r + Ea - Z - \nu h \leq 1$ gives $h > 0$. Thus $|\hat{q}_1| = 1 + \beta h \leq p$ implies $p > 1$, and $P_0 = -j - \beta g \geq -1$ gives $j + \beta g \leq 1$, in particular $j < 1$. Put $s = p - 1$ and $\epsilon = r - 2$. Initially $0 < s \leq \beta \leq 1$, and the same first column yields

$$k \leq p + Xj, \quad k \leq 1 + Xj + g, \quad k \leq 1 + s + X(1 - \beta s).$$

For the last inequality substitute $j \leq 1 - \beta g$: the function $\min(s, g) - \beta Xg$ is maximized at $g = s$ on $[0, 1]$. Also $k \leq 1 + X + (1 - \beta X)g \leq 2 + (1 - \beta)X \leq 3 - \beta$. Consequently $\beta < 1$, $p < 2$, and $0 < \epsilon \leq s(1 - \beta) < s$. The middle-column original bound gives $r + Ea \leq 1 + Z + \nu h \leq 2 + \beta - T$, whence $T \leq \beta - \epsilon$. Adding this bound for r to $r \leq 3 - \beta$ proves $r < 5/2$.

The inequality $O_{11} = r - ta - Y + \Gamma h \leq 1$ gives $ta > 0$, hence $t > 0$ and $a > 0$. The stage bound $S_{01} \geq -p$ gives $a + X(1 + \beta h) \leq p$, so $a < p$. If $X \leq 1/(1 + \beta)$, the preceding bound for k gives

$$k \leq 1 + s + \frac{1 - \beta s}{1 + \beta} = 2 + \frac{s - \beta}{1 + \beta} \leq 2,$$

a contradiction. In particular $X > \delta$. Since $Xj + g \geq k - 1 > 1$, we have $j + g > 1$. Now $Z > T$ and $O_{20} = -Ek - Zj - Tg \geq -1$ imply $Ek < 1 - T$. Thus $E < 1/2$. Finally,

$$(1 + \beta)Z - 1 - T = (r - 2) + Ea + (1 - O_{21}) + \nu(1 - h) > 0.$$

Together with $pZ \leq 1 + T$, this proves the strict inequality $p \leq (1 + T)/Z < 1 + \beta$.

Here is a positive-sum proof of the stronger row-scale budget. Set

$$A = r - p, \quad z_* = \frac{1 + \epsilon + Ts/\beta}{p}, \quad C_0 = 1 - Ek - \frac{Z}{X}(k - p) - Ts.$$

The exact identities

$$\begin{aligned} C_0 &= (1 + O_{20}) + T(1 - O_{00}) + \frac{\Delta}{X}(p - S_{00}) \geq 0, \\ B_R &:= 1 - Er - Z(r - p) - Ts \\ &= C_0 + (E + Z)(k - r) + Z(1/X - 1)(k - p) \geq 0, \\ \beta p(Z - z_*) &= \beta(Ea + 1 - O_{21}) + \nu(s - \beta h) \geq 0 \end{aligned}$$

show that $C := 1 - Er - z_*A - Ts = B_R + A(Z - z_*) \geq 0$. Define

$$\begin{aligned} m &= \frac{r}{2} - s - \frac{As}{\beta p} = \frac{(1-s)(2s-\epsilon)}{2p} + \frac{A(\beta-s)}{\beta p} > 0, \\ B &= 3s(1+s)(1-\beta) + 4(\beta-s) + (1+s)(\beta-s)(1-\beta), \\ N &= 2(\beta-s)\epsilon^2 + B\epsilon + 2\beta(1-\beta)(1-s^2) > 0. \end{aligned}$$

All displayed signs follow from $0 < \epsilon < s \leq \beta < 1$. Direct expansion, using $p = 1 + s$ and $r = 2 + \epsilon$, gives

$$W := 3 - r - 2E - T = \frac{N}{\beta pr} + \frac{2}{r}\{C + m(\beta - \epsilon - T)\} > 0.$$

This proves the row-scale budget without changing a matrix coordinate.

For $\Delta > \tau$, use $\Delta - \tau = \delta(Z - Y) + X(\Gamma + \nu)$. If $Y \geq Z$, then

$$h(\Delta - \tau) - X\epsilon = X\{Ea + 1 - Y + 1 - O_{21}\} + X\Gamma h + (X - \delta h)(Y - Z) \geq 0.$$

If $Y \leq Z$, the same difference equals

$$\delta h(Z - Y) + X\Gamma h + X\{Ea + 1 - Z + 1 - O_{21}\} \geq 0.$$

Thus $\Delta - \tau \geq X\epsilon/h > 0$. Let

$$\begin{aligned} B_0 &= (1 + O_{20}) + T(1 - O_{00}) \geq 0, \\ J &= \mu(1 + O_{02}) + \Delta(1 - O_{12}) + \tau(1 + O_{22}) \geq 0. \end{aligned}$$

Expansion at the same complete source gives

$$\begin{aligned} \Delta b(t - V) - \frac{\Delta + \tau}{2}(F - 4) &= \Delta(1 - V) + \tau\{B_0 + \Delta j + (E + T)(k - b)\} \\ &\quad + J + \frac{\Delta - \tau}{2}(r + w) > 0. \end{aligned}$$

The strict sign follows already from $\tau\Delta j > 0$. Since $b \geq 0$, it implies $b > 0$ and $t > V$. Therefore $Q_o = \delta(t - V) + \beta\tau > 0$ and $\Omega = (t - V)\Delta + (E + T)\tau > 0$. For the last direction use

$$\zeta h - (t + E)(r - 2) = E(1 - Y) + t(1 - Z) + E(1 - O_{11}) + t(1 - O_{21}) \geq 0.$$

The tail signs follow from two more slack identities:

$$\begin{aligned} 4 - W + (EX + Z)q - F &= (1 + O_{22}) + E(1 + O_{02}) \\ &\quad + (E + T)(1 - v) \geq 0, \\ 4 - T - W + (EX + Z)P_2 - F &= (1 + O_{22}) + E(1 + O_{02}) + E(1 - v) \\ &\quad + E\beta Xv + \nu v \geq 0 \quad (v \geq 0). \end{aligned}$$

The first, with $W > 0$, proves $q > 0$; the second proves $P_2 > 0$ after Lemma B.4 has supplied $v > 0$. That lemma uses only the preceding budgets, not this last positive-prefix conclusion.

For the quantitative assertions assume $F \geq 33/8$, so $\epsilon \geq 1/16$. Since $r \leq 3 - \beta$, $1 - \beta \geq \epsilon$, and $s \leq \beta$ gives $1 - s \geq \epsilon$. Keeping only the constant term of N in the formula for W yields

$$W \geq \frac{2(1-\beta)(1-s)}{r} > \frac{2(1/16)^2}{5/2} = \frac{1}{320}.$$

The already proved direction estimates give $\Delta - \tau \geq X\epsilon/h > \epsilon/2 \geq 1/32$, and $t - V > (F - 4)/(2b) > 1/48$, using $b \leq k < 3$. The middle and head bands also give

$$r + Ea \leq 1 + Z(1 + \beta h) - Th \leq 1 + pZ - Th \leq 2 + T(1 - h),$$

so $T > \epsilon \geq 1/16$. Finally $EX + Z < 3/2$, and the two tail slack identities imply

$$q > \frac{1/8 + 1/320}{3/2} = \frac{41}{480}, \quad P_2 > \frac{1/8 + 1/16 + 1/320}{3/2} = \frac{61}{480}.$$

□

Lemma B.3 (The nonpositive-receiver tail face). *On the complete canonical source of Definition B.1, the conditions $P_2 = 1$ and $v \leq 0$ imply*

$$r < \frac{33}{16}, \quad F < \frac{33}{8}.$$

Proof. We give the analytic reduction and all the scalar polynomials used in its finite sign check. This lemma is restricted to the canonical $u_0 = 1$ source already under discussion; it does not need a further normalization of the first head coordinate.

Suppose instead that $r \geq R := 33/16$, and put $\epsilon = 1/16$, $\rho = 17/16 = R - 1$. Then $k \geq r + Ea \geq r > 2$. Directly from the first-column bands, $g > 0$ and $j > 0$: use $O_{00} = k + XP_0 - \delta g \leq 1$ with $P_0 \geq -1$ for the former; $j \leq 0$ would give $O_{00} = k - Xj - g \geq k - 1 > 1$. Also $O_{21} = r + Ea - Z - \nu h \leq 1$ implies $h > 0$, so $p \geq 1 + \beta h > 1$. Consequently $0 < s := p - 1 \leq \beta \leq 1$. Write $f = -v \in [0, 1]$, so the tail-face equality gives $q = 1 + \beta f \geq 1$. If $t \leq 0$, then $O_{12} = r - tb + Yq - Vf \geq r + Y - V > 1$, a contradiction. Thus $0 < t \leq 1$. These facts use only $r > 2$, not the height conclusion being proved.

The first-column slack identity

$$\begin{aligned} V(p + Y) - \beta Y(p - 1) - Vtk &= \Gamma(p - S_{10}) + \beta Y(1 - O_{10}) \\ &\quad + VY(1 + P_0) \geq 0 \end{aligned}$$

retains the prefix bound at this same matrix. It gives

$$tk \leq 1 + s + Y - \frac{\beta Y s}{V}. \tag{54}$$

Let $d_0 = tX - Y$ be the physical head difference. The tail original bounds give

$$r \leq 1 + t + d_0q + (V - t)f.$$

If $d_0 \leq 0$, then $q \geq 1$ and $0 \leq f \leq 1$ bound this expression by $1 + t + \max(0, V - t) \leq 2$. Hence $d_0 > 0$. Moreover $\tau = Y - XV > 0$ gives

$$t\delta - \Gamma = t - V - \beta d_0 > \frac{d_0\delta}{X} > 0.$$

Substituting $q = 1 + \beta f$ therefore yields

$$r \leq 1 + t + d_0 - (t\delta - \Gamma)f \leq 1 + t + d_0, \quad d_0 \geq \rho - t > 0, \quad Y \leq t - d_0.$$

The middle-column receiver is denoted here by $\chi = -\hat{q}_1 = 1 + \beta h$, to avoid confusing it with the first-column parameter j . Its two row-0 bounds give $a + X\chi \leq \min(1 + s, 1 + h)$. Thus

$$r \leq 1 + t + t \min(s, h) - d_0\chi - Vh. \quad (55)$$

From $L_2 \leq 1$, $O_{21} \leq 1$, and $Ea \geq 0$, we also have $r \leq 1 + h + Z(\chi - ph)$. The factor $\chi - ph$ is positive, since otherwise $r \leq 2$. Using $Z \leq 1$, we obtain

$$\chi \geq \rho + sh, \quad \epsilon \leq (\beta - s)h, \quad 0 < s \leq \beta - \epsilon \leq 15/16.$$

For the moment put $d = d_0$, and define

$$A = \rho(1 + d) - t > 0, \quad B = ts - A.$$

Equation (55) implies both

$$A \leq (t - V - ds)h, \quad (V + ds)h \leq B.$$

In particular $t - V - ds > 0$, $B > 0$. Comparing the two bounds on h , and using $A + B = ts$, gives

$$V \leq \frac{B - ds^2}{s}, \quad (\beta - s)B \geq \epsilon(V + ds).$$

Let

$$H = 1 + s + Y - tR, \quad J = B + \epsilon d > 0, \quad K = B - ds^2 > 0, \quad C = BH - \epsilon Ys.$$

From (54) and $k \geq R$, $VH \geq \beta Ys > 0$. Combining this with the preceding inequality gives $VC \geq Ys^2J > 0$, hence $C > 0$. Now $V \leq K/s$ and $V(Y + d) < tY$, the latter being the cofactor condition $V < Y/X$, imply

$$Q := HK - Ys(\epsilon + s^2) \geq 0, \quad P := tC - s^2J(Y + d) \geq 0. \quad (56)$$

For the first implication use the exact identity $KC - Ys^3J = BQ$, with $B > 0$. For the second, $VC \geq Ys^2J$ and $V < tY/(Y + d)$ even give $P > 0$; the weaker form above suffices.

We next lower the auxiliary difference to $d = \rho - t$. This is a relaxation of necessary inequalities, not a new physical matrix. The left side A decreases, $t - V - ds$ increases, $V + ds$ decreases, and B increases. The constraints $\epsilon \leq (\beta - s)h$ and $VH \geq \beta Ys$ do not change; $V(Y + d) < tY$ and $Y \leq t - d$ become weaker. Thus the same s, t, Y, V, β, h satisfies this relaxed system,

and the preceding eliminations may be repeated at the new d . All letters A, B, H, J, K, C, Q, P henceforth use $d = \rho - t$. Since

$$K = t(1 + s + \rho + s^2) - \rho(1 + \rho + s^2) > 0,$$

we have

$$t > \ell(s) := \frac{17(16s^2 + 33)}{16(16s^2 + 16s + 33)}.$$

The coefficient of t is positive, and $t \leq 1$, so $0 < K \leq K|_{t=1} = s - 2\epsilon - \epsilon^2 - \epsilon s^2 < s - 1/8$. Every hypothetical source therefore supplies

$$\begin{aligned} \frac{1}{8} < s \leq \frac{15}{16}, \quad \ell(s) < t \leq 1, \\ 0 < Y \leq Y_m := 2t - \rho, \quad H, K > 0, \quad Q, P \geq 0. \end{aligned} \tag{57}$$

Here is the complete finite exclusion of this scalar domain. Put $H_* = 1 + s - tR$ and $u_*(s) = 16(1 + s)/33$. Both Q and P are affine in Y . If $H_* \leq 0$, write $Q = H_*K + Y\{K - s(\epsilon + s^2)\}$. Since $Q \geq 0$, its slope in Y must be nonnegative; therefore $Q \leq Q(s, t, Y_m)$. The finite sign checks below give $Q(s, t, Y_m) < 0$ throughout $[1/8, 15/16] \times [\ell(s), 1]$, a contradiction. If $H_* > 0$, then $t < u_*(s)$. The same checks give $u_* < \ell$ on $[1/8, 7/10]$, so $s > 7/10$. They also give $P(s, t, 0) < 0$ and $P(s, t, Y_m) < 0$ on $[7/10, 15/16] \times [\ell(s), u_*(s)]$. Affineness in Y then contradicts $P \geq 0$ on the whole interval $[0, Y_m]$. This proves $r < R$, and $F \leq 2r < 33/8$.

To specify the finite checks without an external polynomial dictionary, write $f(s, t) = \sum_{i=0}^n f_i(s)t^i$, and define

$$\mathcal{B}_k[f; a_0, a_1, D_0] = \sum_{j=0}^k \frac{\binom{k}{j}}{\binom{n}{j}} (a_1 - a_0)^j \sum_{i=j}^n \binom{i}{j} f_i a_0^{i-j} D_0^{n-i}. \tag{58}$$

Here a_0, a_1, D_0 are polynomials in s . The polynomial $\mathcal{B}_k$ is D_0^n times the k -th Bernstein coefficient of f on $[a_0/D_0, a_1/D_0]$. Its formula divides by no interval length and remains valid at coincident endpoints. Let

$$\begin{aligned} D_s &= 16s^2 + 16s + 33, & a &= 17(16s^2 + 33), \\ d_Q &= 16D_s, & d_P &= 528D_s, & b_P &= 256(1 + s)D_s. \end{aligned}$$

All these denominators are strictly positive for $s \geq 0$. The eleven coefficient polynomials and the twelfth polynomial are specified exactly by

$$\begin{aligned} Z_{Qi} &= -\mathcal{B}_i[Q(s, t, 2t - \rho); a, d_Q, d_Q], & 0 \leq i \leq 2, \\ Z_{0i} &= -\mathcal{B}_i[P(s, t, 0); 33a, b_P, d_P], & 0 \leq i \leq 3, \\ Z_{mi} &= -\mathcal{B}_i[P(s, t, 2t - \rho); 33a, b_P, d_P], & 0 \leq i \leq 3, \\ Z_u &= 33a - b_P. \end{aligned}$$

The degrees in t used in (58) are respectively 2, 3, and 3. Also $u_* - \ell = -Z_u/d_P$. For any univariate polynomial $z(s) = \sum_{i=0}^N z_i s^i$, its Bernstein

coefficients on $[L, H]$ are explicitly

$$b_k = \sum_{j=0}^k \frac{\binom{k}{j}}{\binom{N}{j}} (H - L)^j \sum_{i=j}^N z_i \binom{i}{j} L^{i-j} \quad (0 \leq k \leq N). \quad (59)$$

Applying this rational formula to the twelve displayed polynomials gives the following thirteen checks. In every row, every coefficient is strictly greater than 1 under the stated common-denominator normalization.

Polynomial	Degree	Interval
Z_{Q0}, Z_{Q1}	7	$[1/8, 15/16]$, each
Z_{Q2}	7	$[1/8, 17/32]$ and $[17/32, 15/16]$
Z_{00}	7	$[7/10, 15/16]$
Z_{01}	10	$[7/10, 15/16]$
Z_{02}, Z_{03}	11	$[7/10, 15/16]$, each
Z_{m0}	9	$[7/10, 15/16]$
Z_{m1}	10	$[7/10, 15/16]$
Z_{m2}, Z_{m3}	11	$[7/10, 15/16]$, each
Z_u	3	$[1/8, 7/10]$

These inequalities are fixed finite rational arithmetic statements: the formulas above define every coefficient to be checked. Positive Bernstein coefficients imply strict positivity because the Bernstein basis is nonnegative and sums to 1 on the closed unit interval. Thus $Z_{Qi}, Z_{0i}, Z_{mi} > 0$ give the required strict negative t -coefficients, and $Z_u > 0$ gives the endpoint ordering. Empty t -intervals impose no condition; degenerate ones are covered by the same formulas. The two intervals for Z_{Q2} cover the whole required s -range. This is a fixed thirteen-check certificate, with no adaptive subdivision or unreported numerical optimization. $\square$

Lemma B.4 (Positive tail receiver). *On the complete canonical source, $v \leq 0$ implies $F < 33/8$. The required tail-face bound is Lemma B.3, whose fixed polynomial-positivity certificate is displayed above.*

Proof. Only $F > 4$ needs consideration. If $v \leq 0$, then $O_{02} = -b + XP_2 + \delta v \geq -1$ gives $b \leq 2 < k$. The preceding budgets give $q > 0$, $t > V$, and $r + E < 3$. From $O_{12} - tO_{02} = r - (tX - Y)q + (V - t)v$ we obtain $r \leq 1 + t + (tX - Y)q + (t - V)v$; thus $tX - Y > 0$. Increase v, b together by

$$0 \leq \lambda \leq \min\{-v, (1 - P_2)/\beta\}.$$

This fixes p, r, w, F , the head, E , and both other columns. The changes in the last original column are $(0, -(t - V)\lambda, (E + T)\lambda)$, while P_2 increases by $\beta\lambda$. Throughout the segment $v' \leq 0$, $P'_2 \leq 1$, and $0 \leq b' \leq 2 < k$. The original lower bound in row 1 is safe because $O'_{12} \geq r - 2 - V > -1$; its upper bound improves. The stage bounds satisfy $S'_{02} = O_{02} - v' \geq -1$ and $S'_{12} = r - tb' + Yq > 0$, while their upper bounds improve. Both row-2 lower

bounds improve, and

$$S'_{22} = w + Eb' + Zq \leq w + Ek + Zp < w + 2 < 1, \quad O'_{22} \leq S'_{22}.$$

Here $Ek < 1 - T$, $pZ \leq 1 + T$, and $w + 2 = r - F + 2 < 1/2$. The core entry $r - tb'$ stays positive; $w + Eb'$ lies between $w \geq -k$ and $S'_{22} < k$. All remaining changed bounds are either fixed or improved. Hence the whole segment is physical.

Its endpoint has $P'_2 = 1$ or $v' = 0$. In the latter case, if $P'_2 < 1$, then $0 < q = P'_2 < 1$. Increase q by $0 \leq \sigma \leq 1 - q$, increase b' by $X\sigma$, and keep $v' = 0$. The original changes are $(0, -(tX - Y)\sigma, (EX + Z)\sigma)$. We still have $b'' \leq 1 + Xq'' \leq 2$; now $O''_{12} \geq r - 2 > 0$, $O''_{22} < w + 2 < 1$, and $S''_{i2} = O''_{i2}$. The same core estimates apply. This second physical segment ends at $P''_2 = 1, v'' = 0$. The finite safe-tail exclusion therefore applies to a complete source with the original r, F , proving the claim. No rank assumption is added and no cofactor changes in either segment. $\square$

Lemma B.5 (Normalization of the second pivot). *Every complete source with $F > 4$ can be replaced, at the same head, E, D, F , by one with $p = \bar{p} := (1 + T)/Z$. This replacement preserves the entire original tail block O and all prefix coordinates P_j .*

Proof. The preceding proof gives $p \leq \bar{p} < 1 + \beta < 2$. Increase only p along $[p, \bar{p}]$. The $\hat{q}, S$ bounds are relaxed. All tail and core coordinates are fixed. For the head column, its lower bounds follow from $p'x_i - u_i \geq -u_i \geq -1$; its upper bounds follow from

$$p'X < 2, \quad p'Y < (1 + \beta)Y < 1 + V, \quad p'Z \leq 1 + T.$$

The middle strict inequality uses $1 + V - (1 + \beta)Y = 1 - Y + \Gamma > 0$. Also $0 < p' - \beta < 1$. Thus every point of the segment is a complete physical source, and at its endpoint $L_2 = 1$. $\square$

Lemma B.6 (The complete twelve first-column capacities). *Fix $0 < \beta, V, T, X, Y, Z \leq 1$, all six positive cofactors, $0 \leq E \leq 1$, $V < t \leq 1$, and $1 < p = \bar{p} < 1 + \beta < 2$. These assumptions hold for a complete high-value head by Lemmas B.2 and B.5; the recovery assertion does not require an already completed middle or last column. Define*

$$\begin{aligned} j_L(k) &= \max\{(k - p)/X, (tk - p)/Y, (tk - 1 - V)/Y\}, \\ j_U(k) &= \min\{(1 + \beta - \beta k)/\delta, (1 + T - (E + T)k)/\Delta, \\ &\quad (V + \beta - \beta tk)/\Gamma, (V + T - (VE + Tt)k)/\mu\}. \end{aligned}$$

For $k > 2$, a complete first column exists if and only if $j_L(k) \leq j_U(k)$. A single witnessing pair is

$$j = j_L(k), \quad g = \max\{k - 1 - Xj, (tk - 1 - Yj)/V\}. \quad (60)$$

Write the three lower affine functions as $\lambda_i k - \eta_i$ and the four upper functions as $\xi_j - \kappa_j k$, where

$$(\lambda_i, \eta_i) = (1/X, p/X), \quad (t/Y, p/Y), \quad (t/Y, (1 + V)/Y),$$

$$(\xi_j, \kappa_j) = \begin{pmatrix} ((1 + \beta)/\delta, \beta/\delta), ((1 + T)/\Delta, (E + T)/\Delta), \\ ((V + \beta)/\Gamma, \beta t/\Gamma), ((V + T)/\mu, (VE + Tt)/\mu) \end{pmatrix}.$$

Then the complete first-column capacity is

$$K = \min_{1 \leq i \leq 3, 1 \leq j \leq 4} K_{ij}, \quad K_{ij} := \frac{\eta_i + \xi_j}{\lambda_i + \kappa_j}. \quad (61)$$

Every denominator is positive. Raising an existing $k > 2$ to K , while reconstructing the first column, preserves the head, E, p, r, w, F and the other two columns.

Proof. The first-column constraints require

$$\begin{aligned} j &\geq (k - p)/X, \quad j \geq (tk - p)/Y, \\ g &\geq \max\{k - 1 - Xj, (tk - 1 - Yj)/V\}, \\ g &\leq \min\{1, (1 - j)/\beta, (1 - Ek - Zj)/T\}. \end{aligned}$$

Comparing the two lower bounds for g with its last two upper bounds gives exactly the four terms of j_U . The first lower bound for g is at most $p - 1 < 1$; the second is at most 1 precisely when $j \geq (tk - 1 - V)/Y$. This proves necessity and gives (60) when the intervals intersect.

For completeness, the omitted physical bounds follow from this same pair. We have $j > 0$, and the first upper bound in j_U gives $j < 1$, since $\beta(1 + X - k) < 0$. Then $g \geq k - 1 - Xj > k - 2 > 0$, while the interval comparisons give $g \leq 1$. Thus $O_{00} = k - Xj - g > k - 2 > 0$ and $O_{10} = tk - Yj - Vg > t(k - g) - 1 > -1$, using $t > V$. Their upper bounds have already been imposed. The stage entries S_{00}, S_{10} lie above -1 and below p . Also $-1 \leq O_{20} < 0$ gives $-1 \leq S_{20} < 0$; $P_0 = -j - \beta g \in [-1, 0)$, and the receiver bounds follow from $0 < j < 1 < p$, $0 < g \leq 1$. Finally the first core column obeys $|D_{i0}| \leq k$ because $0 < t \leq 1$, $0 \leq E \leq 1$. This is a complete column.

Every lower-upper comparison is $(\lambda_i + \kappa_j)k \leq \eta_i + \xi_j$, proving (61). At $k = K > 2$ the displayed construction is therefore feasible. Interpolate linearly between the old and new triples (k, j, g) : all first-column conditions are affine at the fixed head and E, p , while increasing k relaxes all other core bounds and $a, b \leq k$. The other two columns and the height stay fixed. The reconstructed first column itself gives $k \leq 1 + Xj + g \leq 2 + (1 - \beta)X < 3$ and $j + g > 1$; together with $Z > T$ and $Ek + Zj + Tg \leq 1$, this proves $Ek < 1 - T$ without assuming the existence of either of the other columns. $\square$

Lemma B.7 (Strict gaps at an actually completed high-value capacity). *Every complete source with $F > 4$ satisfies*

$$O_{00} < 3 - F/2, \quad g < 3 - F/2, \quad F + \beta < 5.$$

Consequently, at a complete high-value source with $k = K$, all eight capacities other than $K_{13}, K_{14}, K_{23}, K_{24}$ are strictly larger than K . In particular

$$K_{13} = \frac{p\Gamma + X(V + \beta)}{\Gamma + \beta tX}, \quad K_{23} = \frac{p\Gamma + Y(V + \beta)}{tV},$$

$$K_{14} = \frac{p\mu + X(V + T)}{\mu + X(VE + Tt)}, \quad K_{24} = \frac{p\mu + Y(V + T)}{V(tZ + EY)}.$$

These four capacities may replace the twelve at this completed source and in any neighborhood where the strict gaps persist. This statement does not assert that deleting eight inequalities from an arbitrary uncompleted or low-value first-column problem is harmless.

Proof. With the nonnegative tail slack J used above,

$$\begin{aligned} \mu(1 - O_{00}) &= \tau(F - 4) + (\Delta - \tau)(r - 2) + \Omega(k - b) \\ &\quad + \Delta(1 - O_{10}) + \tau(1 + O_{20}) + J. \end{aligned}$$

Since $r - 2 \geq (F - 4)/2$ and $0 < \mu = V\Delta - T\tau < \Delta + \tau$, this yields $1 - O_{00} > (F - 4)/2$. Furthermore

$$\begin{aligned} \beta X(1 - g) &= (k - 2) + (p - S_{00}) + X(1 + P_0) \\ &\quad + (1 + \beta - p) + (1 - \beta)(1 - X) \geq k - 2, \end{aligned}$$

and $0 < \beta X < 1$ gives $1 - g > k - 2 \geq F/2 - 2$. Finally

$$O_{00} - (k - 2 + \beta) = (1 - X)j + (1 + P_0) + (1 - \beta)(1 - g) > 0,$$

so $k \geq F/2$ proves $F + \beta < 5$.

If a first-column capacity is attained, its lower and upper affine bounds for j coincide and equal the actual j . For columns 1 or 2 of the capacity array, equality forces the O_{00} lower bound on g to equal a prefix or row-2 upper bound, hence $O_{00} = 1$, which is impossible. For row 3 of the capacity array, equality forces the O_{10} lower bound for g to be 1, hence $g = 1$, also impossible. The excluded indices are precisely the complement of $\{13, 14, 23, 24\}$. All inequalities are strict at the complete source, so continuity preserves them locally. $\square$

Lemma B.8 (Recovering the omitted first-column bounds from a common tail). *Fix $0 < \beta, V, T, X, Y, Z \leq 1$, all six positive cofactors, $0 \leq E \leq 1$, and $V < t \leq 1$. Assume $\Delta > \tau$, $\Omega > 0$, and $1 < p = \bar{p} < 1 + \beta < 2$. Suppose*

$$2 < k \leq K_{13}, K_{14}, K_{23}, K_{24}, \quad 2 < r \leq k, \quad |w| \leq r, \quad F = r - w > 4,$$

and suppose a tail choice $0 \leq b \leq k$ has

$$J_{\text{tail}} := \mu + \Delta + \tau - \Delta r + \tau w + \Omega b \geq 0. \quad (62)$$

Then the first column has the complete physical reconstruction

$$j = \max\{(k - p)/X, (tk - p)/Y\}, \quad g = (tk - 1 - Yj)/V. \quad (63)$$

In particular $O_{10} = 1$, and the original twelve-capacity test holds. If $k = \min\{K_{13}, K_{14}, K_{23}, K_{24}\}$, then $k = K$ as defined in (61).

Proof. The four capacity inequalities say that each of the two lower bounds in (63) is at most each of the last two upper bounds in j_U . Substitution therefore gives

$$S_{00} \leq p, \quad S_{10} \leq p, \quad O_{10} = 1, \quad j + \beta g \leq 1, \quad Ek + Zj + Tg \leq 1.$$

At this point no sign or receiver bound for g is presumed. The first-column gap identity used in Lemma B.7, being algebraic, now gives

$$\mu(1 - O_{00}) = \tau(F - 4) + (\Delta - \tau)(r - 2) + \Omega(k - b) + \tau(1 + O_{20}) + J_{\text{tail}} > 0.$$

Thus $O_{00} < 1$ follows from the common tail slack, without assuming that the unfinished first column is already physical. We have $j > 0$ because $k > 2 > p$. If $g \leq 0$, then $j + \beta g \leq 1$ implies $O_{00} = k - Xj - g \geq k - X - \delta g \geq k - 1 > 1$, a contradiction. Hence $g > 0$ and $j < 1$. Combining $k - Xj \leq p$ with $j \leq 1 - \beta g$ gives

$$\beta Xg \leq p + X - k < \beta + X - 1 \leq \beta X,$$

so $g < 1$. Now $O_{00} > k - 2 > 0$, $O_{10} = 1$, and $-1 \leq O_{20} < 0$. The stage entries satisfy $S_{00} > k - 1 > 0$, $S_{10} = 1 + Vg > 0$, and $-1 \leq S_{20} < 0$; their upper bounds were already imposed. The prefix, receiver, and first-core-column bounds follow as in Lemma B.6. This is a complete first column. Consequently $k \leq K$. Conversely K is the minimum of all twelve capacities and hence at most their four-term minimum, which proves the final assertion. The argument makes no claim about a four-capacity point lacking the common positive-height tail slack. $\square$

Lemma B.9 (The complete middle-column capacity). *Fix head coordinates in $(0, 1]$ with all six cofactors positive. Assume*

$$\begin{aligned} 0 \leq E \leq 1, \quad V < t \leq 1, \quad X > \delta, \quad \zeta = t\nu - E\Gamma > 0, \\ p = \bar{p}, \quad 1 < p < 1 + \beta < 2. \end{aligned}$$

Fix $k > 2$ for which a complete first column has been supplied; in particular one may use Lemma B.8 or $k = K$ above. No pre-existing complete high-value matrix is assumed in the converse. Put $H = (p - 1)/\beta \in (0, 1)$, and define

$$\begin{aligned} A_1 &= 1 + Y + t(1 - X), & B_1 &= Q_o, \\ A_2 &= \frac{t(1 + Z) + E(1 + Y)}{t + E}, & B_2 &= \frac{\zeta}{t + E}, \\ C_1 &= 1 + Y + t(p - X), & D_1 &= \Gamma + t\beta X, \\ C_2 &= \frac{tk + E(1 + Y)}{t + E}, & D_2 &= \frac{E\Gamma}{t + E}. \end{aligned}$$

Then $B_1, B_2, D_1 > 0$, $D_2 \geq 0$, and the middle column can carry a prescribed $r > 2$ if and only if

$$r \leq \mathcal{R}_{\text{mid}} := \min \left\{ \frac{C_1, C_2, A_1 + B_1 H, A_2 + B_2 H, \frac{B_i C_j + D_j A_i}{B_i + D_j}}{(1 \leq i, j \leq 2)} \right\}. \quad (64)$$

For every such r , the single reconstruction is

$$\begin{aligned} h &= \max\{0, (r - A_1)/B_1, (r - A_2)/B_2\}, \\ a &= \frac{r - 1 - Y + \Gamma h}{t}, \quad \hat{q}_1 = -1 - \beta h. \end{aligned} \quad (65)$$

It has $O_{11} = 1$ and $O_{01} = -1$ or $O_{21} = 1$. It uses exactly the p, k, E of the reconstructed first column.

Proof. First consider an existing middle column. Lower a to $a_* = (r - 1 - Y + \Gamma h)/t > 0$. Since $O_{11} \leq 1$, this does not increase a ; it raises O_{11} to 1. The potentially increasing O_{01} remains below zero because $O_{01} = -a - X + \delta h < \delta - X < 0$. The stage entry S_{01} rises but stays negative. Both O_{21} and D_{21} decrease and remain positive, because $r > 2$, $Z \leq 1$, and $\nu h < 1$. At the endpoint $D_{11} = 1 + Y - \Gamma h \in (0, 2)$ and $S_{11} = 1 - Vh \in (0, 1]$; their lower bounds improve and their upper bounds hold throughout. Thus the segment is physical and fixes every other column and r, w .

With $O_{11} = 1$, substitution of $a = a_*$ gives exactly

$$\begin{aligned} t(1 + O_{01}) &= A_1 + B_1 h - r, \\ t(1 - O_{21}) &= (t + E)(A_2 + B_2 h - r), \\ t(p + S_{01}) &= C_1 - D_1 h - r, \\ t(k - r - Ea) &= (t + E)(C_2 - D_2 h - r). \end{aligned}$$

Together with $0 \leq h \leq H$, these are the four line tests $r \leq A_i + B_i h, C_j - D_j h$. Their interval intersection is nonempty exactly under the eight comparisons in (64). This remains valid when $E = 0, D_2 = 0$, because every $B_i + D_j$ stays positive. Its smallest admissible h is (65). Since $A_2 \leq 2 < r$, this h is positive and at least one of the two increasing lines is attained.

Conversely use this h, a . The four identities impose the two original bounds, the stage bound, and $D_{21} \leq k$. We have $a > 0$, while the stage bound gives $a + X(1 + \beta h) \leq p$, hence $a < p < 2 < k$. The other core entry is $D_{11} = 1 + Y - \Gamma h \in (0, 2)$. The stage entries are $S_{11} = 1 - Vh \in (0, 1]$ and $S_{21} = r + Ea - Z(1 + \beta h) \geq r - p > 0$; also $S_{21} \leq O_{21} \leq 1$. The remaining original bounds follow from $O_{01} < \delta - X < 0$ and $O_{21} > r - 2 > 0$. Receiver and prefix bounds follow from $0 < h \leq H < 1$, $1 + \beta h \leq p$, and $P_1 = -1$. All head quantities, the first column, and the last column remain unchanged. This proves simultaneous physical reconstruction, rather than independent existence claims for different columns. $\square$

B.3. The height polygon and simultaneous wall reconstruction.

Throughout this subsection put $H_0 = 33/8$, and fix the seven-coordinate head $\eta = (t, \beta, V, T, X, Y, Z)$. Cofactors always have the meanings in (49). We use Γ , not the height threshold γ , for the cofactor $V - \beta Y$. Define

$$Q_o = t\delta - \Gamma, \quad \ell = \nu - E\delta, \quad \Omega = t\Delta + E\tau - \mu, \quad \zeta = t\nu - E\Gamma.$$

The common-column results above give, in the high-value domain, $t > V$, $\Delta > \tau$, $Q_o > 0$, $\Omega > 0$, $p = (1 + T)/Z < 1 + \beta < 2$, and $X > 1/(1 + \beta)$. They also provide one first column at the complete twelve-capacity minimum $k = K(E)$, and one middle column for every $2 < r \leq \mathcal{R}_{\text{mid}}(E)$. These are simultaneous completions at the same η, E, p, k .

Lemma B.10 (The positive tail slope gap). *A complete canonical source with $F \geq H_0$ has $\ell > 0$.*

Proof. Suppose instead that $\ell \leq 0$. Write $\epsilon = r - 2 \geq 1/16$, $s = p - 1$, and, for this proof only, $A = (1 + E)\epsilon$, $B = \beta$, $D_B = 1 - B^2$. The common-column budgets give $0 < s \leq B < 1$, $E < 1/2$, and $k \leq 1 + s + X(1 - Bs)$. The middle-column identity

$$t(a - \epsilon) = \epsilon(1 - t) + (1 - Y) + \Gamma h + (1 - O_{11}) > 0$$

gives $a > \epsilon$. Since $\nu > 0$ and $\nu \leq E\delta$, also $E > 0$, and hence $k \geq r + Ea > 2 + A$. It follows that

$$A \leq s(1 - B) \leq B(1 - B), \quad X \geq \frac{1 + A - s}{1 - Bs}.$$

The difference between the last lower bound and its value at $s = B$ is

$$\frac{(B - s)(1 - B - BA)}{(1 - Bs)(1 - B^2)} \geq 0,$$

where $1 - B - BA \geq (1 - B)(1 - B^2) > 0$. Consequently

$$0 < \delta \leq \bar{\delta} := \frac{1 - B - BA}{1 - B^2}.$$

Two further same-matrix identities are

$$\begin{aligned} E\delta - (1 + A - Z) &= (E\delta - \nu) + \nu(1 - h) \\ &\quad + E(a - \epsilon) + (1 - O_{21}) > 0, \end{aligned}$$

$$1 - Ek - \frac{Z}{X}(k - p) - Ts = (1 + O_{20}) + T(1 - O_{00}) + \frac{\Delta}{X}(p - S_{00}) \geq 0.$$

Using $X \leq 1$, $k > 2 + A$, $s \leq B$, $Z > T$, and $T = BZ - \nu \geq BZ - E\delta$, they imply

$$E\bar{\delta} > A, \quad 1 > E(2 + A - B\bar{\delta}) + (1 + A - E\bar{\delta})(1 + A - B + B^2).$$

The substitution $\delta \leq \bar{\delta}$ in the second inequality has the stated direction: its right-hand side before substitution has coefficient $-E(1 + A + B^2) < 0$ in δ .

Set

$$\begin{aligned} \mathfrak{F}(A, E, B) &= (1 - B^2)\{E(2 + A - B\bar{\delta}) \\ &\quad + (1 + A - E\bar{\delta})(1 + A - B + B^2) - 1\}, \\ \mathfrak{G}(A, E, B) &= E(1 - B) - (1 - B^2 + EB)A. \end{aligned}$$

Thus $\mathfrak{F} < 0 < \mathfrak{G}$. Direct expansion gives

$$\mathfrak{F}_A = (1 - B^2 + EB)(2A + B^2 - B + 2) > 0, \quad \mathfrak{G}_A = -(1 - B^2 + EB) < 0.$$

Replacing A by $A_0 = (1 + E)/16$ preserves these necessary inequalities. Put $C = B^2 - 17B + 15$, $D = 1 - B^2$. Then $16\mathfrak{G}(A_0, E, B) = CE - BE^2 - D > 0$, so $C > 0$, $E > D/C$, and $C > 2D$. Since $3B^2 - 17B + 13$ decreases on $[0, 1]$ and is negative at $11/12$, we obtain $0 < B < 11/12$.

Writing $\mathfrak{F}_0(E, B) = \mathfrak{F}(A_0, E, B)$, expansion gives

$$256(\mathfrak{F}_0)_E = 3BE^2 + (32B^3 - 34B^2 + 68B + 2)E + H_4(B),$$

where

$$H_4 = -16B^4 + 288B^3 - 802B^2 + 273B + 290.$$

The Bernstein coefficients of H_4 on $[0, 1]$ are $290, 1433/4, 1757/6, 663/4, 33$, all positive. Therefore $\mathfrak{F}_0$ increases in $E \geq 0$. Moreover

$$\mathfrak{F}_0(D/C, B) = \frac{(1-B)^2(1+B)P_7(B)}{256C^3},$$

$$\begin{aligned} P_7(B) = & -256B^7 + 13072B^6 - 233488B^5 + 1641728B^4 \\ & - 3447057B^3 + 3056607B^2 - 1206016B + 176640. \end{aligned}$$

On the three intervals $[0, 3/8]$, $[3/8, 1/2]$, and $[1/2, 11/12]$, its degree-seven Bernstein coefficients have respective positive minima

$$\frac{52615035}{16384}, \quad \frac{31756901}{17920}, \quad \frac{43980727}{559872}.$$

These three finite coefficient calculations can be recovered directly from the displayed polynomial: after substituting $B = a + (b-a)z$, if the power coefficients are c_j , the Bernstein coefficient of index i is $\sum_{j=0}^i c_j \binom{i}{j} / \binom{7}{j}$. Thus $P_7 > 0$ on the entire required interval, and $\mathfrak{F}_0(E, B) > \mathfrak{F}_0(D/C, B) > 0$, a contradiction. This proof uses four fixed univariate positive-coefficient certificates; it does not use an adaptive multidimensional tree. $\square$

For the reconstruction statements below, the head satisfies the common-column entrance inequalities just listed, and $0 \leq E < \min\{1, \nu/\delta\}$; in particular $\ell > 0$. The quantities K and $\mathcal{R}_{\text{mid}}$ are the complete capacities, with $K > 2$ and $\mathcal{R}_{\text{mid}} > 2$. When a reconstruction is applied at $r > 2$, its first and middle columns are supplied by those capacity lemmas before the tail is constructed. Put

$$\begin{aligned} b_0 &= 1 + X, & a_T &= 1 - Y + \Gamma b_0/\delta, & c_T &= 1 + Z + \nu b_0/\delta, \\ R_T(b) &= a_T + Q_o b/\delta, & C_T(b) &= c_T - \ell b/\delta, & C_0 &= C_T(b_0) = 1 + Z + E b_0, \\ A_c &= \mu + \Delta + \tau, & B_c &= \Delta + \nu + \delta, \\ \mathcal{C} &= A_c + \Omega K, & \mathcal{D} &= \ell a_T + Q_o c_T. \end{aligned} \tag{66}$$

Here R_T, C_T are scalar tail capacities, not matrices. Two useful cofactor identities are

$$\Delta Q_o - \tau \ell = \Omega \delta, \quad \Delta \mathcal{D} = \ell A_c + \Omega B_c.$$

Lemma B.11 (Anchor bounds and lowering the shared coordinate). *One has $a_T < 2 < c_T$. At fixed head, decreasing E preserves a completed first and middle column at any $r > 2$. After replacing the first column by its new complete maximum $K(E)$, the same r remains feasible. In particular $\mathcal{R}_{\text{mid}}(E)$ is nonincreasing on its high-value range, in the precise sense that*

$$E_1 \leq E_2, \quad 2 < r \leq \mathcal{R}_{\text{mid}}(E_2) \implies r \leq \mathcal{R}_{\text{mid}}(E_1).$$

This statement concerns the first two columns only.

Proof. From $Q_o > 0$, $t \leq 1$, and $\tau > 0$, one obtains $\Gamma < \delta$ and $\Gamma X < \delta Y$. If $Y \geq X$, then $a_T < 2 + X - Y \leq 2$; if $Y \leq X$, then $a_T < 1 + Y/X \leq 2$. Also $p < 1 + \beta$ gives $\nu + Z > 1$, whence $c_T > 1 + Z + \nu > 2$.

For the monotonicity assertion retain all completed variables except E . In the first column the only affected entries are $-Ek$, $-Ek - Zj$, and $-Ek - Zj - Tg$. They increase toward zero when E decreases, and remain negative because $j, g > 0$; all their absolute-value bounds are preserved. In the middle column the affected entries are $r + Ea$, $r + Ea - Z(1 + \beta h)$, and $r + Ea - Z - \nu h$. Their upper bounds improve. Their lower bounds remain valid since

$$r + Ea - Z(1 + \beta h) \geq r - Zp = r - 1 - T > 0,$$

and the last original entry is this stage entry plus $Th \geq 0$. Here $1 + \beta h \leq p$, $pZ = 1 + T$, and $r > 2$. The core entry $r + Ea$ remains positive. These are all occurrences of E in the two columns. Finally each original first-column capacity has a positive constant numerator and a positive affine denominator with nonnegative slope in E ; hence K cannot decrease when E is lowered. Enlarging the permitted core bound from the old k to this new K , and using the common-column reconstruction, preserves the middle column. No assertion about preserving the tail column under this operation has been used. $\square$

Lemma B.12 (Tail slacks at one complete matrix). *Write $x_r = r$, $y = -w$, and define*

$$\begin{aligned} J_a &= \mu(1 + O_{02}) + \Delta(1 - O_{12}) + \tau(1 + O_{22}), \\ J_b &= \Delta(1 - P_2) + \nu(1 + O_{02}) + \delta(1 + O_{22}). \end{aligned}$$

Then the following exact identities hold:

$$\begin{aligned} J_a &= A_c + \Omega b - \Delta x_r - \tau y, \\ J_b &= B_c - \ell b - \delta y, \\ \ell J_a + \Omega J_b &= \Delta(\mathcal{D} - \ell x_r - Q_o y), \\ C_0 - y &= (1 + O_{22}) + E(1 + O_{02}) + (EX + Z)(1 - P_2) + \ell v. \end{aligned} \tag{67}$$

At a complete source of height at least H_0 , all displayed slacks are nonnegative, and $C_0 - y > 0$.

Proof. Expand $O = S + u\hat{v}^T$, $S = D + x\hat{q}^T$, and $P = \beta\hat{v} + \hat{q}$, and cancel terms by the six cofactor definitions. Every weight in J_a, J_b is positive. In the last identity all terms are nonnegative and $\ell v > 0$, by Lemmas B.10 and B.4. $\square$

The height polygon is the following closed set of scalar pairs:

$$\begin{aligned} 0 \leq y \leq x_r \leq \mathcal{R}_{\text{mid}}, \quad y \leq C_0, \\ \Delta x_r + \tau y \leq \mathcal{C}, \quad \ell x_r + Q_o y \leq \mathcal{D}. \end{aligned} \tag{68}$$

Every complete high-value source maps to this polygon after normalizing p, k : the first oblique inequality follows from $b \leq K$ and $J_a \geq 0$, the second from their positive combination in (67), and $y \leq C_0$ from the last identity. Also $y = F - r > 0$, because $r < 5/2 < F$. This argument does not assume that the original matrix already has any of the three tail contacts.

Lemma B.13 (Complete recovery on the wall). *Fix normalized common first and middle columns, and suppose $r \geq H_0/2$. There is a complete tail at $w = -r$ if and only if*

$$r \leq \mathcal{R}_{\text{mid}}(E), \quad r \leq \mathcal{W}(E), \quad \mathcal{W}(E) := \min \left\{ C_0, \frac{\mathcal{C}}{\Delta + \tau}, \frac{\mathcal{D}}{\ell + Q_o} \right\}.$$

The sufficiency assertion uses the complete twelve-capacity first column. It does not treat four selected capacities as a standalone relaxation. If a complete wall of height $m \geq f > H_0$ exists at this head and E , another complete wall of height exactly f exists there, with all columns reconstructed at common p, K, E .

Proof. Necessity is (68) at $x_r = y = r$. For sufficiency first suppose $r \leq R_T(K)$. Set

$$b = \max\{b_0, \delta(r - a_T)/Q_o\}, \quad v = (b - b_0)/\delta, \quad q = 1 - \beta v.$$

Then $b_0 \leq b \leq K$, $R_T(b) \geq r$, and $C_T(b) \geq r$. For the last inequality use $r \leq C_0$ when $b = b_0$, and $(\ell + Q_o)r \leq \mathcal{D}$ otherwise. The capacity $K \leq b_0 + \delta(p - 1)$ gives $0 \leq v \leq p - 1 < 1$; hence $0 < q \leq 1$, $P_2 = 1$, and $O_{02} = -1$. The two remaining original entries obey $O_{12} = 1 + r - R_T(b) \leq 1$ and $O_{22} = -r + C_T(b) - 1 \geq -1$. Their opposite bounds follow from

$$O_{12} > r - K > -1, \quad O_{22} \leq -r + EK + Zq + Tv < 2 - r < 0.$$

The strict upper estimate uses $EK < 1 - T$, $Zq \leq 1$, and $Tv \leq T$. For the stage bounds, $S_{02} = -1 - v \geq -p$, $S_{12} > r - K > -1 > -p$, and $S_{22} = O_{22} - Tv \geq -1 - Tv \geq -p$. All stage upper bounds follow from $S_{i2} = O_{i2} - u_i v \leq 1 < p$. The core entries satisfy $r - tb \geq r - K > -K$, $-r + Eb \geq -r \geq -K$; their upper bounds follow from the corresponding original entries after subtracting the nonnegative completion terms. Also $b \leq K$. These verify every tail band in (48).

If instead $r > R_T(K)$, set

$$b = K, \quad q = \frac{1 + V + (t - V)K - r}{\tau},$$

$$v = \frac{Xr - (tX - Y)K - X - Y}{\tau}.$$

Then $O_{02} = -1$, $O_{12} = 1$, and $O_{22} \geq -1$ is exactly $(\Delta + \tau)r \leq \mathcal{C}$. At $r = R_T(K)$ these formulas agree with the preceding formulas at $b = K$; increasing r increases v and decreases P_2 . Thus $v > 0$ and $P_2 < 1$. The missing

upper bound for v must not be assumed. It follows from the independent identity

$$\begin{aligned}\mu(p + S_{02}) &= X(V + T)(r - 2) \\ &\quad + [X(VE + Tt) + \mu](K - b) + \mu(p - S_{00}) \\ &\quad + XV[(1 + O_{20}) + (1 + O_{22})] \\ &\quad + XT[(1 - O_{10}) + (1 - O_{12})] > 0.\end{aligned}$$

All quantities on its right have already been verified, and its first term is strictly positive. Since $S_{02} = -1 - v$, it gives $v < p - 1 < 1$. Now $q = (K - 1 - v)/X > 0$, so $0 < P_2 < 1$ and $q < 1$. The remaining original, stage, receiver, and core bounds follow exactly as in the preceding construction; in particular $S_{12} = 1 - Vv > 0$ and $S_{22} \geq -1 - Tv \geq -p$.

In both cases first and middle columns are those given by Lemmas B.6 and B.9 at the same p, K, E, r . Hence this is a single matrix, not a combination of separately optimized completions. Finally, decreasing the wall coordinate from $m/2$ to $f/2$ preserves every inequality of (68): the two oblique coefficients are positive and all other bounds are upper bounds. The two explicit constructions apply again at $r = f/2 > 2$. $\square$

Lemma B.14 (Exactness at high polygon maxima). *If the maximum of $x_r + y$ on (68) exceeds H_0 , it is attained by one complete canonical source at the fixed η, E . Thus the polygon computes the exact maximum on this high-value range; no assertion about recovery of arbitrary low-value polygon points is made.*

Proof. Take a maximizing point. The first and middle columns are reconstructed at $k = K$ and its actual $r = x_r$. Since $x_r \geq (x_r + y)/2 > 2$, the first and middle columns alone give $k \leq 3 - \beta$ and $x_r \leq 2 + \beta - T$. Thus $x_r \leq \mathcal{R}_{\text{mid}} < 5/2$ and $y > 0$. A wall point was handled in Lemma B.13. At a nonwall point with $y = C_0$, the second oblique inequality gives $x_r \leq R_T(b_0)$; take $b = b_0, v = 0, q = 1, w = -C_0$. The same entrywise checks apply, and $O_{22} = -1$. This would contradict the positive-receiver lemma at high value, so this alternative cannot actually occur.

If $x_r > y$ and $y < C_0$, the second oblique inequality must be an equality. Otherwise a sufficiently small move in the direction $(-\tau, \Delta)$ preserves the first oblique constraint, improves the bound on x_r , preserves all strict bounds, and increases the objective by $\Delta - \tau > 0$. Set

$$b = (B_c - \delta y)/\ell, \quad v = (b - b_0)/\delta, \quad q = 1 - \beta v, \quad w = -y.$$

The inequalities and the two cofactor identities give $b_0 < b \leq K$, $R_T(b) = x_r$, and $C_T(b) = y$. Thus $P_2 = 1$, $O_{02} = -1$, $O_{12} = 1$, $O_{22} = -1$. The bounds $v \leq p - 1$, $q > 0$, and all stage bounds are as in the first wall construction. For the last core entry use $-y + Eb \geq -y \geq -x_r \geq -K$ and $-y + Eb \leq O_{22} = -1 < K$. This proves complete recovery. $\square$

B.4. Why a maximum at fixed head has a wall representative. The next finite statement is an explicit computational input. Its analytic role is separated from the existence and derivative argument. For $i = X, Y$ set

$$c_X = \frac{\mu}{VX} + \frac{Tt}{V}, \quad c_Y = \frac{tZ}{Y}, \quad D_i = E + c_i, \quad \sigma = Q_o - \ell,$$

and define

$$\begin{aligned} N_{i1} &= (Q_o + \delta E)D_i b - \sigma(D_i a + k), \\ N_{i2} &= Q_o \nu(t + E)D_i b - \sigma[t\nu D_i a + \zeta k]. \end{aligned} \quad (69)$$

Proposition B.15 (Finite exclusion of the nonwall stationary source). *Consider a complete source with $F \geq H_0$, $\Gamma > \nu$, $\sigma > 0$, $r + w > 0$, and the simultaneous contacts*

$$\begin{aligned} O_{10} &= 1, \quad O_{20} = -1, \quad D_{21} = k, \quad pZ = 1 + T, \quad O_{11} = 1, \\ P_2 &= 1, \quad O_{02} = -1, \quad O_{12} = 1, \quad O_{22} = -1, \\ (S_{00} &= p \text{ or } S_{10} = p), \quad (O_{01} = -1 \text{ or } O_{21} = 1). \end{aligned} \quad (70)$$

Then all four numbers $N_{X1}, N_{Y1}, N_{X2}, N_{Y2}$ are positive.

Certificate reduction and verification obligation. The identity

$$N_{i2} = \zeta N_{i1} + E\Gamma\sigma D_i(b - a)$$

reduces the assertion to $N_{X1} > 0$ and $N_{Y1} > 0$. Here $\zeta = t\ell + EQ_o > 0$, and $b = 1 + X + \delta\nu > 1 + X$, whereas $a < p - X < 2 - X$; thus $b - a > 0$. The two choices in each of the last line of (70) give four contact systems. Each is combined separately with $-N_{X1} \geq 0$ and $-N_{Y1} \geq 0$, giving eight closed necessary systems.

Their algebraic elimination is explicit. The first column is

$$\begin{aligned} S_{00} = p : \quad k &= \frac{p\mu + X(V + T)}{\mu + X(VE + Tt)}, \quad j = (k - p)/X, \\ S_{10} = p : \quad k &= \frac{p\mu + Y(V + T)}{V(tZ + EY)}, \quad j = (tk - p)/Y, \end{aligned}$$

with $g = (tk - 1 - Yj)/V$. The middle column is respectively

$$\begin{aligned} O_{01} = -1 : \quad r &= \frac{Q_o k + E[\delta(1 + Y) + \Gamma(1 - X)]}{Q_o + \delta E}, \\ h &= (r - 1 - Y - t(1 - X))/Q_o, \quad a = 1 - X + \delta h; \\ O_{21} = 1 : \quad h &= (k - 1 - Z)/\nu, \quad r = k - Ea, \\ a &= (k - 1 - Y + \Gamma h)/(t + E). \end{aligned}$$

In either case

$$\begin{aligned} b &= \frac{\delta(r - 1 + Y) - \Gamma(1 + X)}{Q_o}, \quad v = (b - 1 - X)/\delta, \\ q &= 1 - \beta v, \quad y = 1 + Z + Eb - \nu v. \end{aligned}$$

Every denominator displayed is positive. The archived finite systems retain these as denominator-cleared equations and retain defined auxiliaries, so the

maximum input degree is three; no auxiliary is free to vary independently of its defining equation.

For precision, their common closed root box is

t	$[1/16, 1]$	E	$[1/512, 1/2]$
β, T	$[1/16, 7/8]$	V	$[1/16, 1]$
X, Z	$[1/2, 1]$	Y	$[1/32, 1]$
δ	$[1/8, 1]$	Γ, ν	$[1/16, 1]$
$Q_o + \delta E$	$[0, 2]$	k	$[33/16, 3]$
a	$[1/8, 3/2]$	b	$[3/2, 3]$
r	$[33/16, 5/2]$	y	$[13/8, 5/2]$
p	$[17/16, 15/8]$	g	$[1/8, 15/16]$
j	$[3/16, 1]$	h	$[1/16, 1]$
v	$[0, 1]$	D_i	$[0, 48]$.

The less immediate bounds follow without numerical root searching. Put $\epsilon = r - 2 \geq 1/16$. The middle column gives $\nu h \geq \epsilon$, hence $\nu \geq 1/16$; $\Gamma > \nu$, $V > \Gamma$, and $Y > XV$ give the displayed lower bounds for Γ, V, Y . The strict first-column gap gives

$$g - 1/8 = (k - 33/16) + (1 - Xj) + (15/16 - O_{00}) > 0.$$

Also $p < 15/8$ and $k - Xj \leq p$ imply $j > 3/16$. The contact $O_{20} = -1$ gives $Ek - \nu g = (1 - Z) + Z(1 - j - \beta g) \geq 0$, so $E > 1/384 > 1/512$. Further,

$$(t - E)a - 2\epsilon = (1 - Y) + (1 - Z) + (\Gamma - \nu)h + (1 - O_{21}) > 0,$$

giving $a > 1/8$, and $T(1 - h) - \epsilon - Ea = (1 - O_{21}) + Z(p - 1 - \beta h) \geq 0$ gives $T > 1/16$. Finally $k \leq 3 - \beta$ and $r \leq 2 + \beta - T$ imply $r < 5/2$, whence $y = F - r > 13/8$. The definitions bound $D_X < 1/2 + 32 + 14 < 48$ and $D_Y < 1/2 + 32 < 48$. Other endpoints are weakened forms of the common-column and first-column height bounds.

The eight proof trees, grouped by stage contact, middle contact, and tested numerator X, Y , have respective node counts

$$(17317, 9111), \quad (42915, 22627), \quad (8781, 4825), \quad (20223, 10711).$$

They contain 136510 nodes and 68259 contradictory leaves, with no open leaf. Each stored model has 18 equalities and 60 nonnegative polynomial rows, including the one tested negative numerator. Their coefficients, root boxes, and immutable payload identities are specified in the supplementary source index under the nonwall stationary exclusion (historical identifier C19/V20). Acceptance reconstructs those rows and checks every leaf with outward integer interval arithmetic; covered binary splits propagate emptiness to each root by finite induction. This finite acceptance, rather than the preceding algebra alone, establishes the two strict positivities. The displayed numerator identity then establishes the other two. $\square$

Theorem B.16 (Same-head maximum on the wall). *If the complete canonical fibre at a fixed head has a source of height greater than H_0 , its maximum is attained, and every value of E attaining that maximum has a complete*

wall representative of the same height. Consequently every attainable target $H_0 < f < 5$ has a same-head wall representative of height exactly f .

Proof. Attainment follows directly from the full physical domain. Restrict to heights $F \geq f > H_0$ below a known attainable height. Then $r \geq f/2$, $k \geq r$, $p \geq k/2 \geq f/4$, and $p \leq 2, k \leq 4$; all other inner variables are bounded by the physical bands. At a fixed head the six positive cofactors are constant. The remaining constraints are closed, and the pivots have uniform positive lower bounds. This high-value fibre is compact.

Normalize p, k and use the exact polygon maximum. Write $\sigma = Q_o - \ell$ and $R_m = \mathcal{R}_{\text{mid}}$. For $\sigma \geq 0$, elimination of $y = F - x_r$ from the polygon gives its high maximum as the minimum of

$$\begin{aligned} U_1 &= 2R_m, & U_2 &= 2\mathcal{C}/(\Delta + \tau), & U_3 &= R_m + C_0, \\ U_4 &= [\mathcal{C} + (\Delta - \tau)C_0]/\Delta, \\ U_5 &= (\mathcal{D} + \sigma R_m)/Q_o, & U_6 &= a_T + c_T + \sigma K/\delta. \end{aligned}$$

For example, set $s_0 = R_m - x_r$, $s_1 = x_r - y$, $s_2 = C_0 - y$, $s_3 = \mathcal{C} - \Delta x_r - \tau y$, and $s_4 = \mathcal{D} - \ell x_r - Q_o y$. Their slack identities are

$$\begin{aligned} U_1 - F &= 2s_0 + s_1, & U_2 - F &= [2s_3 + (\Delta - \tau)s_1]/(\Delta + \tau), \\ U_3 - F &= s_0 + s_2, & U_4 - F &= [s_3 + (\Delta - \tau)s_2]/\Delta, \\ U_5 - F &= (s_4 + \sigma s_0)/Q_o, & U_6 - F &= [(\Delta - \tau)s_4 + \sigma s_3]/(\Omega\delta). \end{aligned}$$

The converse follows by comparing the lower and upper bounds for x_r , so these are complete polygon budgets, not a selection of numerically observed active constraints.

Suppose there is no wall maximum at a maximizing E_* . If $\sigma \leq 0$, the direction $(-1, 1)$ preserves height, improves both oblique bounds and the upper bound on x_r , and reaches either the wall or $y = C_0$. The latter has a complete high-value recovery with $v = 0$, already excluded. Hence $\sigma > 0$. The first four budget equalities force, respectively, a wall point or $y = C_0$. They therefore have strict slack at this maximum. Only U_5, U_6 can be active.

We next verify the increase of U_6 where needed. By the strict first-column gap, the only locally relevant capacities are the four in (74); two are constant in E , and the others have the form $n/(E + c_i)$, $i = X, Y$. For these last two branches

$$\frac{d}{dE} \left(a_T + c_T + \frac{\sigma n}{\delta(E + c_i)} \right) = \frac{n[\delta c_i - (Q_o - \nu)]}{\delta(E + c_i)^2} > 0.$$

Here is a direct proof of the sign. The inequalities $X > \delta$, $\nu + Z > 1$ imply $\Delta > X\delta$: for $Z \geq X$ use $\Delta - X\delta = \delta(Z - X) + X\nu$; for $Z \leq X$ use

$$\Delta - X\delta = X(\nu + Z - 1) + X(1 - X) + (X - \delta)(X - Z).$$

Thus $\delta Z/X - (Q_o - \nu) = \Delta/X - Q_o > 0$. If $d = tX - Y \geq 0$, both $c_X, c_Y \geq Z/X$. If $d \leq 0$, then $c_X \geq c_Y$, and positivity for c_Y follows from $\Gamma + \nu + t\delta(Z/Y - 1) > 0$ when $Z \geq Y$, or from

$$\Gamma + (\nu + Z - 1) + (1 - Y) + (1 - t\delta/Y)(Y - Z) > 0$$

when $Z \leq Y$; here $t\delta < Y$. Constant capacities likewise give a positive derivative. All finitely many active right branches of U_6 increase.

It follows that U_5 must be active. Its slack identity gives $x_r = R_m$ and $\ell x_r + Q_o y = \mathcal{D}$. The nonwall construction in Lemma B.14 therefore gives all four tail contacts in (70). The common middle reconstruction gives $O_{11} = 1$, with $O_{01} = -1$ or $O_{21} = 1$.

On an active differentiable branch of R_m ,

$$U'_5 = b + \frac{\sigma}{Q_o} R'_m, \quad b = \delta(R_m - a_T)/Q_o > 1 + X > a, \quad 0 < \sigma/Q_o < 1.$$

For the five middle capacities independent of K , the derivatives are zero or, at their actual contacts,

$$-\frac{ta}{t+E}, \quad -\frac{(\Gamma + t\beta X)a}{\nu + \Gamma + t\beta X + \beta X E}.$$

Each is at least $-a$, so it cannot prevent U_5 from increasing. For a constant active K , the two core intersection derivatives are likewise $-Q_o a/(Q_o + \delta E)$ and $-ta/(t+E)$. Thus a blocking right branch must use an E -dependent first capacity and an active middle core capacity. These force $O_{20} = -1$, $D_{21} = k$. The strict first-column gap excludes $O_{00} = 1$ and $g = 1$; hence they also force $O_{10} = 1$ and $S_{00} = p$ or $S_{10} = p$.

For completeness the sign region of this source is determined by three identities. Put $G = g - v$, $A = k - b$, $s_P = 1 + P_0$. The contacts just obtained give

$$\begin{aligned} \Gamma G &= (r - 2) + tA + Y s_P > 0, & (t - E)k &= (Y + Z)j + (V + T)g > 0, \\ (\Gamma - \nu)G &= (r + w) + (t - E)A + (Y + Z)s_P > 0. \end{aligned}$$

Thus $\Gamma > \nu$. In particular the region $\Gamma \leq \nu$, including equality, requires no finite nonwall exclusion.

The two possible variable- K core derivatives are

$$R'_{12} = \frac{Q_o(K'_i - a)}{Q_o + \delta E}, \quad R'_{22} = \frac{\zeta K'_i}{\nu(t + E)} - \frac{ta}{t + E}, \quad K'_i = -\frac{k}{D_i}.$$

Multiplying U'_5 by the respective positive denominators gives exactly N_{i1} and N_{i2} in (69). At a local maximum at least one common active right branch must have a nonpositive numerator; otherwise the finite minimum of all active budgets increases. Proposition B.15 excludes this.

No differentiability assumption at a capacity tie is needed: take all active right branches of the finite family of rational functions. There is a right neighbourhood since a high source has $E < 1/2$ and $\ell > 0$. At $E = 0$, the bare middle capacity C_2 ties the two core intersection capacities at K . Its right derivative is $K'_i - a_0$, where $a_0 = (r - 1 - Y)/t$; the actual reconstructed $a = a_0 + \Gamma h/t > a_0$ supplies an intersection branch with no greater derivative. Thus it introduces no missing obstruction. The hypothetical absence of a wall representative is impossible. Finally lower a maximum wall to any prescribed $f > H_0$ by Lemma B.13. $\square$

B.5. The head entrance used by the near-peak argument.

Proposition B.17 (Finite nonpositive-difference exclusion). *A complete canonical source with $d = tX - Y \leq 0$ has $F \leq H_0$. This includes $d = 0$.*

Analytic reduction to eight finite systems. Suppose $F > H_0$. Theorem B.16 supplies a same-head maximum wall of height $m > H_0$. At a normalized wall feasibility is exactly $r \leq \mathcal{R}_{\text{mid}}(E), \mathcal{W}(E)$. The first capacity is nonincreasing in its high range. When $d \leq 0$, the second is nondecreasing: C_0 increases, and

$$\left(\frac{\mathcal{D}}{\ell + Q_o} \right)' = \frac{\delta Q_o(c_T - a_T)}{(\ell + Q_o)^2} > 0.$$

For the complete first-column terms $N/(A + BE)$, differentiation of ΩK_i gives a positive multiple of $\tau/X - (t - V) = -d/X$, or $\tau t/Y - (t - V) = -Vd/Y$, when $B > 0$; when $B = 0$ the derivative is positive. Hence their minimum is nondecreasing.

These directions allow a maximum wall with $\mathcal{R}_{\text{mid}}(E) = \mathcal{W}(E) = m/2$. Indeed $\mathcal{W}(0) \leq 1 + Z \leq 2 < m/2$, whereas at an actual maximum both capacities are at least $m/2$. Continue the two continuous capacities to $E_{\text{max}} = \min\{1, \nu/\delta\}$. At the right endpoint $\mathcal{R}_{\text{mid}}$ cannot remain above $H_0/2$ while $\mathcal{W} \geq m/2$: near-endpoint wall recovery and a convergent subsequence of bounded inner variables would give a complete high wall with either $\ell = 0$ or $E = 1$, contrary to the proved high-value bounds. The intermediate value theorem gives an interior intersection at height at least $m/2$; maximality gives equality.

At this intersection the C_0 capacity cannot be active, by the strict last slack of Lemma B.12. The two other tail-capacity slack identities force

$$O_{02} = -1, \quad O_{12} = 1, \quad O_{22} = -1, \quad P_2 = 1 \text{ or } b = k.$$

Moreover

$$\frac{k-p}{X} - \frac{tk-p}{Y} = \frac{(Y-tX)(k-p) + pX(1-t)}{XY} \geq 0.$$

The common first-column completion therefore gives $S_{00} = p$, $O_{10} = 1$, and $P_0 = -1$ or $O_{20} = -1$. The middle completion gives $O_{11} = 1$ and $O_{01} = -1$ or $O_{21} = 1$. Together with $pZ = 1 + T$, $w = -r$, $r \geq 33/16$, $d \leq 0$, these are eight exhaustive contact systems. Ties belong to multiple systems and are not discarded.

The finite certificates for these eight systems have, in the order ($P_0 = -1, O_{20} = -1$), then ($O_{01} = -1, O_{21} = 1$), then ($P_2 = 1, b = k$), node counts

$$21341, 31141, 101533, 313597, 21593, 30615, 106077, 284869.$$

The total is 910766 nodes, 455387 contradictory leaves, and zero open leaves. The exact source systems and payload identities are indexed as the nonpositive head-difference exclusion (historical identifier D22/V22). For the $P_2 = 1$

systems there are 24 equalities and 50 nonnegative rows; for the $b = k$ systems there are 23 equalities and 49 nonnegative rows. All input polynomials have degree at most three. These finite acceptance results are computational premises of this proposition.

We record the source of their compact root boxes, to distinguish coverage from a box chosen around numerical samples. Write $\epsilon = r - 2 \geq 1/16$. The contacts give

$$(t - V)v = \epsilon + (1 - t) + (Y - tX)q, \quad tp = 1 + Vg + (Y - tX)j,$$

$$\mu(g - v) = (Y + Z)\epsilon + (tZ + YE)(k - b) + Y(1 + O_{20}) + Y(1 + O_{22}),$$

$$\mu(1 - p + g) = (\Delta + \tau)\epsilon + \Omega(k - b) + \tau(1 + O_{20}) + \tau(1 + O_{22}).$$

The common first-column budget gives $k \leq 9/4$ and $16X^2 - 17X + 4 \geq 0$. This polynomial is negative on $[1/2, 45/64]$, so $X > 45/64$. Since $v \leq p - 1 < 7/8$, the first identity gives $t > 17/30 > 9/16$, $t - V > 1/14$, and $v > \epsilon$. The middle column gives $\nu \geq 1/16$, $T > 1/16$, then $\beta > 1/8$, $Z > 1/2$. With $Y \geq tX$, this yields $Y > 25/64$, $V > 3/64$, $\tau > 3/64$, $\mu = \Gamma Z + \nu Y > 3/128$, and $\Delta > \tau + 1/32 > 5/64$. The strict first-column gap gives $g > 1/8, j > 3/16$. On the wall

$$Eb - \epsilon = (1 + O_{22}) + (1 - Z) + Z(1 - P_2) + \nu v > 0,$$

so $E > 1/36 > 1/64$. The other upper bounds are the physical boxes, $a < p - X < 3/2$, $b \leq k \leq 9/4$, and the first-column height bounds of Lemma B.2, equation (53), and Lemma B.7. The archive uses the following closed weakenings:

t	$[9/16, 1]$	E	$[1/64, 1/2]$
β	$[1/8, 7/8]$	V	$[3/64, 1]$
T	$[1/16, 7/8]$	X	$[45/64, 1]$
Y	$[25/64, 1]$	Z	$[1/2, 1]$
δ	$[1/8, 1]$	Γ	$[0, 1]$
ν	$[1/16, 1]$	τ	$[3/64, 1]$
Δ	$[5/64, 1]$	μ	$[3/128, 1]$
k, r	$[33/16, 9/4]$	a	$[0, 3/2]$
b	$[3/2, 9/4]$	p	$[17/16, 15/8]$
g	$[1/8, 15/16]$	j	$[3/16, 1]$
h	$[1/16, 1]$	v	$[1/16, 7/8]$
q	$[1/16, 1]$		

The stated lower bound for q uses the high-value positive-tail bound in (53) of Lemma B.2, namely $q > 41/480 > 1/16$; this is one of the root-entry dependencies and is not inferred merely from $q > 0$. All six cofactor auxiliaries remain bound to their definitions. As for Proposition B.15, full integer interval acceptance and covered split induction exclude each closed necessary system. Hence a complete source with $d \leq 0, F > H_0$ cannot exist. $\square$

Corollary B.18 (The two remaining head entrance inequalities). *At a head which admits height $f > H_0$, put $p = (1 + T)/Z$, $d = tX - Y$. Then*

$$d > 0, \quad tp + d - 1 > 0, \quad 1 + t + d < 33/16.$$

Proof. The first statement follows from Proposition B.17. At a normalized complete representative $O_{10} = 1$, and

$$tp + d - 1 = (O_{10} - 1) + t(p - S_{00}) + Vg + d(1 - j) > 0.$$

Let $r_0 = 1 + t + d$ and take the same-head wall of height f , with $r = f/2$. If $r \leq r_0$, replace only its tail by $b = b_0 = 1 + X, v = 0, q = 1, w = -r$. The resulting original entries are

$$O_{02} = -1, \quad O_{12} = 1 + r - r_0, \quad O_{22} = C_0 - r - 1.$$

Here $O_{12} > 0$ because $r > 2$, $t \leq 1$, and $0 < d < 1$, while $O_{12} \leq 1$ by the supposed bound on r . Also $C_0 > r$ by the strict wall slack, and $Eb_0 \leq EK < 1 - T$, giving $-1 < O_{22} < 0$. As $b_0 \leq 2 < K$, all remaining stage, receiver, prefix, and core bounds follow as in the wall construction. This would be a complete source of height $f > H_0$ with $v = 0$, impossible. Thus $r > r_0$.

If nevertheless $r_0 \geq 33/16$, keep the same E, p, K and first column, reconstruct the middle column at $r' = r_0$, and use the same zero-receiver tail with $w = -r_0$. The middle reconstruction is valid because $2 < r_0 < r \leq \mathcal{R}_{\text{mid}}$; the tail check now uses $C_0 > r > r_0$. It gives a complete source of height $2r_0 \geq H_0$ and $v = 0$, again impossible. Therefore $r_0 < 33/16$. $\square$

B.6. From a wall representative to a single target endpoint. We now isolate the last finite input to the representative theorem. It concerns a specified set of complete matrices, not all matrices having one of the listed contacts. Let $\mathcal{R}_*$ be the subset of $\mathcal{K}$ satisfying

$$\begin{aligned} w = -r, \quad pZ = 1 + T, \quad O_{10} = O_{11} = 1, \\ (O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \\ (S_{00} = p \text{ or } S_{10} = p), \quad (P_0 = -1 \text{ or } O_{20} = -1), \\ (O_{01} = -1 \text{ or } O_{21} = 1), \quad (P_2 = 1 \text{ or } b = k). \end{aligned} \tag{71}$$

Every contact here is evaluated in the same matrix (47).

Proposition B.19 (Finite exceptional-contact exclusion). *For a complete source in $\mathcal{R}_*$,*

$$S_{00} = p \text{ or } P_0 = -1 \implies F < \Lambda = 1653/400.$$

This is a computational proposition, established by the six exact exclusion certificates described below.

Source reduction and certificate acceptance. Suppose $F \geq \Lambda$. The analytic entrance results above apply since $\Lambda > 33/8$. The possible exceptional contacts are covered by the following six systems; no disjointness is required at ties.

System	First-column contact	Stage contact	Middle contact
PX1	$P_0 = -1$	$S_{00} = p$	$O_{01} = -1$
PX2	$P_0 = -1$	$S_{00} = p$	$O_{21} = 1$
PY1	$P_0 = -1$	$S_{10} = p$	$O_{01} = -1$
PY2	$P_0 = -1$	$S_{10} = p$	$O_{21} = 1$
EX1	$O_{20} = -1$	$S_{00} = p$	$O_{01} = -1$
EX2	$O_{20} = -1$	$S_{00} = p$	$O_{21} = 1$

In each system retain all complete physical inequalities and all common contacts in (71). The last disjunction is encoded by $(1 - P_2)(k - b) = 0$; both factors are nonnegative. The trigger is the exact integer inequality $800r - 1653 \geq 0$. Thus it is not rounded to the earlier threshold $33/8$.

For explicit source-to-box inclusion put $\epsilon = r - 2$, $A = \delta h$, $B = \epsilon + Ea$, and $u_1 = 1 + O_{01}$. These three additional coordinates and the three cofactors below are bound to their defining polynomials. The frozen 24-coordinate root boxes are

Coordinate	Interval	Coordinate	Interval
t	$[15/32, 1]$	E	$[1/64, 1/2]$
β	$[1/8, 7/8]$	V	$[0, 1]$
T	$[1/16, 7/8]$	X	$[45/64, 1]$
Y	$[0, 1]$	Z	$[1/2, 1]$
δ	$[1/8, 1]$	Γ	$[0, 1]$
ν	$[1/16, 1]$	k	$[33/16, 9/4]$
a	$[0, 3/2]$	b	$[1, 9/4]$
r	$[33/16, 9/4]$	p	$[17/16, 15/8]$
g	$[1/8, 15/16]$	j	$[3/16, 1]$
h	$[1/16, 1]$	v	$[0, 7/8]$
q	$[1/16, 1]$	A	$[0, 1]$
B	$[1/16, 1/4]$	u_1	$[0, 1]$

Here $\epsilon > 1/16$, $r \leq k \leq 9/4$, and $\beta < 5 - F < 7/8$. The normalized middle-column bands imply

$$\nu h \geq \epsilon + Ea, \quad \beta h \leq s,$$

$$\epsilon + Ea \leq \nu h - (1 - Z) \leq \nu s / \beta - (1 - Z) = T(\beta - s) / \beta < T \leq s = p - 1.$$

Consequently $\nu, T, h > 1/16$, $\beta > 1/8$, $17/16 < p < 15/8$, and $Z = (1 + T)/p > 1/2$. Also $\delta > 1/8$, $g < 3 - r < 15/16$, and $k \leq p + Xj$ gives $j > 3/16$. The first-column bound $O_{00} < 3 - r$, together with $Xj < 1$, gives $g = k - Xj - O_{00} > 1/8$.

To obtain the stated lower bound for X , use $X > 1/(1 + \beta) > 1/2$ and the common first-column budget $k \leq 1 + \beta + X(1 - \beta^2)$. The identity

$$X(1 + X - k) + \frac{1}{4} = X[1 + \beta + X(1 - \beta^2) - k] + (X\beta - \frac{1}{2})^2 \geq 0$$

gives $16X^2 - 17X + 4 \geq 0$. This convex quadratic is negative at both endpoints of $[1/2, 45/64]$, and hence throughout that interval. Thus $X >$

45/64. The contact $O_{02} = -1$ gives $b = 1 + Xq + v > 1$ and $v \leq p - 1 < 7/8$. The contact $O_{12} = 1$ gives $r < 1 + (9/4)t$, so $t > 17/36 > 15/32$. The positive wall slack gives $Eb > \epsilon$, hence $E > 1/36 > 1/64$; the common budget $Ek < 1 - T$ gives $E < 1/2$. The middle stage gives $a < p - X < 3/2$, and $r + Ea \leq k$ gives $1/16 < B \leq 1/4$. Since $X > \delta$ and $h \leq 1$, $O_{01} = -a - X + \delta h < 0$; therefore $0 \leq u_1 < 1$. Finally $q > 41/480 > 1/16$, while $P_2 \leq 1$ gives $q < 1$. The remaining bounds follow directly from (48). Closed endpoints in the model enlarge this necessary system and do not remove any strict physical source.

For completeness, the additional polynomial inequalities used in the six models have explicit same-source payments. Put $L = 1 + B - X > 0$. The normalized bottom prefix and the middle bands give

$$Z\{h(\beta - s) - B\} = (1 - O_{21}) + (1 - Z)(1 - h + B) \geq 0, \quad h \leq s + u_1, \quad \delta s \geq L.$$

Define

$$\begin{aligned} F_1 &= (1 + Xu_1)^2 - 4XA - 4X^2B, \\ F_2 &= A - 4XAL - 4X^2BL, \\ F_3 &= (1 + Xu_1)^2(1 - 4XL) - 4X^2B. \end{aligned}$$

Their nonnegativity follows from

$$\begin{aligned} F_1 &= (2Xh - 1 - Xu_1)^2 + 4X^2[h(\beta - s) - B] + 4X^2h(s + u_1 - h), \\ AF_2 &= [2\delta(A + XB) - A]^2 + 4XA(A + XB)(\delta s - L) \\ &\quad + 4X\delta^2(A + XB)[h(\beta - s) - B], \\ F_3 &= 4XF_2 + (1 - 4XL)F_1. \end{aligned}$$

Indeed $A > 0$, and $F_2 \geq 0$, $B, L > 0$ imply $1 - 4XL > 0$. These identities use $A = \delta h$ and $u_1 = 1 - a - X + A$. If $O_{01} = -1$, so $u_1 = 0$, $F_3 \geq 0$, $B > 1/16$, and $X > 1/2$ further imply $X \geq 4/5$, because

$$[4X(1 + B - X) + 4X^2B - 1]_{B=1/16} = -\frac{1}{4}(3X - 1)(5X - 4).$$

This last strengthening is used only in the three systems with $O_{01} = -1$. All equalities and inequalities remain polynomial conditions on one 24-coordinate point.

Two further strict rows in the model also come from this same source:

$$\begin{aligned} \mu(g - v) &= (Y + Z)\epsilon + (tZ + YE)(k - b) + Y(1 + O_{20}) > 0, \\ (t - E)k &= O_{10} + O_{20} + (Y + Z)j + (V + T)g > 0. \end{aligned}$$

The first uses the three tail contacts and $O_{10} = 1$; the second uses $O_{10} = 1$, $O_{20} \geq -1$, and positive receivers. Thus $g > v$ and $t > E$ are necessary rows, not extra assumptions.

The model files specify the complete coefficient lists and their necessary-row derivations. The exact acceptor rebuilds interval bounds and polynomial derivative enclosures. An interval leaf exhibits an empty necessary interval. At a combination leaf, after normalizing the current box to $z_i \in [-1, 1]$, it

reconstructs necessary rows $b_j + \sum_i a_{ji} z_i \geq 0$ and verifies rational weights $\lambda_j \geq 0$ with

$$\sum_j \lambda_j b_j + \sum_i \left| \sum_j \lambda_j a_{ji} \right| < 0.$$

The left side bounds the weighted sum above on the entire box, contradicting the necessary rows. Each split covers its closed parent; induction through every branch therefore proves root emptiness. The six accepted trees contain respectively 28073, 38039, 42425, 49053, 29989, 40963 nodes, and together 114274 contradiction leaves with no open leaf. Their exact files and replay entry are identified in Supplementary Index S3. The argument above supplies the source inclusion and the logical meaning of acceptance; the finite verification of all those leaves is a separate part of the computational proof. $\square$

Lemma B.20 (Strict selection of the first-column capacity). *Every source in $\mathcal{R}_*$ with $F \geq \Lambda$ satisfies*

$$S_{10} = p, \quad O_{20} = -1, \quad S_{00} < p, \quad P_0 > -1, \quad k = K_{24} < K_{13}, K_{14}, K_{23},$$

where the capacities are those in (74).

Proof. Proposition B.19 and the disjunctions in (71) give the four entry conditions. Write $K_i = N_i/D_i$, with each $D_i > 0$. Direct expansion in the common matrix gives

$$\begin{aligned} N_{13} - D_{13}k &= \Gamma(p - S_{00}) + XV(1 + P_0) + \beta X(1 - O_{10}) > 0, \\ N_{14} - D_{14}k &= \mu(p - S_{00}) + XV(1 + O_{20}) + XT(1 - O_{10}) > 0, \\ N_{23} - D_{23}k &= \Gamma(p - S_{10}) + VY(1 + P_0) + \beta Y(1 - O_{10}) > 0, \\ N_{24} - D_{24}k &= \mu(p - S_{10}) + VY(1 + O_{20}) + YT(1 - O_{10}) = 0. \end{aligned}$$

All strict signs follow from the positive cofactors and positive X, Y, V . Dividing by the positive denominators proves the claim. $\square$

B.7. The prescribed-target representative and its equivalence. The preceding analytic lemmas and the explicitly identified finite exclusions give the following representative theorem. It asserts existence at the same head, not that every original source already has the representative contacts.

Proposition B.21 (Prescribed-target representative). *Put $\Lambda = 1653/400$. Fix a seven-coordinate head $\eta = (t, \beta, V, T, X, Y, Z)$ for which a complete source in $\mathcal{K}$ can have height at least f , where $\Lambda \leq f < 5$. There is a complete matrix at this same head, with height exactly f , satisfying*

$$\begin{aligned} w &= -r, \quad r = f/2, \quad pZ = 1 + T, \quad P_1 = -1, \quad P_2 = 1, \\ O_{10} &= O_{11} = 1, \quad (O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \\ S_{10} &= p, \quad O_{20} = -1, \quad S_{00} < p, \quad P_0 > -1. \end{aligned} \tag{72}$$

Its cofactor and physical inequalities remain those of Definition B.1. It also satisfies the high-value qualifications recorded in (77) below. Conversely, a target representative at this head is a complete physical source of height f .

Proof. Fix the head η and a target $\Lambda \leq f < 5$, and assume that some complete source has $F \geq f$. Compactness is applied before imposing the strict endpoint qualifications. On the high-value part of the fixed-head fibre,

$$r \geq F/2 \geq f/2, \quad k \geq r + Ea \geq r, \quad p \leq 1 + \beta \leq 2, \quad k \leq p(1 + X) \leq 4.$$

Thus $p \geq f/4 > 1$, $k, r > 2$; all remaining inner coordinates are bounded by the original and stage bands. The six positive cofactors are fixed with the head. The polynomial weak physical inequalities and the uniform positive pivot bounds make this fibre compact. Height therefore attains a maximum $m \geq f$.

Theorem B.16 supplies a complete wall representative at the same head and height m . The wall polygon and reconstruction in Lemma B.13 then give height exactly f : lowering its two equal wall coordinates from $m/2$ to $f/2$ preserves every positive-coefficient upper inequality of the polygon. This use of reconstruction is confined to the high range $f/2 > 2$. The set of such complete walls of height f is a nonempty closed subset of the compact fibre. Choose one minimizing E , and write $r = f/2$.

Normalize its common first and middle columns by Lemmas B.5, B.6, and B.9, without changing the head, E, r, w . The resulting first-column capacity is the full $K(E)$. The high-value entrance inequalities give $v, \ell > 0$, $Ek < 1 - T$, and $Eb > r - 2$; hence $0 < E < 1/2$, strictly within the admissible E -interval. In the notation of Lemma B.13, feasibility requires $r \leq \mathcal{R}_{\text{mid}}(E), \mathcal{W}(E)$. If $\mathcal{W}(E) > r$, continuity preserves that strict inequality when E decreases a little. Lemma B.11 preserves first- and middle-column feasibility. The full wall reconstruction would therefore yield the same height at smaller E , contradicting minimality. Thus $\mathcal{W}(E) = r$.

On this wall $y = r$, and the last identity in (67) gives $C_0 - y > 0$, since $\ell v > 0$ and all other terms are nonnegative. Thus $C_0 > r$, so the active tail bound is not C_0 . If the active bound is $\mathcal{C}/(\Delta + \tau)$, its slack $J_a + \Omega(K - b) = 0$ forces $J_a = 0$ and $b = K$. If it is $\mathcal{D}/(\ell + Q_o)$, its slack $\ell J_a + \Omega J_b = 0$ forces $J_a = J_b = 0$. All the coefficients in these sums are positive. It follows that

$$(O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \quad P_2 = 1 \text{ or } b = k.$$

The common column reconstructions also give the first- and middle-column alternatives in (71). Consequently this one complete representative belongs to $\mathcal{R}_*$. Proposition B.19 and Lemma B.20 give

$$S_{00} < p, \quad P_0 > -1, \quad S_{10} = p, \quad O_{20} = -1, \quad k = K_{24} < K_{13}, K_{14}, K_{23}.$$

Suppose now that $P_2 < 1$. Then $b = k$, $J_a = 0$, and $J_b = \Delta(1 - P_2) > 0$. The only active tail bound is $\mathcal{C}/(\Delta + \tau)$; the other two have strict slack. Lemma B.7 excludes the other eight original capacities. Together with the three strict comparisons above, this makes the full $K(E)$ locally equal to

$K_{24}(E)$. Its numerator $N_{24} = p\mu + Y(V + T) > 0$ is independent of E , and direct differentiation gives

$$\frac{d(\Omega K_{24})}{dE} = -\frac{N_{24}V\mu(tX - Y)}{[V(tZ + EY)]^2} < 0.$$

Decreasing E therefore improves the active tail capacity; continuity preserves the other tail and first-column gaps, and Lemma B.11 preserves the middle-column test. Reconstruction again contradicts minimality. Thus $P_2 = 1$.

The equalities $J_a = J_b = 0$ now give

$$\ell(r - a_T) + Q_o(r - c_T) = 0, \quad r = a_T + Q_o b / \delta.$$

Since $a_T < 2 < r$, solving them is legitimate and yields $E = L_D$ and $b = \delta(r - a_T)/Q_o$ in (75). The strict C_0 slack gives $E > L_C$. The tail receiver is positive, so $b > 1 + X$; the complete core retains $b \leq k$. This proves all contacts and qualifications claimed for the prescribed-target representative.

Here is also the reduction to the five simultaneous middle tests, including its converse. Write the two increasing lines as $I_i(h) = A_i + B_i h$, the decreasing lines as $J_i(h) = C_i - D_i h$, and put $R_{ij} = (B_i C_j + D_j A_i)/(B_i + D_j)$. The complete middle test is the minimum of

$$I_1(H), I_2(H), C_1, C_2, R_{11}, R_{21}, R_{12}, R_{22}.$$

The common normalization has $H = s/\beta$, and

$$\begin{aligned} R_{11} &= A_1 + B_1 s, & I_1(H) - R_{11} &= Q_o s(1 - \beta)/\beta > 0, \\ C_1 - R_{11} &= (t - Q_o)s > 0. \end{aligned}$$

Also $A_2 \leq 2 < r$ and

$$B_2(C_2 - R_{22}) = D_2(R_{22} - A_2).$$

Thus the five retained tests in (76) imply the three omitted ones, and the reverse implication is immediate. All denominators are positive; at ties the same closed inequalities remain valid. If the minimum capacity is written before selecting K_{24} , the two relevant intersections increase with K :

$$\partial_K R_{12} = \frac{Q_o}{Q_o + \delta E} > 0, \quad \partial_K R_{22} = \frac{t\nu - E\Gamma}{\nu(t + E)} > 0.$$

Hence the strict capacity qualifications retain every original first-column comparison rather than discarding it.

Conversely, a head satisfying the stated entrance qualifications, (75), and (76) supplies the formal tail $v = (b - 1 - X)/\delta > 0$, $q = 1 - \beta v$, $w = -r$. The formulas for E, b give all three original tail contacts by substitution. In particular

$$Eb - (r - 2) = (1 - Z) + \nu v > 0,$$

so $E > 0$ before any assertion of complete physicality. First apply the five-to-eight line-interval argument purely algebraically and choose its smallest admissible h , then $a = (r - 1 - Y + \Gamma h)/t > 0$. The fourth line inequality gives

$r + Ea \leq k$, hence $r < k$. The formal tail contacts also give $J_{\text{tail}} = J_a = 0$. These facts, $b \leq k$, and the four capacity qualifications are exactly the hypotheses of Lemma B.8; they restore the full twelve-capacity first-column condition without assuming a pre-existing complete matrix. Lemma B.9 now supplies the complete middle column, and Lemma B.13 supplies the tail. Their common realization is exactly the one printed below (76), with every initial, stage and core band satisfied. Its height is $2r = f$. No step selects incompatible completions of different columns, assumes that all maxima share these contacts, or uses the final global height bound. $\square$

For clarity, the exact endpoint qualifications used in this proposition are included here. They can either be retained in these coordinates or replaced by the equivalent scale predicate in Proposition B.23. For the fixed head put

$$p = (1 + T)/Z, \quad s = p - 1, \quad Q_o = t\delta - \Gamma, \quad \ell = \nu - E\delta.$$

The head entrance includes

$$\begin{aligned} 0 < t \leq 1, \quad 0 < V < 1, \quad 0 < T, X, Y, Z \leq 1, \\ t > V, \quad \tau > 0, \quad \Delta > \tau, \quad \mu > 0, \\ \max\{0, p - 1, 1/X - 1, T/Z\} < \beta < \min\{1, V/Y, 5 - f\}, \\ d = tX - Y > 0, \quad tp + d - 1 > 0, \quad 1 + t + d < 33/16. \end{aligned} \tag{73}$$

Thus $Q_o = \delta(t - V) + \beta\tau > 0$. Define

$$\begin{aligned} K_{13} &= \frac{p\Gamma + X(V + \beta)}{\Gamma + \beta tX}, \quad K_{14} = \frac{p\mu + X(V + T)}{\mu + X(VE + Tt)}, \\ K_{23} &= \frac{p\Gamma + Y(V + \beta)}{tV}, \quad K_{24} = \frac{p\mu + Y(V + T)}{V(tZ + EY)}. \end{aligned} \tag{74}$$

With

$$a_T = 1 - Y + \Gamma(1 + X)/\delta, \quad c_T = 1 + Z + \nu(1 + X)/\delta,$$

the endpoint and its complete qualifications are

$$\begin{aligned} E = L_D &:= \frac{\nu(r - a_T) + Q_o(r - c_T)}{\delta(r - a_T)}, \quad L_C := \frac{r - 1 - Z}{1 + X}, \\ E > L_C, \quad E < 1, \quad \ell > 0, \quad k = K_{24} > 2, \quad k < K_{13}, K_{14}, K_{23}, \\ b &= \frac{\delta(r - a_T)}{Q_o}, \quad 1 + X < b \leq k. \end{aligned} \tag{75}$$

Here $a_T < 2 < r$, so no denominator in (75) vanishes. All four capacity denominators are positive on the declared head. The three strict comparisons in (75) are essential.

The five remaining middle-column tests are also joint tests at this same (η, E, k) . Set

$$\begin{aligned} A_1 &= 1 + Y + t(1 - X), & B_1 &= Q_o, \\ C_1 &= 1 + Y + t(p - X), & D_1 &= \Gamma + t\beta X, \\ A_2 &= \frac{t(1 + Z) + E(1 + Y)}{t + E}, & B_2 &= \frac{t\nu - E\Gamma}{t + E}, \\ C_2 &= \frac{tk + E(1 + Y)}{t + E}, & D_2 &= \frac{E\Gamma}{t + E}, & H &= s/\beta. \end{aligned}$$

Their denominators and B_1, B_2, D_1, D_2 are strictly positive. The tests are

$$\begin{aligned} A_2 + B_2 H &\geq r, & A_1 + B_1 s &\geq r, \\ \frac{B_2 C_1 + D_1 A_2}{B_2 + D_1}, & \frac{B_1 C_2 + D_2 A_1}{B_1 + D_2}, & \frac{B_2 C_2 + D_2 A_2}{B_2 + D_2} &\geq r. \end{aligned} \quad (76)$$

For every head with the declared entrance conditions, the target reduction says that existence of a complete source of height at least f is equivalent to (75) and (76) at $E = L_D$. A common realization is

$$\begin{aligned} g &= s/V, & j &= (tk - p)/Y, & h &= \max\{0, (r - A_1)/B_1, (r - A_2)/B_2\}, \\ a &= (r - 1 - Y + \Gamma h)/t, & v &= (b - 1 - X)/\delta, & q &= 1 - \beta v, & w &= -r. \end{aligned}$$

The proof uses the exact complete-column reconstruction, not separate optimal completions of different columns.

Collecting Lemmas B.2, B.4, B.7, B.10, and Corollary B.18, together with the early-pivot bound, gives the necessary inequalities used subsequently:

$$\begin{aligned} 0 < g, j, h &\leq 1, & v, q, \ell &> 0, & 1 < p < 1 + \beta < 2, & a < p, & k &\leq 9/4, \\ 0 < t &\leq 1, & t &> V, & \Delta &> \tau, & X &> 1/(1 + \beta), & Ek &< 1 - T, \\ F + \beta &< 5, & g &< 3 - F/2, & d &> 0, & tp + d - 1 &> 0, & 1 + t + d &< 33/16. \end{aligned} \quad (77)$$

They apply to complete high-value representatives in this positive chamber; they are not imposed as necessary conditions on arbitrary signed X matrices.

Definition B.22 (Admissible target representative). An admissible target representative is a source in $\mathcal{K}$, at a height $\Lambda \leq F < 5$, with all contacts (72). In using its high-value necessary inequalities we retain the proved qualifications (77). No conditions $t = X = Z = 1$, $b = k$, or $h = s$ are included in this definition.

B.8. A complete row-scale coordinate system. For an admissible target representative with $\Lambda \leq F < 5$, write $F = 2r = 2(1 + R)$, $p = 1 + s$, and use $(\beta, V, s, Y, Z, v, h, b, R)$ as coordinates. Define

$$\begin{aligned} T &= (1 + s)Z - 1, & q &= 1 - \beta v, \\ \Gamma &= V - \beta Y, & \nu &= \beta Z - T, & \mu &= VZ - TY, \\ A &= R + Y + \Gamma v, & C &= R - Z + \nu v, \\ W &= R(Y + Z) + \mu v, & N &= V(Y + Z) + s\mu, & \lambda &= \frac{N}{VW}. \end{aligned} \quad (78)$$

The complete reconstruction is

$$\begin{aligned} t &= A/b, \quad E = C/b, \quad X = (b-1-v)/q, \quad k = \lambda b, \\ g &= s/V, \quad j = (\lambda A - p)/Y, \quad a = \frac{b(R-Y+\Gamma h)}{A}, \quad w = -r. \end{aligned} \quad (79)$$

Inserting these expressions into (47) recovers one matrix. In particular the products appearing in different columns retain the same β, p, k, u, x .

To derive the dictionary, the tail contacts give $tb = A$, $Eb = C$, and $b = 1 + v + Xq$. The first-column contacts give $Vg = s$, $tk = p + Yj$, and $Ek + Zj + Tg = 1$. Since $YC + ZA = W$, elimination in these three equations gives $k/b = N/(VW)$. The equation $O_{11} = 1$ then gives the formula for a . These are exact identities with the same variables.

Here is the full domain needed for the converse reconstruction. Define the fixed-coordinate predicate BASE by

$$\begin{aligned} 0 &< \beta < 1, \quad 0 < V < 1, \quad 0 < T, Y, Z \leq 1, \\ 1 &< p < 1 + \beta, \quad F + \beta < 5, \quad 0 < v \leq s, \\ \Gamma, \nu, \mu &> 0, \quad 0 < g \leq 1, \quad \lambda \geq 1, \\ 0 &< j < 1 - \beta g, \quad pY - V \leq 1, \\ D_h &:= A\nu - CT = \mu + R(\nu - \Gamma) > 0, \\ n_h &:= A(R - Z) + C(R - Y), \quad 0 < h_0 := n_h/D_h \leq H := s/\beta. \end{aligned} \quad (80)$$

Throughout this appendix's height range, $R > 1$. Under BASE, $q > 0$, $A > C > 0$, and all denominators in (79) are positive once $b \geq A$. Put

$$\begin{aligned} B_0 &= 2 + (1 - \beta)v, \quad B_\tau = 1 + v + qY/V, \\ \sigma_0 &= \lambda q - j = \frac{(V+T)(gR-v)}{W} > 0, \quad B_F = \frac{pq - j(1+v)}{\sigma_0}, \\ I &= [A, \infty) \cap (1+v, \infty) \cap (-\infty, B_0] \cap (-\infty, B_\tau) \cap (-\infty, B_F). \end{aligned} \quad (81)$$

The five conditions respectively mean $t \leq 1$, $X > 0$, $X \leq 1$, $\tau > 0$, and $S_{00} < p$. The interval may be empty or a valid singleton. A singleton at a closed endpoint is retained; a point on either of the last two open endpoints is excluded.

Define

$$\begin{aligned} \mathbf{a} &= R(1+q) + Vv, \quad \mathbf{c} = V + \beta R, \\ L(h) &= \frac{rA + C(R-Y+\Gamma h)}{A\lambda}, \\ U_O(h) &= \frac{A[B_0 + (1+\beta)h]}{\mathbf{a} + \mathbf{c}h}, \quad U_S(h) = \frac{A[pq + (1+v)(1+\beta h)]}{\mathbf{a} + \mathbf{c}h}. \end{aligned} \quad (82)$$

Proposition B.23 (Complete scale fibre). *Under BASE, a complete representative with the specified fixed coordinates and target height exists if and only if*

$$h_0 \leq h \leq H, \quad b \in I, \quad L(h) \leq b \leq \min\{U_O(h), U_S(h)\} \quad (83)$$

has a solution (b, h) . Every such solution reconstructs a matrix satisfying all the original physical bands and all six positive cofactors. Its pivots are $1, p, k, r, -2r$.

Proof. The defining identities give

$$A(1 - O_{21}) = D_h h - n_h, \quad (84)$$

$$Aq(1 + O_{01}) = A[B_0 + (1 + \beta)h] - b(\mathbf{a} + \mathbf{c}h), \quad (85)$$

$$Aq(p + S_{01}) = A[pq + (1 + v)(1 + \beta h)] - b(\mathbf{a} + \mathbf{c}h). \quad (86)$$

The inequality $r + Ea \leq k$ is $L(h) \leq b$. The receiver inequality $|-1 - \beta h| \leq p$ is $h \leq H$. Together with (81), these prove necessity.

For sufficiency, $b \geq A > C > 0$ gives $0 < t \leq 1$ and $0 < E < 1$, and $b \in I$ gives $0 < X \leq 1$, $\tau > 0$, and $S_{00} < p$. The other cofactors follow from $\Gamma, \nu, \mu > 0$, $\delta = 1 - \beta X > 0$, and $\Delta = Z - TX \geq Z - T > 0$. The signs and the opposite sides of the physical bands are not discarded. They follow as follows:

- The head bounds use $p - \beta \in (0, 1)$, $L_2 = 1$, $L_0 = pX - 1 \in (-1, 1)$, and $L_1 = pY - V \in (-1, 1]$.
- The first receiver column has $0 < g \leq 1$, $0 < j < 1 - \beta g < 1 < p$, and $P_0 = -\beta g - j \in (-1, 0)$. Its exact contacts are $S_{10} = p$, $O_{10} = 1$, $O_{20} = -1$. Also $S_{20} = Tg - 1 \in [-1, 0]$. From $k \geq r + Ea > r > 2$ and $Xj < 1$, $S_{00} = k - Xj > 1$. Thus $0 < O_{00} = S_{00} - g < p - g < 1$, using $g = s/V > s$.
- The middle receiver has $0 < h \leq H < 1$ and $-p \leq \hat{q}_1 < -1$, with $P_1 = -1$. Equations (85)–(86) give its two nontrivial lower bands. Their opposite inequalities hold from the reconstructed signs; in particular $S_{01} = -a - X(1 + \beta h) < 0$ and $O_{01} = S_{01} + h < 1$. Furthermore $S_{11} = 1 - Vh \in (0, 1)$, $O_{11} = 1$, and $S_{21} = r + Ea - Z(1 + \beta h) > r - 2 > 0$. Together with $O_{21} = S_{21} + Th \leq 1$, this establishes both sides of the remaining middle-column bands.
- The tail has $P_2 = 1$, $(O_{02}, O_{12}, O_{22}) = (-1, 1, -1)$, and

$$(S_{02}, S_{12}, S_{22}) = (-1 - v, 1 - Vv, -1 - Tv).$$

The condition $v \leq s = p - 1$ establishes all three p -bands.

- The core bounds use $a < p < 2 < k$, $b \leq k$, $tk \leq k$, $Ek < k$, and $r + Ea \leq k$. Here $a > 0$ follows from $R > 1 \geq Y$, and $a < p$ from $S_{01} \geq -p$. The remaining entries satisfy $D_{11} = 1 + Y - \Gamma h \in (0, 2)$, $D_{12} = 1 - Yq - Vv \in (-1, 1)$, and $D_{22} = -r + C \in (-r, 0)$.

All entries are those of the same reconstructed matrix. These identities establish the complete bands, not only a subset of necessary inequalities. They also recover the strict first-column capacity comparisons in (75) from the first-column residual identities and $S_{00} < p$, $P_0 > -1$. $\square$

Why no division or endpoint is hidden. The interval I in (81), rather than its closure, is used in the equivalence. The quantities $V, Y, q, A, \lambda, D_h, \sigma_0, \mathbf{a} +$

ch are positive in their stated domains. A zero denominator outside this domain is not an infeasibility proof: such a point has not met the declared qualification. Later, on the certified near-peak box, these qualifications have uniform margins, so its closure does not introduce zero-cofactor or excluded-endpoint cases.

B.9. Common necessary inequalities and the finite localization.

Put $\epsilon = R - 1 > 0$. Two slacks of the same physical completion have numerators

$$\mathcal{T} = (s - Vv)\mu - \epsilon V(Y + Z) = VW(\lambda - 1) \geq 0,$$

$$\mathcal{P} = \epsilon V[(Y + Z) - v(\nu - \Gamma)] - (s - Vv)D_h = VW(1 - j - \beta g) > 0.$$

Their sum is

$$\mathcal{T} + \mathcal{P} = (sR - Vv)(\Gamma - \nu). \quad (87)$$

Because $v \leq s$ and $V < 1 < R$, the first factor on the right is positive. Therefore $\Gamma > \nu$ throughout the qualified high-value completion domain. This is a common-slack identity, not a sign inferred from the candidate matrix.

The two middle capacities satisfy

$$U_O - U_S = \frac{Aq(h - s)}{\mathbf{a} + ch}, \quad U'_O > 0, \quad U'_S < 0, \quad L' > 0.$$

For example

$$U'_O = \frac{2Aq(R - V)}{(\mathbf{a} + ch)^2}, \quad U'_S = -\frac{Aq[\beta R(p - 1 - v) + V(p + 1 + v)]}{(\mathbf{a} + ch)^2}.$$

It follows that every feasible (b, h) satisfies $A \leq b \leq U_O(s) = U_S(s)$, without setting $h = s$. Define

$$\begin{aligned} d_1 &= 1 - V - \beta\epsilon, & a_h &= \epsilon(1 + q) - (1 - V)v, \\ d_S &= V + \beta(\epsilon - v), & s_h &= pq + 1 + v - \mathbf{a}, \\ h_A &= a_h/d_1, & h_S &= s_h/d_S. \end{aligned} \quad (88)$$

Feasibility forces $d_1 > 0$: indeed

$$B_0 + (1 + \beta)h - \mathbf{a} - ch = d_1(h + v) - 2\epsilon q$$

and $b \geq A$, $b \leq U_O(h)$ make the left side nonnegative. Also $d_S > 0$, since $v \leq V$, $\beta < 1$, and $\epsilon > 0$. The two middle inequalities then imply $h_A \leq h \leq h_S$. Equivalently, with

$$\mathcal{M} = (1 - V)(s + v) - \epsilon[2 + \beta(s - v)],$$

one has $\mathcal{M} \geq 0$, $h_A = s - \mathcal{M}/d_1$, and $h_S = s + \mathcal{M}/d_S$.

For reference we specify the necessary system used in the finite localization. This makes the connection from the physical domain to its rational certificate explicit. In the following table all expressions are evaluated at

the same six coordinates (β, V, s, Y, Z, v) and the same fixed R ; they are not independently selected bounds. In addition to the definitions above, put

$$\begin{aligned} u_0 &= rA + C(R - Y), \quad u_1 = C\Gamma, \quad B_s = B_0 + (1 + \beta)s, \quad D_s = \mathfrak{a} + \mathfrak{c}s, \\ c_0 &= NA^2B_0 - VWu_0\mathfrak{a}, \\ c_1 &= NA^2(1 + \beta) - VW(u_0\mathfrak{c} + u_1\mathfrak{a}), \\ c_2 &= -VWu_1\mathfrak{c} < 0. \end{aligned}$$

Thus $c_0 + c_1h + c_2h^2 = VW\Pi(h)$, where

$$\Pi(h) = A^2\lambda[B_0 + (1 + \beta)h] - (u_0 + u_1h)(\mathfrak{a} + \mathfrak{c}h).$$

The unconditional necessary inequalities are $g_i \geq 0$, $0 \leq i \leq 34$, with the following exact polynomial dictionary.

i	g_i
0–6	$\beta - s, T, \nu, \Gamma, \mu, s - v, V - s$, in this order.
7	$B_0 - A$.
8	$(s - Vv)\mu - \epsilon V(Y + Z)$.
9	d_1 .
10	$(1 - V)(s + v) - \epsilon[2 + \beta(s - v)]$.
11	D_h .
12	$\beta s - s^2 - \epsilon\beta$.
13	$3 - 2R - \beta$.
14	$(1 + \beta)Y - V$.
15	$1 - pY + V$.
16	$sD_h - \beta n_h$.
17	$s_h D_h - n_h d_s$.
18	$NAB_0D_h - VW(u_0D_h + u_1n_h)$.
19	$NAB_0d_1 - VW(u_0d_1 + u_1a_h)$.
20	$\mathcal{P}$.
21	$Nq - (V + T)(sR - Vv)$.
22	$pqVW - NqA + [Nq - (V + T)(sR - Vv)](A - 1 - v)$.
23	$qY + V(1 + v) - VA$.
24	$NA^2B_sD_h - VW D_s(u_0D_h + u_1n_h)$.
25	$AB_s - (1 + v)D_s$.
26	$c_1^2 - 4c_0c_2$.
27	$9VW - 4NA$.
28	$(9A - 4u_0)D_h - 4u_1n_h$.
29	$(1 - T)VW - CN$.
30	$(3 - r)V - s$.
31	$17q - 16R - 16Vv + 16v(1 + \beta)$.
32	$B_0(R + Vv) - A(1 + v)$.
33	$(R + Vv - q)B_0 + A(pq - 1 - v)$.
34	$\Gamma - \nu$.

The table is a necessary system, not the definition of a complete physical source. Its rows follow by clearing positive denominators in the scale conditions, the common bounds on h , and (77). To indicate all distinct sources: rows 0–17 use BASE, $A \leq b \leq B_0$, the tail capacity, and $h_0 \leq H, h_S$; rows 18–19 use $L(h_0), L(h_A) \leq B_0$; rows 20–23 use the strict first-column bands and $\tau > 0$. Rows 24–25 use the shared middle bound at s , row 26 the discriminant of the same concave quadratic, rows 27–30 the original pivot and head budgets, and rows 31–33 the three original geometric entrance inequalities. For the latter, the identities are

$$d = \frac{R + Vv - A(1 + v)/b}{q}, \quad 1 + t + d = 1 + \frac{R + Vv}{q} - \frac{Av(1 + \beta)}{qb},$$

$$tp + d - 1 = \frac{(R + Vv - q)b + A(pq - 1 - v)}{bq}.$$

The bounds $A \leq b \leq B_0$ give the displayed necessary rows with their stated directions. Row 34 is (87).

There is one conditional row, not an additional unconditional assumption:

$$g_{36} := c_1 + 2c_2s \geq 0 \implies g_{35} := c_0 + c_1s + c_2s^2 \geq 0. \quad (89)$$

If a feasible $h \leq s$ exists, the nonnegative derivative at s makes the concave quadratic increasing up to s , so $\Pi(s) \geq \Pi(h) \geq 0$. If $h \geq s$, then $L(s) \leq L(h) \leq U_S(h) \leq U_S(s) = U_O(s)$, with the same conclusion. Any certificate row using g_{35} must first establish $g_{36} \geq 0$ on its entire current box.

Set

$$f_0 = \frac{4132517}{10^6}, \quad f_1 = \frac{103313}{25000}, \quad R_0 = \frac{2132517}{2000000}, \quad R_1 = \frac{53313}{50000}. \quad (90)$$

The root enclosure for height f_0 is

$$\mathcal{B}_0 = [1/4, 7/8] \times [1/16, 1] \times [1/16, 7/8] \times [0, 1] \times [1/2, 1] \times [0, 7/8]. \quad (91)$$

Indeed the common middle and core bounds give

$$\epsilon + Ea \leq \frac{T(\beta - s)}{\beta} \leq \frac{s(\beta - s)}{\beta} \leq \frac{\beta}{4}.$$

Thus $s > \epsilon > 1/16$ and $\beta > 4\epsilon > 1/4$. Also $s \leq V < 1$, $v \leq s < \beta < 5 - f_0 < 7/8$, and $Z > 1/(1 + \beta) > 1/2$. Hence every relevant six-tuple lies in $\mathcal{B}_0$. Its zero endpoints are only outer relaxations.

Proposition B.24 (Finite localization and upper-target exclusion). *At $R = R_0$, every point of $\mathcal{B}_0$ satisfying the 35 unconditional inequalities and the conditional implication (89) lies in*

$$Q = [0.617, 0.6181] \times [0.7788, 0.7795] \times [0.4525, 0.4540] \\ \times [0.97352, 0.97361] \times [0.99998, 1] \times [0.1944, 0.1950]. \quad (92)$$

All finite decimals here and below denote exact rational numbers. At $R = R_1$, the same necessary system is infeasible in the box obtained from Q by replacing its last interval by $[0.1944, 0.19501]$.

Finite proof objects and acceptance rule. The first statement is the repaired complete localization certificate: 152,467 records, including 35,069 binary splits, 82,328 contraction nodes, 3,338 direct contradiction terminals, 31,729 nonnegative combination terminals, and three terminals wholly contained in Q . The contraction nodes contain 208,274 individually checked thin-slice exclusions. The second statement is the repaired upper-target tree, with 169 nodes and 85 contradiction terminals. Neither statement uses the earlier, superseded localization or upper-target tree counts.

Their small mathematical acceptance rule is as follows. On a current box use exact outward rational derivative and Hessian enclosures to obtain affine upper bounds $g_j(x) \leq b_j + \sum_i a_{ji}z_i$, $|z_i| \leq 1$. If $g_j \geq 0$, the resulting affine row is necessary. Nonnegative rational weights w_j produce a contradiction when

$$\sum_j w_j b_j + \sum_i \left| \sum_j w_j a_{ji} \right| < 0.$$

The same inequality applied to an edge slice justifies a contraction. Both children of each binary split cover their parent; the contracted child contains every previously possible point. At a localization terminal, containment of the entire current box in Q is checked coordinate by coordinate. This proves the statement for continuous real boxes; the rational mesh records enclosure endpoints, not a discrete sample of possible source points. The complete finite payloads and entry points are identified in §B.13.

B.10. Transport of the target and a compact nine-variable model.

The fixed-target statement alone would not suffice. Fix the old head η of a high representative. Its tail identities give

$$tb = R + Y + \Gamma v, \quad b = 1 + X + (1 - \beta X)v,$$

so

$$v(R) = \frac{R + Y - t(1 + X)}{Q_o}, \quad Q_o = t(1 - \beta X) - \Gamma > 0. \quad (93)$$

Choose its height- f_0 representative using Proposition B.21. Its six-tuple lies in Q . Since $b \geq A$, $b \leq B_0$, and $X \leq 1$,

$$Q_o \geq (A/B_0)(1 - \beta) - \Gamma > 1/5.$$

The last strict inequality is a rational enclosure on the entire height- f_0 box, not just its center. It follows that

$$0 \leq v(R) - v(R_0) \leq 5(R_1 - R_0) = \frac{3}{400000} < 10^{-5} \quad (R_0 \leq R \leq R_1). \quad (94)$$

The coordinates β, V, s, Y, Z do not change with the target, since $s = (1 + T)/Z - 1$ is fixed by the old head. If a source had height at least f_1 , it would have a target representative at f_1 with this same head. Equations (93)–(94) would place it in the upper-target exclusion box, a contradiction. This proves $F < f_1$ for every high representative, including those initially

specified above f_1 . The comparison is between representatives at two targets; it does not assert existence of a continuous physical flow in E .

The bounds $A \leq b \leq B_0$ and $h_A \leq h \leq h_S$ now put every height- $F \geq f_0$ completion in

$$\begin{aligned} \mathcal{B} = & Q_{\beta,V,s,Y,Z} \times [0.1944, 0.19501]_v \times [0.4513, 0.4566]_h \\ & \times [2.0741, 2.0748]_b \times [R_0, R_1]_R. \end{aligned} \quad (95)$$

The relevant verified margins and enclosure statements are gathered in the next proposition so that the inverse-function argument has an explicit physical domain.

Proposition B.25 (Uniform qualifications). *Throughout $\mathcal{B}$, the rational expressions in (78)–(88) have positive denominators, and the following strict inequalities hold:*

$$\begin{aligned} q, \Gamma, \nu, \mu, A, C, W, N, D_h, d_1, d_S, j, g, T &> 0, \\ T < 1, \quad g < 1, \quad E < 1, \quad 0 < v < s < \beta < 1, \quad h < H, \\ 1 - j - \beta g &> 1/1000, \quad p - S_{00} > 1/200, \\ pY - V < 1, \quad F + \beta < 5, \quad X > 0, \quad b > 1 + v, \\ 0 < \beta, V, Y &< 1. \end{aligned}$$

For any point in this box with $X \leq 1$, also

$$\delta \geq 1 - \beta > 0, \quad \tau \geq Y - V > 0, \quad \Delta \geq Z - T > 0.$$

In the enlarged six-coordinate box and height interval, the same rational enclosures give

$$A > 2.0741, \quad B_0 < 2.0748, \quad h_A > 0.4513, \quad h_S < 0.4566.$$

At the low target the transport denominator has the lower bound $(A/B_0)(1 - \beta) - \Gamma > 1/5$.

These are 42 finite rational interval checks, including the matching of the nine-dimensional Jacobian box with (95). The expression for $1 - j - \beta g$ is enclosed using $\mathcal{P}/(VW)$; this preserves the common-slack cancellation in (87). The three last cofactor lower bounds are conditional on $X \leq 1$, which will be supplied by an explicit nonnegative slack, rather than being claimed on the whole outer box.

B.11. Eight physical slacks and a uniform Jacobian. On the domain of the reconstruction define the following eight slacks, all with the feasible

direction $G_i \geq 0$:

$$\begin{aligned}
G_1 &= b - A, & G_2 &= B_0 - b, \quad G_3 = \lambda - 1, \\
G_4 &= \lambda b - r - \frac{C(R - Y + \Gamma h)}{A}, \\
G_5 &= A[B_0 + (1 + \beta)h] - b(\mathbf{a} + \mathbf{c}h), \\
G_6 &= A[pq + (1 + v)(1 + \beta h)] - b(\mathbf{a} + \mathbf{c}h), \\
G_7 &= D_h h - n_h, & G_8 &= 1 - Z.
\end{aligned} \tag{96}$$

Their physical meanings, in the same order, are

$$\begin{aligned}
t &\leq 1, \quad X \leq 1, \quad b \leq k, \quad r + Ea \leq k, \\
O_{01} &\geq -1, \quad S_{01} \geq -p, \quad O_{21} \leq 1, \quad Z \leq 1.
\end{aligned}$$

For example $G_1 = b(1 - t)$, $G_2 = q(1 - X)$, and $G_3 = (k - b)/b$; the identities (84)–(86) give $G_5 = Aq(1 + O_{01})$, $G_6 = Aq(p + S_{01})$, and $G_7 = A(1 - O_{21})$. All these factors are positive.

Lemma B.26 (Exact feasible set in the localized box). *Let $\mathcal{C}$ be the set of all complete scale representatives at heights $F \geq f_0$, with the source and qualifications of Propositions B.21 and B.23. Then*

$$\mathcal{C} = \{z \in \mathcal{B} : G_i(z) \geq 0 \ (1 \leq i \leq 8)\}. \tag{97}$$

In particular, $\mathcal{C}$ is compact and every member has $F < f_1$.

Proof. The forward inclusion is the localization and target-transport argument above, followed by the physical interpretation of the slacks. For the reverse inclusion, all the strict constant qualifications in BASE follow from Proposition B.25; its non-strict tail capacity $\lambda \geq 1$ is $G_3 \geq 0$. To check explicitly the only remaining condition involving h_0 , observe that $R > 1$, $Y < 1$, and $Z \leq 1$ imply $n_h > 0$. Thus $G_7 \geq 0$ gives $0 < h_0 \leq h < H$, using $D_h > 0$. The constraints $G_1, G_2 \geq 0$ give $A \leq b \leq B_0$. The other endpoints of I follow from $X > 0$, $\tau > 0$, and $p - S_{00} > 1/200$; the cofactor bound for τ here uses precisely $G_2 \geq 0$. Hence $b \in I$, with both strict endpoints respected. Finally $G_4, G_5, G_6 \geq 0$ are exactly $L(h) \leq b \leq \min\{U_O(h), U_S(h)\}$. Proposition B.23 now supplies every original physical band of one complete matrix. The inherited high-value necessary bounds apply to this complete representative as specified in §B.13.

All eight functions are continuous on a neighborhood of the closed box because their denominators are bounded away from zero there. The right side of (97) is therefore a closed subset of a compact box. No previously strict cofactor or first-column condition has been replaced by a boundary case. The independent upper-target exclusion already proves $F < f_1$ on this set. $\square$

Write $\zeta = (\beta, V, Y, Z, v, h, b, R)$, so that $z = (\beta, V, s, Y, Z, v, h, b, R)$ and $J = \partial(G_1, \dots, G_8)/\partial\zeta$ at fixed s .

Proposition B.27 (Uniform slack coordinates). *For every $z \in \mathcal{B}$, $J(z)$ is invertible. In the local coordinates $(s, G_1, \dots, G_8)$,*

$$\frac{\partial R}{\partial G_i} < -\frac{1}{250} \quad (1 \leq i \leq 8). \quad (98)$$

Finite rational verification of the proposition. The repaired certificate specifies a fixed rational matrix C_0 , a rational vector y_0 , and positive rational radii ρ_i . With the rational midpoint of $\mathcal{B}$ as center, exact interval Hessians enclose every entry of J by its central value plus its variation over all nine coordinates. The verifier checks, for the entire box,

$$\mathcal{E} = I - C_0^T J^T, \quad \|\mathcal{E}\|_\infty < \frac{7}{50} < 1. \quad (99)$$

Thus $C_0^T J^T$, and hence J , are invertible. Let $e_R = (0, 0, 0, 0, 0, 0, 0, 1)^T$. The same verifier bounds the centered residual $C_0^T(e_R - J^T y_0)$ and proves componentwise that

$$\left| C_0^T(e_R - J^T y_0) \right|_i + \sum_j |\mathcal{E}_{ij}| \rho_j < \rho_i.$$

Consequently the affine map $y \mapsto y + C_0^T(e_R - J^T y)$ maps $y_0 + [-\rho, \rho]$ strictly into itself, uniformly in z . Its unique fixed point solves $J^T y = e_R$, so $y_i = \partial R / \partial G_i$. In the stated order the exact upper bounds $y_{0,i} + \rho_i$ are

$$\begin{array}{cccc} -\frac{31023639}{10^9}, & -\frac{184717889}{10^9}, & -\frac{47423419}{500000000}, & -\frac{5888171}{25000000}, \\ -\frac{380657}{25000000}, & -\frac{834537}{62500000}, & -\frac{9580477}{200000000}, & -\frac{4195499}{500000000}. \end{array}$$

Each is strictly smaller than $-1/250$. These are interval conclusions on the full nine-variable box, not gradients at its center or a finite set of sampled points. The precise certificate data and verifier are identified in §B.13.

Lemma B.28 (A supercritical maximum has all eight contacts). *If $\mathcal{C}$ contains a point of height greater than α , it has a maximizing point z_* with $f_0 < F(z_*) < f_1$, and every such maximizing point satisfies $G_1(z_*) = \dots = G_8(z_*) = 0$.*

Proof. Compactness gives a maximum. Under the stated hypothesis its height is greater than $\alpha > f_0$; the upper-target exclusion makes it less than f_1 . This existence argument does not require an additional numerical or rational witness.

Suppose $G_i(z_*) > 0$. The inverse-function theorem, using Proposition B.27, supplies a local smooth inverse in the eight slack coordinates at fixed s . Decrease this one slack by a positive amount u , leaving the other seven fixed. For sufficiently small u , all eight slacks remain nonnegative. At $u = 0$, (98) gives $dR/du > 1/250$, so a short arc strictly increases height. All the other strict qualifications persist by continuity. The reconstruction theorem therefore makes this a feasible arc in the original complete scale model, with height still above f_0 .

The artificial faces of $\mathcal{B}$ are not additional physical constraints. In constructing this short arc one works in an open neighborhood on which the reconstruction denominators and strict qualifications stay positive, and imposes only the eight physical slacks. If a point of the arc appeared to leave $\mathcal{B}$, the already proved global localization, applied to that original feasible point, would force it back into $\mathcal{B}$. Thus the arc belongs to $\mathcal{C}$, contradicting maximality. This proves all eight equalities. $\square$

B.12. The contact carrier and its exact elimination. The contact family is a consequence of the preceding maximum argument. Indeed $G_1 = G_2 = G_8 = 0$ gives $t = X = Z = 1$, whence $T = s$. The equations $G_3 = G_4 = 0$ give $b = k = r + Ea$. Also

$$G_5 - G_6 = Aq(h - s),$$

so $h = s$, and the last middle contacts are $O_{01} = -1$, $S_{01} = -p$, and $O_{21} = 1$. These contacts use the original shared matrix variables.

Put $\epsilon = r - 2 = R - 1$, and let $\mathcal{L} = (2 - s)(1 + s)$. We give the elimination from these contacts to the carrier explicitly. The top middle contact gives $a = (1 - \beta)s$; the bottom middle contact gives $\epsilon + Ea = s(\beta - s)$. Combining the latter with $b = r + Ea$ and the top tail contact gives

$$b = 2 + s(\beta - s), \quad (1 - \beta)v = s(\beta - s).$$

The bottom tail contact is $Eb = \epsilon + (\beta - s)v$. Substitute these expressions into $(b - r)b = a(Eb)$. After clearing only $1 - \beta > 0$, the equation reduces to $2s(\beta - s) = \epsilon\mathcal{L}$. Thus

$$\begin{aligned} \beta &= s + \frac{\epsilon\mathcal{L}}{2s}, & v &= \frac{s(\beta - s)}{1 - \beta}, & b &= 2 + s(\beta - s), & a &= (1 - \beta)s, \\ Y &= 1 - \frac{\epsilon(s - v)}{s + v}, & V &= 1 - \frac{\epsilon[2 + \beta(s - v)]}{s + v}. \end{aligned} \tag{100}$$

For completeness, the last two formulas are the solution of the middle and tail receiver equations

$$R - a = Y(1 + \beta s) - Vs, \quad b - R = Yq + Vv.$$

Their determinant is $v(1 + \beta s) + sq = s + v > 0$. The remaining first-column relation, from $Eb + Zj + Tg = 1$, $j = (b - p)/Y$, and $g = s/V$, is

$$\epsilon + (\beta - s)v + \frac{b - 1 - s}{Y} + \frac{s^2}{V} = 1. \tag{101}$$

To specify a compact integer polynomial, define

$$\begin{aligned} d_* &= 2s(1 - s) - \epsilon\mathcal{L}, & B_* &= 2s^2 + \epsilon\mathcal{L}, \\ Y_n &= s(1 - s)(1 - \epsilon) + \epsilon^2\mathcal{L}, & Y_d &= s(1 - s), \\ V_n &= 2s^2(1 - s) - 2\epsilon d_* - \epsilon B_* s(1 - s) + \epsilon^2 B_* \mathcal{L}, & V_d &= 2s^2(1 - s). \end{aligned}$$

The subscript on d_* distinguishes it from the geometric entrance quantity $d = tX - Y$. Direct substitution in (100) gives $Y = Y_n/Y_d$ and $V = V_n/V_d$. Define

$$G(s, \epsilon) = [2d_*(1 - \epsilon) - \epsilon^2 \mathcal{L}^2]Y_n V_n - d_*(2 - 2s + \epsilon \mathcal{L})Y_d V_n - 4d_* s^4(1 - s)Y_n. \quad (102)$$

This polynomial has degree 10 in s and degree 7 in ϵ . With β, v, b, Y, V given by the same carrier dictionary, its exact relation to the residual is

$$\epsilon + (\beta - s)v + \frac{b - 1 - s}{Y} + \frac{s^2}{V} - 1 = -\frac{G(s, \epsilon)}{2d_* Y_n V_n}. \quad (103)$$

Here $0 < s < \beta < 1$, $d_* = 2s(1 - \beta) > 0$, $Y_n = YY_d > 0$, and $V_n = VV_d > 0$. Thus no zero-denominator component has entered the carrier equation: (101) is equivalent to $G(s, \epsilon) = 0$ in the proved physical domain.

Lemma B.29 (Stationarity on the attained contact curve). *At the supercritical maximizing point of Lemma B.28,*

$$G(s, \epsilon) = 0, \quad G_s(s, \epsilon) = 0.$$

Proof. The same invertible Jacobian makes all eight coordinates ζ smooth functions of s when the eight slacks are held at zero. For s in a two-sided open interval about its maximizing value, all strict physical qualifications persist. The contact equalities preserve the eight physical constraints, and the reconstruction therefore gives a curve of complete physical representatives. As in Lemma B.28, global localization handles any artificial face of the recorded box. Since the maximum is strictly above f_0 , neither an auxiliary height endpoint nor an original physical endpoint prevents this two-sided curve. Maximality gives $\epsilon'(s) = R'(s) = 0$. Differentiating the identity $G(s, \epsilon(s)) = 0$ now gives $G_s = 0$. This argument does not assume $G_\epsilon \neq 0$; the full eight-slack Jacobian has already supplied the local parameter. $\square$

Proposition B.30 (Exact elimination identity). *For the polynomial (102) and the normalization of P_{61} in (2),*

$$\text{Res}_s(G, G_s) = \frac{\epsilon^{34}(\epsilon - 2)(\epsilon - 1)^2(\epsilon + 1)^7(\epsilon^2 - 2\epsilon + 2)^2}{131072} \cdot P_{61}(4 + 2\epsilon). \quad (104)$$

This is a polynomial identity over $\mathbb{Q}$, checked by computing the resultant and verifying that the cleared polynomial difference is identically zero. Its degree in ϵ is 109; the factor outside P_{61} has degree 48. The carrier dictionary and (103) are checked separately, so the identity is connected to the complete physical maximum before root identification is used.

At the maximum,

$$\frac{132517}{2000000} < \epsilon < \frac{3313}{50000}.$$

Every factor outside P_{61} in (104) is nonzero there. The two equations of Lemma B.29 imply that the resultant vanishes, and therefore $P_{61}(F) = 0$.

Section 2 has already proved that α is the unique root of this polynomial in $(4, 5)$, which contains $[f_0, f_1]$. Consequently the maximizing height must be α , contradicting the assumed supercritical point. The representative reduction then proves Theorem 6.2 for every source in $\mathcal{K}$. Neither a stationary-point calculation nor the root-isolation certificate alone supplies this global bound.

B.13. Source dependencies and the finite proof payloads. The analytic representative and reconstruction arguments are proved in §B.2–§B.7 above. The historical labels in the following crosswalk identify their sources and the frozen finite certificates still used by those proofs. The source-to-model arguments, the exact certificate acceptance, and the final composition have separate responsibilities; a localization tree does not by itself establish the preceding representative theorems.

Responsibility	Proof in this article and archived origin
High-value signs and quantitative budgets	Lemma B.2 and Lemma B.4; V11/V13 safe-tail argument and V14 budgets.
Common first-column reconstruction	Lemmas B.6–B.8; V15 and V19.
Simultaneous middle column	Lemma B.9; V15/V18, with one shared head, row scale and pivot.
Tail polygon and wall reconstruction	Lemmas B.10–B.14; V16–V19, including zero-scale and boundary qualifications.
Same-head wall maximum	Theorem B.16; V20, with the separate finite C19 exclusion of Proposition B.15.
Head entrance	Proposition B.17 and Corollary B.18; V22’s finite difference exclusion and analytic reconstruction.
Exceptional contacts and prescribed target	Proposition B.19, Lemma B.20, and Proposition B.21; V23/V24.
Row-scale equivalence	Proposition B.23; V25.
Localization and exact canonical height	The later sections of this appendix; the repaired V28 localization, transport, peak-neighborhood and algebraic modules.

The finite representative exclusions have three distinct scopes: eight C19 systems in Proposition B.15, eight difference systems in Proposition B.17, and six exceptional-contact systems in Proposition B.19. Their continuous source coverage is derived above before their accepted trees are used. The safe-tail polynomial positivity checks are a separate fixed finite certificate. The Supplementary Index, S3, identifies the nested V24 replay with its full-chain option and the recorded acceptance of all 22 exclusion trees. Neither the V28 localization entry nor the final global composer re-executes these upstream trees automatically.

The reproducibility unit for the localization and peak-height statements in the later part of this appendix is `rho5_v28_exact.zip`, with internal root `rho5_v28_exact/`. Its top-level report, Sections 1–5, specifies the repaired localization and its interfaces. The `RH05_V26_REBUILT/REPORT.md` retained within it supplies the mathematical argument in its Theorem 6.1 and Sections 7–10. The latter report’s superseded tree counts and old Jacobian contraction bound are not the data used in this appendix.

The order of the mathematical dependencies is essential:

- (1) The same-head wall maximum and the simultaneous reconstruction first give a complete representative at the prescribed attainable height. Compactness and minimization in E are applied to the complete fixed-head fiber. Only then do the exceptional-contact exclusions and the strict capacity comparisons force $E = L_D$, $k = K_{24}$, and the five simultaneous middle-column tests. These steps are proved in Proposition B.21; the V25 handoff is a historical restatement, not an additional premise.
- (2) The high-value bounds (77) are used only in their declared positive canonical domain. The common source budgets and Lemma B.7 supply $Ek < 1 - T$, $F + \beta < 5$, and $g < 3 - F/2$ before wall selection; the latter becomes $g < 3 - r$ on the wall. The earlier-pivot bound supplies $k \leq 9/4$. Proposition B.17 and Corollary B.18 supply the three entrance inequalities used in localization rows 31–33. These conclusions are not inferred from an arbitrary signed matrix or from an observed extremal candidate.
- (3) The scale dictionary, BASE, strict scale endpoints, and both directions of the complete fiber equivalence are proved in §B.8. The archived V25 report gives their original derivation and exact symbolic checks. This appendix does not use its later finite elimination of β .
- (4) The repaired V28 certificate supplies continuous root-box coverage, fixed-target transport margins, the upper-target exclusion, the nine-dimensional slack theorem, and the exact carrier/resultant connection. None uses the complete X -domain or negative-diagonal bound as a premise. Root uniqueness and the definition of the exact constant are the separate result of Section 2.

The Supplementary Index identifies the complete replay entry point and the separate localization, symbolic-interface, target-transport, Jacobian, carrier, and resultant checks. Acceptance requires every localization graft and all required modules. Discovery output or summary counts do not replace that finite replay. The representative reductions and their explicit finite inputs have distinct domains: the canonical-height certificate alone does not prove the other sign chambers, degenerate entrances, or the full matrix theorem.

APPENDIX C. DATA AND COORDINATES FOR THE TWO-GAP TRANSPORT

C.1. Complete coordinates and source inequalities. The coordinates of ξ have the order in (35). All entries of the vectors u, x, v, q have indices

0, 1, 2. Matrices D, S, O below use these same indices. The scalar coordinates A, B are entries of D , not the original input matrix. Define

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r - \sigma + cB \\ dk & r - \tau + dA & w + dB \end{pmatrix}, \quad S = D + xq^T, \quad O = S + uv^T. \quad (105)$$

The associated real 5×5 matrix is

$$M(\xi) = \begin{pmatrix} 1 & -e & v^T \\ \beta & p - e\beta & (q + \beta v)^T \\ u & px - eu & O \end{pmatrix}. \quad (106)$$

Its successive leading Schur matrices, before the last scalar, are

$$\begin{pmatrix} p & q^T \\ px & S \end{pmatrix}, \quad D, \quad \begin{pmatrix} r & r - \sigma \\ r - \tau & w \end{pmatrix}. \quad (107)$$

These are algebraic identities for $p, k \neq 0$; elimination at the fourth pivot additionally uses $r \neq 0$. In particular, changing the source coordinates in the transport changes one complete matrix through (106), with all its stage bounds retained.

For a scalar expression f with band $|f| \leq b$, write $f^+ = b - f$ and $f^- = b + f$. The superscripts in the tables below always denote these slacks, not the positive and negative parts of a number. The full polynomial list consists of the following bands, with the identically zero slack $D_{00}^+ = k - k$ omitted:

$$\begin{aligned} e^\pm &= 1 \mp e, & \beta^\pm &= 1 \mp \beta, & \text{head}^\pm &= 1 \mp (p - e\beta), \\ u_i^\pm &= 1 \mp u_i, & x_i^\pm &= 1 \mp x_i, & v_i^\pm &= 1 \mp v_i, & q_i^\pm &= p \mp q_i, \\ L_i^\pm &= 1 \mp (px_i - eu_i), & P_i^\pm &= 1 \mp (q_i + \beta v_i), \\ D_{ij}^\pm &= k \mp D_{ij}, & S_{ij}^\pm &= p \mp S_{ij}, & O_{ij}^\pm &= 1 \mp O_{ij} & (0 \leq i, j \leq 2). \end{aligned} \quad (108)$$

There are 95 nonzero polynomials in (108). The remaining nine are

$$r + w, \quad r - w, \quad p, \quad k, \quad r, \quad \sigma, \quad \tau, \quad r - \sigma, \quad r - \tau. \quad (109)$$

All 104 are required to be nonnegative for a physical source, with $p, k, r > 0$. On the boxes used here the strict positivity of these three pivots follows from the stronger bounds below. These conditions give $\|M\|_{\max} = 1$, successive stage maxima p, k, r , and legal leading complete pivots $1, p, k, r$, by (107). Since $w < 0$ on these boxes, the last pivot is $-F$ with $F > 0$. No strict inequality between competing maximal entries is required, so tied pivots are included.

For each j , the components of $\Phi_{j,p}$ are the twenty slacks in Table 3, in its displayed order, followed by $r + w, \sigma, \tau$. The row index in that table is the index within this selected list; it is not a numbering of all 104 source

polynomials. The Jacobian columns have the order

$$y = (k, r, w, A, B, c, d, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2, \sigma, \tau). \quad (110)$$

Thus J_j is a 23×23 matrix. Its entries remain polynomials in all 24 coordinates, including the parameter p .

Table 3: The twenty selected source slacks in each local chart.

Index	$j = 0$	$j = 1$	$j = 2$	$j = 3$
0	e^+	e^+	e^+	e^+
1	u_0^+	u_0^+	u_0^+	u_0^+
2	x_0^+	x_0^+	x_0^+	x_0^+
3	S_{01}^-	D_{01}^-	S_{01}^-	D_{01}^-
4	O_{01}^-	O_{01}^-	O_{01}^-	O_{01}^-
5	D_{02}^-	S_{02}^-	D_{02}^+	S_{02}^+
6	O_{02}^-	O_{02}^-	O_{02}^+	O_{02}^+
7	P_1^-	P_1^+	x_1^+	x_1^-
8	D_{10}^+	D_{10}^+	L_1^+	L_1^-
9	S_{10}^+	S_{10}^+	P_1^-	P_1^+
10	O_{10}^+	O_{10}^+	O_{10}^-	O_{10}^+
11	O_{11}^+	O_{11}^+	D_{11}^+	O_{11}^+
12	O_{12}^+	O_{12}^+	O_{11}^+	D_{12}^+
13	x_2^+	x_2^-	O_{12}^+	O_{12}^+
14	L_2^+	L_2^-	P_2^-	P_2^+
15	P_2^+	P_2^-	D_{20}^+	D_{20}^+
16	O_{20}^-	O_{20}^+	S_{20}^+	S_{20}^+
17	D_{21}^+	O_{21}^+	O_{20}^+	O_{20}^+
18	O_{21}^+	D_{22}^-	O_{21}^+	O_{21}^+
19	O_{22}^-	O_{22}^-	O_{22}^-	O_{22}^-

C.2. Exact rational centres. The following tables specify every coordinate of the four centres exactly. They are rational reference points for uniform estimates; no optimality or exact contact identity is assumed at a centre. The twelve common coordinates are

$$\begin{aligned}
(c_j)_k &= K_* := \frac{518617093}{2500000000}, & (c_j)_r &= R_* := \frac{2066258539}{10^9}, \\
(c_j)_w &= -R_*, \\
(c_j)_p &= \frac{726612331}{5000000000}, & (c_j)_\beta &= \frac{617532517}{10^9}, \\
(c_j)_{v_0} &= -\frac{581690673}{10^9}, & (c_j)_{q_0} &= -\frac{159528331}{2500000000}, \\
(c_j)_e &= (c_j)_{u_0} = (c_j)_{x_0} = 1, & (c_j)_\sigma &= (c_j)_\tau = 0.
\end{aligned} \quad (111)$$

For a compact exact table of the other twelve coordinates, define

$$\begin{aligned}
a_0 &= \frac{10833981}{62500000}, & a_1 &= \frac{47361589}{10^9}, \\
a_2 &= \frac{194787661}{250000000}, & a_3 &= \frac{226612331}{500000000}, \\
a_4 &= \frac{194712659}{200000000}, & a_5 &= \frac{194705107}{10^9}, \\
a_6 &= \frac{639940483}{500000000}, & a_7 &= \frac{175952653}{200000000}.
\end{aligned} \tag{112}$$

TABLE 4. The remaining coordinates of the four rational centres.

Coordinate	c_0	c_1	c_2	c_3
A	$-a_0$	$-K_*$	$-a_0$	$-K_*$
B	$-K_*$	$-a_0$	K_*	a_0
c	1	1	$-a_1$	a_1
d	$-a_1$	a_1	1	1
u_1	a_2	a_2	a_3	$-a_3$
u_2	a_3	$-a_3$	a_2	a_2
x_1	a_4	a_4	1	-1
x_2	1	-1	a_4	a_4
v_1	a_3	a_5	a_3	a_5
v_2	a_5	a_3	$-a_5$	$-a_3$
q_1	$-a_6$	a_7	$-a_6$	a_7
q_2	a_7	$-a_6$	$-a_7$	a_6

In particular, $(c_j)_r + (c_j)_w = 0$, and $R_* - R = 2065458539/10^9$. A full outer box allows every coordinate, including p , to vary independently through its radius R . Only the evolution from an individual source fixes that source's value of p .

C.3. Finite rational verification of the differential bounds. The companion proof-data table `two_gap_certificate.json` contains the four 23×23 rational matrices C_j , as well as the coordinate order, centres, and selected row labels. Its data schema is `V36_TWO_GAP_FLOW_V1`. Rows of C_j have the order of y in (110), and columns have the order of $\Phi_{j,p}$ in (37). The mathematical role of this finite table is completely specified by the inequalities below; the file name or its hash is not a substitute for those checks.

Each source slack has degree at most two. Thus J_j is affine in ξ , as is $E_j = I - C_j J_j$. For an affine entry $E_{j,ab}$, the exact whole-box bound used is

$$\sup_{\xi \in Q_j} |E_{j,ab}(\xi)| \leq |E_{j,ab}(c_j)| + R \sum_{m=0}^{23} |\partial_{\xi_m} E_{j,ab}|. \tag{113}$$

After summing across each row and taking the maximum, the right side is η in (42) for each of the four charts. The sum in (113) includes the p

coordinate. This is the explicit uniform parameter check, rather than a computation at four parameter samples.

Since $\|I - C_j J_j\|_\infty < 1$, the Neumann series makes $C_j J_j$ invertible. Both matrices are square, so J_j is invertible and

$$J_j^{-1} = (C_j J_j)^{-1} C_j.$$

For $g \in \{\sigma, \tau\}$, put $v_{j,g}^0 = C_j e_g$ and $n_{j,g} = \|v_{j,g}^0\|_\infty$. The exact inverse direction $v_{j,g} = J_j^{-1} e_g$ obeys

$$\|v_{j,g} - v_{j,g}^0\|_\infty \leq \frac{\eta n_{j,g}}{1 - \eta}, \quad \|v_{j,g}\|_\infty \leq \kappa_{j,g} := \frac{n_{j,g}}{1 - \eta}. \quad (114)$$

The last two rows of $J_j v_{j,g} = e_g$ give exactly

$$((v_{j,\sigma})_\sigma, (v_{j,\sigma})_\tau) = (1, 0), \quad ((v_{j,\tau})_\sigma, (v_{j,\tau})_\tau) = (0, 1).$$

Using $F = r - w - \sigma - \tau + \sigma\tau/r$, differentiation gives

$$D_y F v_{j,\sigma} = (v_{j,\sigma})_r - (v_{j,\sigma})_w - 1 + \frac{\tau}{r} - \frac{\sigma\tau}{r^2} (v_{j,\sigma})_r. \quad (115)$$

For $v_{j,\tau}$, interchange σ and τ in the single-gap term. On the full outer box, $r \geq r_{\min} > 0$ and $|\sigma|, |\tau| \leq R$, even when a point of the box is not physical. Consequently the following rational number is a valid upper bound in either case:

$$\begin{aligned} U_{j,g} = (v_{j,g}^0)_r - (v_{j,g}^0)_w - 1 + \frac{2\eta n_{j,g}}{1 - \eta} + \frac{R}{r_{\min}} \\ + \frac{R^2 n_{j,g}}{(1 - \eta) r_{\min}^2}. \end{aligned} \quad (116)$$

For clarity, all eight directional results can be specified by four speed numerators and four height bounds. Set

$$D_v = 9603274520441553881$$

and

$$\begin{aligned} H_0 &= -\frac{28753988041084852657186021690895856248996134979}{102421779166357566006286717151929765002500000000}, \\ H_1 &= -\frac{12333488469648866145304175510636026684107958389}{102421779166357566006286717151929765002500000000}, \\ H_2 &= -\frac{155271525823737135234142655316615127130663766999}{204843558332715132012573434303859530005000000000}, \\ H_3 &= -\frac{4226128882854719189167870582514468491033177797}{12802722395794695750785839643991220625312500000}. \end{aligned}$$

Substitution of the rational data into (114) and (116) gives Table 5. Every speed in the table is less than 3, and each H_i is at most $-1/10$.

It remains to verify the unselected physical inequalities on the entire box. If a source polynomial is $P(\xi) = \sum_{|\nu| \leq 2} a_\nu \xi^\nu$, its quadratic Taylor expansion

TABLE 5. Exact bounds for the eight inverse-coordinate directions.

Chart	Direction	$D_v \kappa_{j,g}$	$U_{j,g}$
0	σ	146686939720000000000	H_0
0	τ	245661530530000000000	H_1
1	σ	123937294620000000000	H_2
1	τ	100000000000000000000	H_3
2	σ	100000000000000000000	H_3
2	τ	123937294620000000000	H_2
3	σ	245661530530000000000	H_1
3	τ	146686939720000000000	H_0

gives the rational lower bound

$$P(\xi) \geq P(c_j) - R \sum_{m=0}^{23} |\partial_{\xi_m} P(c_j)| - R^2 \sum_{|\nu|=2} |a_\nu| \quad (\xi \in Q_j). \quad (117)$$

The 23 selected coordinate polynomials are distinct. For every one of the remaining $104 - 23 = 81$ source polynomials, the right side of (117) is at least μ in (44), in each chart. This supplies the physical interior margin used during continuation. The selected twenty band slacks and $r + w$ do not need an interior margin: their nonnegative initial values are conserved exactly, and the last two selected coordinates are the explicitly decreasing nonnegative gaps. All comparisons in this subsection are comparisons of rational numbers.

C.4. Qualification of a closed source box. For a closed rational enclosure $Q = \prod_{m=0}^{23} [L_m, U_m]$, define

$$d_j(Q) = \max_m \max\{|L_m - (c_j)_m|, |U_m - (c_j)_m|\}.$$

An immediately checkable sufficient guard for this transport is

$$L_r > 0, \quad L_\sigma, L_\tau \geq 0, \quad d_j(Q) + 3(U_\sigma + U_\tau) < R. \quad (118)$$

Every complete physical source represented by such an enclosure satisfies (40). The enclosure itself need not consist of physical points; the conclusion is conditional on the complete source inequalities, and does not infer simultaneous feasibility from separate coordinate intervals.

For a proper negative-diagonal source described instead by (r, s, t) , the conversion is the actual affine substitution $w = -r, \sigma = r - s, \tau = r - t$. For example, the valid enclosing interval for σ is $[L_r - U_s, U_r - L_s]$, and similarly for τ . This conversion may overestimate the image and may therefore fail the sufficient guard; it does not justify replacing the image by an unrelated smaller box. If (118) is not satisfied, including its equality boundary, this transport rule supplies no terminal there. The complete proof graph must retain that region for another qualified rule or for further subdivision. Zero

gap faces are included whenever the strict guard holds, because a zero gap requires no evolution in its corresponding direction.

APPENDIX D. SOURCE ACCOUNTING AND REPRODUCIBILITY

D.1. Three levels of coverage. The numbers 54, 566 and 469 refer to different levels of the proof. A terminal is a local region with an accepted conclusion. A constituent source may require a tree of such terminals. An original parent responsibility may in turn require many constituent sources. In particular, the last 54-terminal anchor of Section 8.2 discharges one constituent-source obligation, not 54 original parents.

The four residual original-root responsibilities have the following bindings. The indices identify nodes in the retained root; the binary paths identify those same nodes independently of the constituent source numbering.

Box	Original index	Original-root path	Constituents
Q_1	338726	10011000010101010	111
Q_2	435003	10111000001100011000010111011	240
Q_3	563285	1101000101111	159
Q_4	675104	1110101001111	56

Accordingly,

$$111 + 240 + 159 + 56 = 566$$

constituent sources discharge four original obligations,

whereas the original-parent accounting is

$$465 \text{ previously discharged obligations} + 4 = 469.$$

There is no proposed bijection between the 566 constituents and the 469 original parents. For each of the four parents, the constituent target-path set must equal its complete frozen source-path set, with no omission or duplicate. Each target also retains its mathematical acceptance and original binding evidence. Counts alone would not establish any of these assertions.

The final anchor within Q_2 contains 48 contradiction terminals, three height or transport terminals, and three canonical-source threshold exclusions. These are the same 54 positions under the terminal semantics of Section 8. Combining them proves safety at α , not that every region is empty at γ . Local paths are interpreted relative to their frozen source and its recorded transformations. In particular, a local path is not silently concatenated with an original-root path to manufacture a source binding.

D.2. Why the original root still contains four open records. The retained root has 749,693 nodes: 374,846 splits, 374,839 contradiction terminals, four α -safe terminals and four original open records. Its replay proves exactly this partial cover. The four source-bound parent proofs above supply the missing conclusions. The final overlay identifies each parent with the corresponding original-root path and closed rational box, and checks

the identities of the accepted parent records. Applying the mixed coverage lemma then gives an effective open count of zero.

The original root is preserved byte for byte. Its four open records are neither deleted nor relabelled as contradiction terminals. This separation permits an independent reader to replay the original tree, replay the additional mathematical arguments and verify their precise attachment points. A root replay that permits the four recorded open positions is therefore an intermediate check, not acceptance of an incomplete final proof.

D.3. Locating the numerical example and its ancestors. The public release provides the 18.29 MB archive `rho5-final-anchor-example.zip`. The input for (33) is `rho5_v53_exact/proof/TERMINAL_DETAILS.json`; the corresponding `SOURCE_CERTIFICATE.json` and `DEPTH19_COMPLETE_SOURCE_PROJECTION.json` bind it to the complete depth-19 high source. The terminal-details file has SHA-256

```
11d3e273fff5673cc9cb3ed9673860f388e685bd7ae3b769de0f89
d39e20ecc1.
```

The archive contains all 49 weighted rows and all 67 coordinate bounds used above, not only the two displayed rows.

The included restoration and joint-verification entry points replay the complete 54-terminal anchor. Their successful result discharges only that anchor; the standalone whole-parent and whole-domain flags remain false. The complete final-cover archive additionally contains the frozen 239-source baseline, the completed parent record and the original-root overlay. The Supplementary Index, S7, identifies these objects and their hashes. Thus the source-to-root chain for the example is

```
depth-19 source and its contradiction
→ complete 54-terminal anchor at 10011101
→ source 240 of 240 in parent 435003
→  $Q_2$  at its recorded original-root position.
```

The arrows denote complete source-cover compositions, not implications from a terminal count or a receipt name.

D.4. Independent verification and the supplementary map. The fixed public release is available at <https://github.com/hkjtsgmc79-boop/rho5-proof/releases/tag/v1.0.1>. Its index links all 24 original release assets; the archive inventory is approximately 4.494 GB. The repository's `docs/DOWNLOADS.md` gives sizes and archive identities, `docs/VERIFY.md` gives the existing executable entry points, and `docs/PROOF_MAP.md` distinguishes the written proof, exact acceptors and the companion Lean statements. Large-file transport parts reconstruct the original archive bytes and do not partition the mathematical responsibility.

The supplementary index is organized by mathematical dependency:

Part	Responsibility and verification material
S1	Exact constant, isolated root, attaining parameters and matrix.
S2	Early pivot bounds, complete lift and saturation, including the zero-pivot qualifications.
S3	Canonical representative, reconstruction and canonical height; the V24–V28 upstream dependencies.
S4	All seven complete- X certificate components and their respective frozen entry points.
S5	Six-variable elimination in the negative-diagonal domain and the source model used by the capacity certificates.
S6	Necessary-row, height and transport acceptance rules, with their distinct thresholds and conclusions.
S7	The preserved root, four parent bindings, the final anchor and source-cover composition.
S8	The analytical, upstream, X and negative-diagonal archive collections, replay commands and scope of reproduction.

Dependency order for a complete reproduction. The following stages follow mathematical dependencies, rather than the chronological labels used during discovery.

- (1) Fix the polynomial and attaining-matrix inputs (S1), and check the analytic source reductions and their zero and tie cases (S2).
- (2) Check the canonical representative and reconstruction inputs, including the nested exclusion trees, and then the complete localization and canonical-height certificates (S3).
- (3) Check all seven complete- X components (S4), using the preceding height theorem for their qualified local exits. Assemble their stated domains into the complete X bound.
- (4) Check the seventeen-coordinate negative-diagonal source model and the acceptance rules (S5–S6). Replay the preserved original root, retaining its four explicitly identified open positions.
- (5) Replay the mathematical contributions to each of the four residual parents: their inherited closed complements, verified continuations, and height or transport certificates. Include the final 54-terminal anchor. Match every contribution to its actual parent source and ancestor transformations.
- (6) Compose the four complete parent covers with the preserved root (S7). Combine the resulting negative-diagonal bound with the complete X bound, saturation and attainment.

The dependency direction in the middle three stages is canonical height, then complete X , then negative-diagonal safety. The final negative-diagonal conclusion is not an input to its own upstream canonical-height proof. The

frozen component manifests must be followed recursively; neither the release’s asset count nor the final composer’s 357 direct file dependencies gives the full transitive mathematical dependency set.

Acceptance conditions. At each stage, retain the input manifest, verifier identity, exact command or API call, software versions, exit status, mathematical conclusion and elapsed time. A contradiction terminal requires its nonnegative weights and rational upper/lower bounds to be checked. A conditional row requires its guard on the whole box, and a height or transport terminal requires its stated source qualifications and threshold. Such a terminal concludes safety at α ; it is not a contradiction at the smaller rational threshold γ .

For the preserved root, the accepted intermediate result has exactly 749,693 nodes, 374,839 contradiction terminals, four α -safe terminals and the four stated open positions. Each completed parent must cover its entire frozen target-path set without omissions or duplicates: respectively 111, 240, 159 and 56 constituent sources. The final composition must match all four parent boxes and paths to the actual root and obtain effective open count zero. Counts are cross-checks on the cover; they do not replace these source and mathematical checks. A missing dependency, failed guard, changed model, incomplete target set or unmatched source invalidates the proposed acceptance at that stage.

Executable scope and implementation trust. The Supplementary Index, S8, distinguishes command-line entries from lower-level mathematical interfaces and from record composers. The small anchor has a self-contained replay entry. The original negative-diagonal root also has a frozen mathematical replay command. The four-parent chain uses separate mathematical interfaces followed by record composition; its historical configurations include absolute workspace paths. Reproduction on another installation requires an explicit location mapping and fresh output locations while preserving all mathematical input identities and source checks. The dated four-parent orchestration described in Section 11 and Supplementary Index S8 supplies such mappings for the recorded negative-diagonal replay. This supplements the historical lower-level interfaces; it does not regenerate their entire upstream dependency chain. There is not yet a tested location-independent driver that performs this entire transitive chain in one fresh environment. This packaging limitation does not negate the documented component runs and final composition; it limits the presently demonstrated portability of their complete re-execution.

The full computational claim trusts the implementations of the exact rational and interval acceptors, their parsers and row/model generation, and the code that checks source-cover composition. In particular, exact arithmetic prevents rounding uncertainty but does not by itself prove that a guard was applied to the correct source or that all branches were covered. The written arguments in this article establish the intended mathematical rules; the frozen Python/C++ programs implement their finite instances.

Independent reimplementations of those rules would reduce reliance on these particular implementations. No discovery optimizer, floating-point search output or search heuristic is accepted as a proof terminal. Python assertion optimization must remain disabled during replay.

The exact Python and C++ acceptors are not asserted to be formally verified programs. Their correctness, the written analytic arguments and the source/domain composition remain part of this proof’s trust boundary. The companion first-phase Lean development records its remaining global safety hypotheses explicitly; a successful build or a certificate hash does not remove those hypotheses.

REFERENCES

- [1] Ankit Bisain, Alan Edelman, and John Urschel. A new upper bound for the growth factor in Gaussian elimination with complete pivoting. *Bulletin of the London Mathematical Society*, 57(5):1369–1387, 2025.
- [2] James Chen, Alan Edelman, and John Urschel. The largest 5th pivot may be the root of a 61st degree polynomial. *IMA Journal of Numerical Analysis*, 2026. Advance article, drag039; published online 7 September 2026. <https://doi.org/10.1093/imanum/drag039>. Theorem numbering follows <https://arxiv.org/abs/2602.20390v2>.
- [3] A. M. Cohen. A note on pivot size in Gaussian elimination. *Linear Algebra and its Applications*, 8(4):361–368, 1974.
- [4] C. W. Cryer. Pivot size in Gaussian elimination. *Numerische Mathematik*, 12:335–345, 1968.
- [5] Jane Day and Brian Peterson. Growth in Gaussian elimination. *The American Mathematical Monthly*, 95(6):489–513, 1988.
- [6] Nick Gould. On growth in Gaussian elimination with complete pivoting. *SIAM Journal on Matrix Analysis and Applications*, 12(2):354–361, 1991.
- [7] Nicholas J. Higham and Desmond J. Higham. Large growth factors in Gaussian elimination with pivoting. *SIAM Journal on Matrix Analysis and Applications*, 10(2):155–164, 1989.
- [8] Rikhav Shah and John Urschel. Entry growth in Gaussian elimination. <https://arxiv.org/abs/2608.19189v4>, 2026. Preprint; version 4, 31 August 2026; first submitted 19 August 2026.

(Qianli Ma) ZHEJIANG UNIVERSITY; WUJIE AI
Email address: qianli.ma@zju.edu.cn

(Chao Wu) ZHEJIANG UNIVERSITY
Email address: chao.wu@zju.edu.cn